Notes for Measure Theory 2nd edition, Donald Cohn

Lingfei Zhang, lfzhang@tamu.edu

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Jun 7, 2026. The following notes cover, with the help of Chat GPT, the introduction chapter, chapters 1–6 except section 4.5, and 10.1–10.4.

Introduction

For more details of Darboux's definition of the Riemann integral, see *Principles of Mathematical Analysis* by Walter Rudin, chapter 6.

For a slightly stronger version of the Fundamental Theorem of Calculus, see Rudin PMA, theorem 6.20.

For "Theorem 2 is valid for the Riemann integral...", see Rudin PMA, theorem 7.16.

After equation (2), "One can check this does not depend on the value of c...": Consider another number $s < c$ such that $f < s$ on $[a, b]$ (for $s > c$, switch the roles of s, c). Consider a partition $\mathcal{P}^c = \{a_i\}_{i=0}^k$ for $[0, c]$. Recall $a_0 = 0$. Let $j \in \{1, ..., k\}$ be the smallest index such that $s \leq a_j$, i.e., $j = \min\{i : a_i \geq s\}$. By defining $b_i = a_i, i = 0, ..., j-1$ and $b_j = s$. We see that all b_i thus defined form a partition $\mathcal{P}^s$ of $[0, s]$. Now $s(f, \mathcal{P}^s) = \sum_{i=1}^j b_{i-1}\text{meas}(B_i) = \sum_{i=1}^j a_{i-1}\text{meas}(B_i) = \sum_{i=1}^{j-1} a_{i-1}\text{meas}(A_i) + a_{j-1}\text{meas}(B_j)$ while $s(f, \mathcal{P}^c) = \sum_{i=1}^k a_{i-1}\text{meas}(A_i) = \sum_{i=1}^{j-1} a_{i-1}\text{meas}(A_i) + a_{j-1}\text{meas}(A_j) + \sum_{i=j+1}^k a_{i-1}\text{meas}(A_i)$. But notice $\sum_{i=j+1}^k a_{i-1}\text{meas}(A_i) = 0$ since no x is such that $f(x) \geq a_j$, so $s(f, \mathcal{P}^c) - s(f, \mathcal{P}^s) = a_{j-1}(\text{meas}(A_j) - \text{meas}(B_j))$. However, by definition (1), $A_j = B_j$. Therefore, $s(f, \mathcal{P}^c) = s(f, \mathcal{P}^s)$.

Now we've shown, for $s < c$, for any partition $\mathcal{P}^c$, we can find a partition $\mathcal{P}^s$ such that $s(f, \mathcal{P}^c) = s(f, \mathcal{P}^s)$, meaning the set of all possible values that $s(f, \mathcal{P}^c)$ can take is included by the set of all possible values that $s(f, \mathcal{P}^s)$ can take, implying $\sup_{\mathcal{P}^c} s(f, \mathcal{P}^c) \leq \sup_{\mathcal{P}^s} s(f, \mathcal{P}^s)$. Conversely, for any partition $\mathcal{P}^s = \{b_i\}_{i=1}^j$ of $[0, s]$, by simply including $b_{j+1} = c$ we have a partition $\mathcal{P}^c$ for $[0, c]$ with

$s(f, \mathcal{P}^c) = s(f, \mathcal{P}^s)$, and by a similar argument we have $\sup_{\mathcal{P}^s} s(f, \mathcal{P}^s) \leq \sup_{\mathcal{P}^c} s(f, \mathcal{P}^c)$. Therefore, $\sup_{\mathcal{P}^s} s(f, \mathcal{P}^s) = \sup_{\mathcal{P}^c} s(f, \mathcal{P}^c)$ and hence "this does not depend on the value of c".

1 Measures

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(d). We check for closure under finite unions. Suppose $A_1, ..., A_n \in \mathcal{A}$. Case 1: some A_i^c is finite, then $\cap_i A_i^c = (\cup_i A_i)^c$ is finite, so $\cup_i A_i \in \mathcal{A}$. Case 2: all A_i^c are infinite, which means all A_i must be finite, implying $\cup_i A_i$ is finite and so $\cup_i A_i \in \mathcal{A}$.

Note that $\mathcal{A}$ is not closed under countable unions. This is because we can divide X into two disjoint infinite sets X_1, X_2 (for countable X, we just put $x_1, x_3, ...$ into X_1 and $x_2, x_4, ...$ into X_2; for uncountable X we divide X' into such X_1', X_2' for a countable subset $X' \subset X$ and let one of X_1', X_2' be united with $X - X'$). Define $A_i = \{y_i\}, i \in \mathbb{N}$ for distinct $y_i \in X_1$. Then $\cup_i A_i$ is not finite, and neither is its complement as the complement includes X_2.

(f). To check for closure under countable union, recall the two cases in (d) above. Consider $A_1, ...$ instead of $A_1, ...A_n$; change "finite" to "countable".

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For proposition 1.1.5, that $\mathcal{B}(\mathbb{R}^d) \subset \mathcal{B}_3$ is not proven. See theorem 1.23, *Probability Theory, Third Edition* by Achim Klenke for a more rigorous and comprehensive treatment regarding the Borel sigma algebra of $\mathbb{R}^d$ and its generators.

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Note the remark "there are Borel sets that belong to none of them". So it is wrong to think that a set from the Borel sigma algebra can be explicitly expressed as (possibly grouped by parentheses) countable unions or intersections of open/closed sets.

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Fig 1.1. For example, we know from the proposition that $\mathcal{F} \subset \mathcal{G}_\delta$. So a countable intersection of sets in $\mathcal{F}$ is also a countable intersection of sets in $\mathcal{G}_\delta$, i.e., $\mathcal{F}_\sigma \subset \mathcal{G}_{\delta\sigma}$, etc.

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(d). Let $A_1, ..., A_n \in \mathcal{A}$ be disjoint. We check for finite additivity. Case 1: some A_i are such that A_i^c is finite $\implies A_i = \mathbb{N} - A_i^c$ is infinite. Note there can be only one such index i, otherwise, say A_j is also s.t. A_j^c is finite, then there exists $N_i, N_j \in \mathbb{N}$ such that all elements in A_i^c (respectively, A_j^c) are smaller than N_i (respectively, N_j). This means A_i^c (A_j^c) $\subset \{1, 2, ..., N_i\}$ ($\{1, 2, ..., N_j\}$) and that $\{N_i, N_i + 1, ...\} \subset A_i, \{N_j, N_j + 1, ...\} \subset A_j$, contradicting the fact that A_i, A_j are disjoint. So

A_k^c is infinite for all indices $k \in \{1, ..., n\} - \{i\}$ meaning all those A_k are finite. With the fact that A_i is infinite, the countable additivity holds.

Case 2: all $A_i, i = 1, ..., n$ are such that A_i^c is infinite. Then by the definition of $\mathcal{A}$, all the A_i are finite, and countable additivity holds.

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Line 11. On constructing B_i from A_i, note if we define $C_n = \cup_{i=1}^n A_i$, then $\cup_n C_n = \cup_i A_i = X$. It is obvious that we also have $C_n = \cup_{i=1}^n B_i$ and $X = \cup_n C_n = \cup_i B_i$.

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Notice line 8, the first term. We need to justify that $\sum_j (b_j - a_j)$ makes sense. We consider an essentially the same example.

Let's say we have a nonnegative, doubly-indexed sequence (like a table extending to infinity in two directions) $(a_{ij})_{i,j \in \mathbb{N}}$, and assume $\sum_j a_{ij} = 1/2^i$.

By theorem 8.3, Rudin's PMA, we know $\sum_j \sum_i a_{ij} = \sum_i \sum_j a_{ij} = 1$. Rudin's theorem 3.55 also says, if series $\sum s_n$ converges absolutely, then any rearrangement of it converges to the same limit.

Now lets do the "diagonal trick" and convert the doubly-indexed sequence (a_{ij}) into just a normal sequence (a_n). Is there a way we can show that $\sum a_n = 1$?

Let $f : \mathbb{N} \to \mathbb{N} \times \mathbb{N}$ be a bijection. Denote the sequence $a_{f(n)}$ by b_n. We want to show $\sum_{n=1}^\infty b_n = 1$. Denote $f(n) = (x_n, y_n)$.

Let the partial sums be $p_m := \sum_{n=1}^m b_n$. It's easy to see each p_m is upper bounded by 1. (Otherwise, it would contradict the fact that $\sum_i \sum_j a_{ij} = 1$.)

Recall the value of a series whose terms are nonnegative is the supremum (also the limit, as the partial sums are increasing) of its partial sums. So we still have to show that 1 is the least upper bound of $\{p_m\}_{m=1}^\infty$.

Let $\mathbb{I}$ denote the indicator function. Pick $0 < \epsilon < 1$. There is an $N \in \mathbb{N}$ such that $\sum_{i=1}^N 1/2^i > 1 - \epsilon/2$. Also, there is an M such that if $m > M$, then $\{b_1, ..., b_m\}$ contains enough terms from $\{a_{i,j}\}_{j=1}^\infty$ such that $1/2^i \geq \sum_{n=1}^m b_n \mathbb{I}\{x_n = i\} > 1/2^i - \epsilon/2N$, for any $i \in \{1, 2, ..., N\}$. Therefore, for $m > M$, $\sum_{n=1}^m b_n \geq \sum_{n=1}^m b_n(\mathbb{I}\{x_n = 1\} + \mathbb{I}\{x_n = 2\} + ... + \mathbb{I}\{x_n = N\}) > \sum_{i=1}^N (1/2^i - \epsilon/2N) = \sum_{i=1}^N (1/2^i) - \epsilon/2 > 1 - \epsilon$.

This shows 1 is the limit of $\{p_m\}_{m=1}^\infty$ and hence the value of $\sum_{n=1}^\infty b_n$.

"It is easy to see that $\lambda^*([a, b]) \leq b - a$": Consider a sequence of open intervals (I_n) where $I_1 = (a - \epsilon/4, b + \epsilon/4)$, and the sum of lengths of $I_2, I_3, ...$ is $\epsilon/2$. This sequence covers $[a, b]$ and the sum of their lengths is $b - a + \epsilon$. So the infimum in the definition of Lebesgue outer measure is $\leq b - a + \epsilon$. Also, $\epsilon > 0$ is arbitrary. So, $\lambda^*([a, b]) \leq b - a$.

"It is easy to check that $b-a \leq \sum_{i=1}^{n}(b_i - a_i)$": I believe there is a way to argue rigorously, but to argue in a not-that-rigorous way, if we have freedom to define n open intervals with fixed lengths, then to cover "as more points as possible", we would want these n open intervals to be disjoint. And if $b-a > \sum_{i=1}^{n}(b_i - a_i)$, then even when all the $(a_i, b_i), i = 1, ..., n$ are disjoint, we cannot cover $[b-a]$.

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"and are such that for each j the interior of K_j is included in some R_i": Don't know how to rigorously show this.

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"It follows that $(t, x]$ is the union of a finite collection of ... in some $(a_n, b_n+\delta_n]$": We saw above that $(t, x] \subset [t, x] \subset \cup_{n=1}^{N}(a_n, b_n+\delta_n)$. Assume for each $n = 1, ..., N$, $(a_n, b_n+\delta_n)$ has nonempty intersection with $(t, x]$ otherwise we can remove $(a_n, b_n+\delta_n)$ from the cover. Now $(t, x] \subset \cup_{n=1}^{N}(a_n, b_n+\delta_n]$, so

$$\begin{aligned}(t, x] =&(t, x] \cap \Big(\cup_{n=1}^{N} (a_n, b_n + \delta_n]\Big)\\ =& \cup_{n=1}^{N} (c_n, d_n],\end{aligned} \tag{p20.1}$$

where $(c_n, d_n] = (t, x] \cap (a_n, b_n + \delta_n]$ is nonempty by our assumption above. Now we claim any union of nonempty half open half closed intervals in the form (p20.1) can be written as the union of **disjoint** nonempty half open half closed intervals in the same form. This clearly holds for $N = 2$ when we consider every possible scenario (disjoint, one included in the other, nonempty intersection and no inclusion relation) of $\cup_{n=1}^{2}(c_n, d_n]$. Suppose the statement holds for $n = 1, ..., k$. For $\cup_{n=1}^{k+1}(c_n, d_n]$, write $\cup_{n=1}^{k}(c_n, d_n]$ as $\cup_{j=1}^{J}(c_j', d_j']$, a union of disjoint nonempty half open half closed intervals. So, $\cup_{n=1}^{k+1}(c_n, d_n]$ equals $\Big(\cup_{j=1}^{J} (c_j', d_j']\Big) \cup (c_{k+1}, d_{k+1}]$ which equals

$$\{\Big(\cup_{j=1}^{J} (c_j', d_j']\Big) \cap (c_{k+1}, d_{k+1}]\} \cup \{\Big(\cup_{j=1}^{J} (c_j', d_j']\Big)^c \cap (c_{k+1}, d_{k+1}]\} \cup \{\Big(\cup_{j=1}^{J} (c_j', d_j']\Big) \cap (c_{k+1}, d_{k+1}]^c\}.$$

We call the three terms inside the three pairs of curly brackets terms (A), (B), and (C), respectively. Clearly the three terms (A), (B), and (C) are disjoint, so it's enough to show each of them, if nonempty, is a union of disjoint nonempty half open half closed intervals. But to verify this for each of (A), (B), and (C) is straightforward, for example, by drawing the real number line and the corresponding intervals involved.

Line 19. We just showed for any cover of $(-\infty, x]$ in the form of $\{(a_n, b_n]\}$, we have $F(x) \leq$

$\sum_{n=1}^{\infty}(F(b_n) - F(a_n)) + 2\epsilon, \forall \epsilon > 0$. This means $F(x) \leq \sum_{n=1}^{\infty}(F(b_n) - F(a_n))$ and hence the infimum of the sums in the form $\sum_{n=1}^{\infty}(F(b_n) - F(a_n))$.

Line 21. Let $B = (-\infty, b]$. We need an equation similar to (5) in the proof of proposition 1.3.7, i.e.,

$$F(b_n) - F(a_n) = \mu^*((a_n, b_n]) = \mu^*((a_n, b_n] \cap B) + \mu^*((a_n, b_n] \cap B^c).$$

For the first equality to hold, we need to show that μ^* assigns each half open half closed interval of the form $(a, c]$ the value $F(c) - F(a)$. Once we show this, since $(a, c] \cap B$ and $(a, c] \cap B^c$ are either (respectively) $(a, c]$ and empty, empty and $(a, c]$, or $(a, b]$ and $(b, c]$, the second equality will also hold.

I feel checking the first equality is doable but would be tedious so I'm skipping this step.

Notice theorem 1.3.10 basically says the mapping defined by $F(\cdot) := \mu((-\infty, \cdot])$ is a bijection between the set of measures $\{\mu : \mu \text{ is defined on } (\mathbb{R}, \mathcal{B}(\mathbb{R})), \mu(\mathbb{R}) < \infty\}$ and the set of all bounded, nondecreasing, right-continuous functions F on $\mathbb{R}$ with $\lim_{x \longrightarrow -\infty} F(x) = 0$.

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Why is $K \subset A$? $C - A \subset U \implies (C - A)^c = C^c \cup A \supset U^c \implies C \cap A = C \cap (C^c \cup A) \supset C \cap U^c = K$.

At the end of page 23, we have shown, for an arbitrary positive ϵ, there is a compact $K \subset A$ with $\lambda(A) - \epsilon < \lambda(K)$. Since $\lambda(A)$ is naturally an upper bound of $\{\lambda(K) : K \subset A, K \text{ compact}\}$, this means $\lambda(A)$ is the supremum of $\{\lambda(K) : K \subset A, K \text{ compact}\}$.

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Line 31. For $i = 1, ..., k_0$, denote the cube in $\mathcal{C}_i$ that includes x by C_i. By the construction, $C_1, ..., C_{k_0-1}$ are not included in $\mathcal{D}$ but C_{k_0} is. This is because, C_{k_0} is included in every one of $C_1, ..., C_{k_0-1}$ so has empty intersection with any other cubes from $\mathcal{C}_1, ..., \mathcal{C}_{k_0-1}$ that are included in $\mathcal{D}$.

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Line 13. We need that V is bounded in order to have $\lambda(V) < \infty$. If $\lambda(V) = \infty$, then as long as $\mu(V - A) = \lambda(V - A) = \infty$, we may have $\mu(A) < \lambda(A)$ while the equality of line 13 still holds.

Line 26. A sequence of open intervals $(R_i)_{i \in \mathbb{N}}$ covers A, i.e., $(R_i)_{i \in \mathbb{N}} \in \mathcal{C}_A$ if and only of $(R_i + x)_{i \in \mathbb{N}}$ covers $A + x$, i.e., $(R_i + x)_{i \in \mathbb{N}} \in \mathcal{C}_{A+x}$, and the volume of R_i equals that of $R_i + x$ for all $i \in \mathbb{N}$. So by the definition of Lebesgue outer measure, $\lambda^*(A) = \lambda^*(A + x)$

B is Lebesgue measurable if and only if for all (we can write any subset of A' of $\mathbb{R}$ in the form of $A - x, A \subset \mathbb{R}$ by taking $A = A' + x$) $A - x \subset \mathbb{R}$, in other words for all $A \subset \mathbb{R}$,

$\lambda^*(A-x) = \lambda^*(A-x\cap B) + \lambda^*(A-x\cap B^c)$. This holds if and only if (by the translate invariant property, adding x to both $A-x$ and B) $\lambda^*(A) = \lambda^*(A\cap B+x) + \lambda^*(A\cap (B+x)^c)$.

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"Each equivalent class under $\sim$ has the form $\mathbb{Q}+x$ for some x and so is dense in $\mathbb{R}$". We recall a basic fact form abstract algebra that if we have an equivalent relation $\sim$ on some set A, then $a\sim b \Leftrightarrow [a]=[b]$. Consider the equivalent class $[x]$, then $y\in[x] \Leftrightarrow y\sim x \Leftrightarrow y = x+r$ for some $r\in\mathbb{Q}$. Therefore, $[x]=\mathbb{Q}+x$.

Recall if we have some topology, then we say $A\subset X$ is dense in X if the closure of A, i.e., the union of A with all limit points of A, equals X. For any $y\in\mathbb{R}$, $y-x$ is either a rational number or the limit of some sequence of rational numbers, due to the fact that $\mathbb{Q}$ is dense in $\mathbb{R}$. So, y is either in $\mathbb{Q}+x$ or the limit of some sequence of elements in $\mathbb{Q}+x$, so $\mathbb{Q}+x$ is also dense in $\mathbb{R}$.

"Each intersects the interval $(0,1)$". If $y\in\mathbb{Z}$, then obviously y can be written as $r+z, r\in\mathbb{Q}, z\in(0,1) \implies z\in[y]$. For any $y\in\mathbb{R}-\mathbb{Z}, y = \lfloor y\rfloor + (y-\lfloor y\rfloor)$, where $\lfloor y\rfloor\in\mathbb{Q}, (y-\lfloor y\rfloor)\in(0,1) \implies (y-\lfloor y\rfloor)\in[y]$.

"Let x be an arbitrary member of (0, 1), and let e be the member of E that satisfies $x\sim e$". Consider $[x]$. We know from the previous discussion that in forming E, we have taken one element $e\in[x]\cap(0,1)$.

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Line 24. "strictly positive". Let A be a nonempty subset of $\mathbb{R}^n$. Then $d(x;A)=0$ if and only if $x\in\bar{A}$. (easy to prove, or see Proposition 1.74, An Easy Path to Convex Analysis and Applications). Here since U^c is closed, $d(x;U^c)=0$ iff $x\in U^c$.

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"G_0 and G_1 are disjoint". Assume instead $x\in G_0\cap G_1$, i.e., $x = r_0+2n_0\sqrt{2} = r_1+(2n_1+1)\sqrt{2}$. Then, $\sqrt{2} = \frac{r_0-r_1}{2(n_1-n_0)+1}$, a contradiction.

"$e_1-e_2+g_0=g_1$ would imply that $e_1=e_2$ and $g_0=g_1$". $e_1-e_2+g_0=g_1 \Leftrightarrow e_1-e_2=g_1-g_0\in G$, meaning $e_1\sim e_2$. By the way E is constructed, $e_1=e_2$.

"It is easy to check that $A^c=E+G_1$". Suppose $x\notin A=E+G_0$. By the way we construct E, there is $e\in E$ such that $[x]=[e], \Leftrightarrow x=e+g$ for some $g\in G$. If $g\in G_0$, this would contradict the fact that $x\notin A=E+G_0$. Therefore, $g\in G_1$, which means $A^c\subset E+G_1$. Conversely, if $x \in E + G_1$, then $x \notin A = E + G_0$, or, $x \in A^c$, otherwise $x = e_1 + g_1 = e_0 + g_0$, where $g_0 \in G_0, g_1 \in G_1, e_0, e_1 \in E$. Then $e_0 - e_1 = g_1 - g_0 \in G$ as G is a group. Therefore, $e_0 \sim e_1 \Rightarrow e_0 = e_1 \Rightarrow g_1 = g_0$, a contradiction as G_0 and G_1 are disjoint.

"A is not Lebesgue measurable". If it were, then let I be the unit cube. As $\mathcal{M}_{\lambda^*}$ is a sigma algebra, both $I \cap A, I \cap A^c$ would be measurable. We would then have $1 = \lambda(I) = \lambda(I \cap A) + \lambda(I \cap A^c)$. So at least one term of the right hand side is positive, and without loss of generality say it's $\lambda(I \cap A)$. Then, by monotonicity of measure, we know $\lambda(A) > 0$. As $A^c = A + \sqrt{2}$, we also have $\lambda(A^c) > 0$.

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Line (15). Notice a symmetric argument. If $B \in \mathcal{A}, B \supset A$, then $\mu(B) \geq \mu(E) = \mu(F)$. This means $\mu(F) = \inf\{\mu(B) : B \in \mathcal{A}, B \supset A\}$.

Line (-9). Note that $F - E = E^c - F^c$.

Line (-5). Any $x \in (\cup_n F_n - \cup_n E_n)$, x belongs to some F_i but not any of E_j. So $x \notin E_i \Rightarrow x \in F_i - E_i \Rightarrow x \in \cup_n (F_n - E_n)$.

Line (-2). "It is an extension of μ". This means $\mu(A) = \bar{\mu}(A), \forall A \in \mathcal{A}$. This is because we saw before that $\bar{\mu}(A) := \mu(E)$ for any E that satisfies (1), and we can take $E = F = A$ in (1).

Check that $\bar{\mu}$ is complete: suppose $B \subset A \in \mathcal{A}_\mu, \bar{\mu}(A) = 0$. Then by (1), there exists $F \in \mathcal{A}, \mu(F) = 0$. Clearly $\emptyset \subset B \subset F$ with $\mu(F - \emptyset) = 0$, so $B \in \mathcal{A}_\mu$.

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Check $\mu_*(A) \leq \mu^*(A)$. If RHS is infinity this holds. Suppose $c = \mu^*(A) < \infty$. Then for any positive ϵ, there exists $B \in \mathcal{A} : B \supset A, \mu(B) < c + \epsilon$. Now for any $C \in \mathcal{A}$ such that $C \subset A$, $\mu(C) \leq \mu(B) < c + \epsilon$. Taking the supremum over all such C, we know $\mu_*(A) \leq c + \epsilon$. Recall ϵ is arbitrary.

Line (-7). Notice if $\mu(F) = \infty$, then since $\mu(E) + \mu(F - E) = \mu(F)$, we must have $\mu(E) = \infty$ so line (-6) would imply $\mu^*(A) = \infty$, a contradiction. So $\mu(F) < \infty$.

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Line 3. Let $A \in \mathcal{R}$, i.e., A satisfies (5) and (6). For $\epsilon > 0$, by (5), there exists U open such that $U \supset A, \mu(U) < \mu(A) + \epsilon/2$. Similarly by (6), there exists C closed, $C \subset A, \mu(C) > \mu(A) - \epsilon/2$. As μ is a finite measure, $\mu(U - C) = \mu(U) - \mu(C) = \mu(U) - \mu(A) + \mu(A) - \mu(C) < \epsilon$. Conversely, if A satisfies (7), then $\mu(A - C) < \epsilon, \mu(U - A) < \epsilon$, and (5) and (6) follow easily.

(8): Notice $U - C = \cup_k (U_k - C) \subset \cup_k (U_k - C_k)$

2 Functions and Integrals

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Example 2.1.2 (b). Suppose the set $S = \{x \in I : f(x) < t\}$ is not empty. Case 1: $S = I$. Case 2: $S \neq I$. In this case, there exists $y \in I - S$. Since $S \neq \emptyset, \exists s \in S, f(s) < t$. As f is non-decreasing, we must have $y > s$. Therefore, the set $Y := I - S$ is bounded below by s. Let $i := \inf Y \geq s$. For any $x \in \mathbb{R}, x < i$, either $x \notin I$, or $x \in S \subset I$. Therefore, depending on whether $i \in S$, S is either $I \cap (-\infty, i]$, or $I \cap (-\infty, i)$, which means S may be a singleton or an interval.

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Line 1. If $f(x) + g(x) < t$, then both $f(x), g(x)$ are upper-bounded, i.e., neither is infinity. There exists a positive number ϵ such that $f(x) + g(x) < t - \epsilon$. There exists a positive number $r < \epsilon$ such that $f(x) + r$ is rational. Then, $f(x) < f(x) + r, g(x) < t - \epsilon - f(x) < t - (f(x) + r)$.

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Line 7. Note that although here the author only says "defined on a subset of X", by the definition of measurability, the domain, call it A, of such a measurable function also $\in \mathcal{A}$. Suppose f is measurable, then for $t \in \mathbb{R}, t > 0$, the preimage of $[t, \infty]$ under f^+ is $f^{-1}([t, \infty]) \in \mathcal{A}$; if $t \leq 0$, the preimage of $[t, \infty]$ under f^+ is A. Therefore, f^+ is measurable, and we can argue similarly for f^-. Conversely, suppose both f^+, f^- are measurable. For $t \in \mathbb{R}, t > 0, f^{-1}([t, \infty]) = f^{+-1}([t, \infty]) \in \mathcal{A}$. For $t = 0$, $f^{-1}([0, \infty]) = f^{+-1}([0, \infty]) \cap f^{--1}(\{0\}) \in \mathcal{A}$. For $t < 0$, $f^{-1}([t, \infty]) = f^{-1}([0, \infty]) \cup f^{-1}([t, 0))$. We already know $f^{-1}([0, \infty]) \in \mathcal{A}$, but note $f^{-1}([t, 0)) = f^{--1}((0, t]) \in \mathcal{A}$.

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Line 17. Check continuity. First suppose $x \in (0, 1]$ and we prove left-continuity. Take a sequence $(x_n) \subset [0, 1], x_n \uparrow x, x_n < x$. Take $\epsilon > 0$. By definition of f and monotonicity, there exists $t \in [0, 1] - K, t < x$ with $f(x) - \epsilon < f(t) \leq f(x)$. There exists $N \in \mathbb{N}$ such that for all $n \geq N, t < x_n \leq x \Rightarrow f(x) - \epsilon < f(t) \leq f(x_n) \leq f(x)$.

Now suppose $x \in [0, 1)$, and suppose right-continuity doesn't hold, namely, there exists $\epsilon > 0, (x_n) \subset [0, 1], x_n \downarrow x, x_n > x$ with $f(x_n) > f(x) + \epsilon$. Recall the definition of f. As f is non-decreasing, we can write $f(x_n) = \sup\{f(t) : t \in [0, 1] - K, x \leq t < x_n\}$, i.e., all the t that are smaller than x doesn't affect the supremum. Now, we have that $f(x_n) - f(x) = \sup\{f(t) : t \in [0, 1] - K, x \leq t < x_n\} - \sup\{f(t) : t \in [0, 1] - K, t < x\} > \epsilon$, which is a contradiction if we recall the definition of f on $[0, 1] - K$ (some drawings may help understand), because as $(x_n) \downarrow x$, the lengths of the intervals $[x, x_n) \downarrow 0$.

Line 18. To see $f(1) = 1$, note that there exists a sequence $(x_n) \subset [0,1] - K, x_n \uparrow x, f(x_n) \uparrow 1$. Then apply continuity.

Line 23. We check the values of g belong to K. If y has the form $m/2^n$, n a natural number, m an odd number less than 2^n, then there exists an open interval $(a,b) \subset [0,1] - K$ such that $f = m/2^n = y$ on (a,b). Notice $a \in K$ and since f is continuous, $f(a) = f(b) = y$. For any $x \in [0,1] - K - (a,b)$, obviously $f(x) \neq y$. Also, for $x \in K - [a,b]$, suppose n_0 is the smallest integer that x becomes an "end point" of K_{n_0}, e.g., for $x = 1/3, n_0 = 1$, and for $x = 2/9, n_0 = 2$. This means we have removed an open interval of the form (c,x) or (x,c) from K_{n_0-1} in constructing K_{n_0}. Then, by continuity of $f, f(x) = m'/2^{n'}$ where $m'/2^{n'}$ is the function value assigned to (c,x) or (x,c), and different than y. This means $f(x) \neq y$. This means the only points in $[0,1]$ that take value y under f are $[a,b]$. Therefore, $g(y) = a \in K$.

Now suppose y does not have the form $m/2^n$. Then, there is no $x \in [0,1] - K$ such that $f(x) = y$. Therefore, $g(y) = \inf\{x \in [0,1] : f(x) = y\} = \inf\{x \in K : f(x) = y\} \in K$ because K is closed.

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Line 9. For each n, $g_{0,n} = g_{1,n}$ holds except for at most a set S_n that is negligible. The set $S = \{x \in X : g_{0,n}(x) = g_{1,n}(x), \forall n\}$ includes $\cap_n(S_n^c)$, so its complement is included in $\cup_n S_n$, which is negligible. On S, we have $g_{0,n} = g_n = g_{1,n}, \forall n \Rightarrow f = \lim g_n = \lim g_{0,n} (= f_0) = \lim g_{1,n} (= f_1)$.

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Line 14. If $f = 0$, then with either representation, its integral is zero. Suppose $f > 0$ on $E \neq \emptyset$ and $f = 0$ on E^c. If we eliminate all sets $A_i : a_i = 0$, and all sets $B_j : b_j = 0$, then the remaining A_i's and B_j's in the representation of f must be such that their corresponding a_i or b_j are > 0, and that A_i, $B_j \subset E$. Therefore, $\cup A_i, \cup B_j \subset E$. Conversely, it's easy to check that $E \subset \cup A_i, E \subset \cup B_j$. So $\cup A_i = \cup B_j = E$.

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Line 16. Notice here $\lim_n \int f_n d\mu$ may be infinity. Recall when we say the limit of the sequence (a_n) exists, we mean $\lim_n a_n = c \in \mathbb{R}$ for some c. When we instead say $\lim_n a_n$ exists, we simply mean $\lim_n a_n$ is well defined and it can be a real number or infinity.

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Line 16. We note that for an $\mathcal{A}$-measurable function f, it is integrable if and only if $|f|$ is integrable. Suppose f is so, which means $\int f^+, \int f^-$ are both finite. Then, $\int |f|^+ = \int |f| = \int f^+ + f^-$, which equals $\int f^+ + \int f^-$ by Proposition 2.3.4, which is finite. Also, $\int |f|^- = 0$ which is finite. Conversely, suppose $\int |f|^+, \int |f|^- = 0$ are both finite, then $\int |f|^+ = \int f^+ + f^- = \int f^+ + \int f^-$ is finite. This means both $\int f^+, \int f^-$ are finite.

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Example 2.3.7, (a). Say f is bounded measurable on X: $|f| \leq M$ for some real M. Then by Proposition 2.3.4, $\int f^+ d\mu \leq \int M d\mu = \int M\chi_X d\mu = M\mu(X) < \infty$. Same argument for $\int f^- d\mu$.

(c). Consider f_n defined this way: $f_n(m) = f(m), m = 1, 2, ..., n$ and $f_n(m) = 0$ elsewhere. For a not necessarily nonnegative f, we have that f^+, f^- integrable iff $\sum_{m=1}^{\infty} \max\{f(m), 0\} < \infty, \sum_{m=1}^{\infty} \max\{-f(m), 0\} < \infty$ iff $\sum_{m\in\mathbb{N}, f(m)\geq 0} f(m) < \infty, \sum_{m\in\mathbb{N}, f(m)<0} -f(m) < \infty$ iff $\sum_{m\in\mathbb{N}} |f(m)| < \infty$.

(d). Denote such a function by f, and say it doesn't vanish on a subset of $E, \mu(E) = 0$. Note that f may take ∞. Say $f = \infty$ on $F \subset E$. Note that $F = \cap_{n=1,2,...}\{x : f(x) > n\}$ measurable. By defining f_n as one that takes n on F and equals f elsewhere, we have a sequence of real valued nonnegative simple functions (f_n). (Or we could simply apply Proposition 2.1.8.) Then Proposition 2.3.3 together with the definition of integral for simple functions implies $\int f d\mu = 0$.

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It's easy to see $f\chi_A$ is integrable as f is so. If $\int f\chi_A > 0$, this means $\int (f\chi_A)^+ > 0$ because $\int (f\chi_A)^+ - \int (f\chi_A)^-$ has to be positive, by definition of integral. However, as $(f\chi_A)^+ = 0$, this is not possible.

Now, since $\int f\chi_A = \int (f\chi_A)^+ = 0$, we must have $\int (f\chi_A)^- = 0$ and therefore $\int |f|\chi_A = 0$, and then we can apply Proposition 2.3.12.

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Line 12. Why is it that $f = \lim_n h_n$? It's easy to see that $\lim_n h_n \leq \lim_n f_n = f$. Suppose $\lim_n h_n \neq f$, i.e., there exists x such that $\lim_n h_n(x) < f(x)$. Then $\lim_n h_n(x)$ cannot be ∞ and there exists $r \in \mathbb{R} : \lim_n h_n(x) < r < f(x)$. As $f_n \to f$, there exists N such that $f_N(x) > r$. As $g_{N,k} \to f_N$, there exists $K > N$ such that $g_{N,K}(x) > r$. Therefore, $h_K(x) \geq g_{N,K}(x) > r \Rightarrow \lim_n h_n(x) \geq r$, contradicting that $\lim_n h_n(x) < r$.

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Line 14. Say $g(x) = h(x)$ except possibly for points in the set E with measure 0. There are obviously only countably many points that are division points, so the set of division points D is of measure 0. Now if something holds on $E^c \cap D^c$, it holds almost everywhere. Say $x \in E^c \cap D^c$ but f is not continuous at x. Then there exists $\epsilon > 0$ and a sequence of points $(x_n) \to x$ with $|f(x_n) - f(x)| > \epsilon$, i.e., either $f(x_n) > f(x) + \epsilon$ or $f(x_n) < f(x) - \epsilon$ (at least one of the two must hold for infinitely many terms in (f_n)). Without loss of generality and up to considering a subsequence of (x_n), suppose $f(x_n) > f(x) + \epsilon, \forall n$. But this would mean that, if we recall the definition of h_n, $h_n(x) > f(x) + \epsilon, \forall n$, while from the definition of g_n, we know $g_n(x) \leq f(x)$. This would mean we cannot have $g(x) = h(x)$, a contradiction.

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Line (-4). Recall "Each of these tagged partitions has mesh less than δ and so satisfies $|\mathcal{R}(f, \mathcal{P}) - L| < \epsilon$." Therefore, suppose $|\mathcal{R}(f, \mathcal{P}_1) - L| = (1-k)\epsilon, k \in (0, 1]$. Since in defining and calculating $l(f, \mathcal{P}_0)$, for each subinterval we use the infimum of f over this subinterval, and there are only finitely many subintervals, say m, we can choose a $\mathcal{P}_1$ such that $f(x_i)(a_i - a_{i-1}) - \inf_{y \in [a_{i-1}, a_i]}\{f(y)\}(a_i - a_{i-1}) < k\epsilon/m, \forall i = 1, ..., m$. Then, we have that $|l(f, \mathcal{P}_0) - \mathcal{R}(f, \mathcal{P}_1)| < k\epsilon$, and so $|l(f, \mathcal{P}_0) - L| < \epsilon$.

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Footnote. Notice for this function, the integrals of the positive part and negative part are both infinity, so the Lebesgue integral is not defined $(\infty - \infty)$. For the improper Riemann integral, recall (see for example, Rudin *Principles of Mathematical Analysis*, exercise 6.8) the definition says that if f is Riemann integrable on $[a, b]$, for all $b > a$ where a is fixed, then $\int_a^\infty f(x)dx := \lim_{b \to \infty} \int_a^b f(x)dx$, provided that this limit exists (and is finite). The improper integral exists because the alternating harmonic series is convergent.

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Remark. After seeing Example 2.6.3, Proposition 2.6.4, we notice that, when we are referring to the definition of measurability in section 2.1, if f defined on X in the measure space $(X, \mathcal{A})$ takes values in $\overline{\mathbb{R}}$, then by saying f is $\mathcal{A}$-measurable, we actually mean f is a measurable mapping from $(X, \mathcal{A})$ to $\big(\overline{\mathbb{R}}, \mathcal{B}(\overline{\mathbb{R}})\big)$. If such f is real-valued, then by saying f is $\mathcal{A}$-measurable, we mean f is a measurable mapping from $(X, \mathcal{A})$ to $\big(\mathbb{R}, \mathcal{B}(\mathbb{R})\big)$.

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Line 1. The product of two measurable complex-valued functions on X is measurable: suppose the two functions are $f = (f_1, f_2)$ and $g = (g_1, g_2)$. Recall that the product of two complex numbers (a, b) and (c, d), namely, of $a + bi$ and $c + di$, is $(ac - bd, ad + bc)$. So the product of the two functions is defined as $(f_1g_1 - f_2g_2, f_1g_2 + f_2g_1)$. Since both $f_1g_1 - f_2g_2$ and $f_1g_2 + f_2g_1$ are measurable, by Proposition 2.1.7, we have that $fg := (f_1g_1 - f_2g_2, f_1g_2 + f_2g_1)$ is measurable, by the discussion in Example 2.6.5.

Line (-7). We have that

$$w^{-1}\int f = R(w^{-1}\int f) = R(\int w^{-1}f) := \int R(w^{-1}f),$$

also note that w^{-1} is of unit length, and recall that left-multiplying a complex number by a unit complex number is equivalent to rotating in the $\mathbb{C}^2$ plane. So the norm of $w^{-1}f$ is the same as $|f|$, but since $w^{-1}f$ might have a nonzero projection on the imaginary axis, the norm of its real part is no more than $|f|$. So, we have $\int R(w^{-1}f) \leq \int |R(w^{-1}f)| \leq \int |f|$.

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Line $(-8), (-3)$. We saw before in Proposition 1.4.4 that $\lambda(B+y) = \lambda(B), \forall B \in \mathcal{B}(\mathbb{R})$. The relation that $\lambda(B) = \lambda(-B)$ should be easy to verify according to the definition of Lebesgue outer measure.

3 Convergence

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Line 8. Notice continuous functions on $[-1,1]$ are Riemann integrable and so Lebesgue integrable. Suppose such $f \in C[-1,1]$ exists, then $f_n \to f$ in mean. By Proposition 3.1.5, f_n converges to f in measure. By 3.1.3, there is a subsequence $f_{n_k} \to f$ a.e. But it's easy to check $f_n \to \chi_{(0,1]}$ pointwise. Therefore, by the definition of "a.e.", $f = \chi_{(0,1]}$ a.e. on $[-1,1]$, i.e., on $[-1,0]$ and $(0,1]$. For any point x in $[-1,0]$, any ϵ-neighborhood of x contains some point $y \in [-1,0]$ such that $f(y) = \chi_{(0,1]}(y) = 0$ (otherwise we would have a contradiction since any interval is of positive Lebesgue measure). Since f is continuous, $f = 0$ on $[-1,0]$. Similarly we can argue that $f = 1$ on $(0,1]$. In other words, $f = \chi_{(0,1]}$. But this function is not continuous on $[-1,1]$.

Proposition 3.2.5. How to get such a sequence? For 1/2, find N_1 such that for $n, m > N_1, |n-m| < 1/2$. Pick $n_1 \geq N_1$. For 1/4, find $N_2 > n_1$ such that for $n, m > N_2, |n-m| < 1/4$. Pick $n_2 \geq N_2$...

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Note that because of (a) and (b), if we consider functions defined on $\mathbb{R}^d$ with the usual Borel sigma algebra and Lebesgue measure, then the definition of $\mathcal{L}^\infty$ only involves the concept of null set.

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We explain Example 3.3.5. If $\sum |a_n|^{q_1} < \infty$, then $|a_n| < 1$ for all but finitely many n. Therefore, $\sum |a_n|^{q_2} < \infty$.

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Line 7. Why is it $\mathcal{N}^p \subset \mathcal{L}^p$? First, $0 \in \mathcal{N}^p$. Second, consider $f, g \in \mathcal{N}^p, a, b \in \mathbb{C}$. When $p < \infty$, we have that $af + bg$ vanishes a.e. because both f and g do, and so by Proposition 2.3.9, by considering the zero function, we see $\int |af + bg|^p = 0$. When $p = \infty$, since $S_1 := \{x \in X : |af(x)|^p > 0\}$ and $S_2 := \{x \in X : |bg(x)|^p > 0\}$ are locally μ-null, and the union of a sequence of locally μ-null sets is locally μ-null, we have $S_1 \cup S_2$ is so as well. $\{x : |af(x) + bg(x)|^p > 0\} \subset S_1 \cup S_2$ therefore is μ-null.

Line 17. It's easy to first check that the two operations are well-defined, i.e., that LHS $\subset$ RHS and vice versa. It's then trivial to check that the ten axioms of a vector space hold. Suppose $f, g \in \mathcal{L}^\infty, f \sim g$, and that $\|f\|_\infty < \|g\|_\infty$. There exists $r \in (\|f\|_\infty, \|g\|_\infty)$ such that $X_1 := \{x \in X : |f(x)| > r\}$ is locally μ-null but $X_2 := \{x \in X : |g(x)| > r\}$ isn't. But

$$X_2 = \{x \in X : |g(x)| > r, g(x) = f(x)\} \cup \{x \in X : |g(x)| > r, g(x) \neq f(x)\},$$

and $\{x \in X : |g(x)| > r, g(x) = f(x)\} \subset X_1$ so is locally null. Therefore, $\{x \in X : |g(x)| > r, g(x) \neq f(x)\}$ must not be locally null, and

$$\{x \in X : |g(x)| > r, g(x) \neq f(x)\} \subset \{x \in X : g(x) \neq f(x)\}$$

so $\{x \in X : g(x) \neq f(x)\}$ is not locally null, but this contradicts the fact that $f \sim g$.

Line (-2). We note that if f_n converges to f in $L^p, 1 \leq p < \infty$, we also have $\|f_n\|_p \to \|f\|_p$. This is because by Minkowski's Inequality, $\|f_n\|_p \leq \|f_n - f\|_p + \|f\|_p$ so $\|f_n\|_p - \|f\|_p \leq \|f_n - f\|_p$. Switching the role of f_n, f and we have $\|f\|_p - \|f_n\|_p \leq \|f_n - f\|_p$. Therefore, we have

$$|\|f\|_p - \|f_n\|_p| \leq \|f_n - f\|_p.$$

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Line 8. After this, to show L^p is complete, we consider a sequence $(F_n) \subset L^p$ which converges absolutely: $\sum \|F_n\|_p < \infty$. This means for each equivalence class F_n, we can take an $f_n \in \mathcal{L}^p$ with $\|F_n\|_p := \|f_n\|_p$ and $\sum \|f_n\|_p < \infty$. Then there is an f s.t. $f_n \to f$ in p-norm, as discussed. Define F as the equivalence class of f, and we want to show that $\lim_n \|\sum_{k=1}^n F_k - F\|_p = 0$. But by definition of the norm $\|\cdot\|_p$ on L^p, $\lim_n \|\sum_{k=1}^n F_k - F\|_p = \lim_n \|\sum_{k=1}^n f_k - f\|_p = 0$.

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Line 11. We first show the analogue of Proposition 3.4.3. Let S denote the set of step functions on $\mathbb{R}$ that vanish outside some bounded interval. The proof goes similarly as the proof of 3.4.3. Note that $S \subset \mathcal{L}^p(\mathbb{R})$. Since the Borel measurable simple functions (in L^p) on $\mathbb{R}$ determine a dense subspace of L^p, it is enough to show that if $f \in \mathcal{L}^p(\mathbb{R})$ is a Borel measurable simple function and if ϵ is a positive number, then there is $g \in S$ such that $\|f - g\|_p < \epsilon$. Since $f \in \mathcal{L}^p$, there is an interval

$[a,b]$ such that $\int_{[a,b]} |f|^p \, d\lambda > \int |f|^p \, d\lambda - (\epsilon/2)^p$ (consider $|f\chi_{[-n,n]}|^p$ and monotone convergence theorem). This means $\|f\chi_{[a,b]^c}\|_p < \epsilon/2$. On $[a,b]$, use the conclusion of 3.4.3 to find step function $g_{[a,b]}$ such that $\|g_{[a,b]} - f\chi_{[a,b]}\|_p < \epsilon/2$. Extend $g_{[a,b]}$ to a function g on $\mathbb{R}$ by letting $g = 0$ outside $[a,b]$. Now $\|g - f\|_p = \|g_{[a,b]} - f\chi_{[a,b]} - f\chi_{[a,b]^c}\|_p \leq \epsilon$.

Now we show the analogue of Proposition 3.4.4. We follow almost the same argument as the original proof. Let C be the set of all continuous functions on $\mathbb{R}$ that vanish outside some bounded interval. Notice $C \subset \mathcal{L}^p(\mathbb{R})$. We just showed S in dense in $L^p(\mathbb{R})$, so it is enough to prove that for each $f \in S$ and each positive number ϵ there is a $g \in C$ that satisfies $\|f - g\|_p < \epsilon$. Suppose f vanishes outside $[a,b]$, then on $[a,b]$ we apply the conclusion of Proposition 3.4.4 and find a continuous function $g_{[a,b]}$ on $[a,b]$ with $\|f\chi_{[a,b]} - g_{[a,b]}\|_p < \epsilon$. However, we can extend $g_{[a,b]}$ to $g \in C$ by letting g vanish outside $[a,b]$. Then, note $\|f - g\|_p = \|f\chi_{[a,b]} - g_{[a,b]}\|_p < \epsilon$.

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Line (-7). The first inequality here uses the fact that $A \triangle B \subset (A \triangle C) \cup (B \triangle C)$. To see this, notice

$$A \triangle B = (A \cap C \cap B^c) \cup (A \cap C^c \cap B^c) \cup (A^c \cap C \cap B) \cup (A^c \cap C^c \cap B)$$

and that $(A \cap C \cap B^c) \subset C - B, (A \cap C^c \cap B^c) \subset A - C, (A^c \cap C \cap B) \subset C - A$, and that $(A^c \cap C^c \cap B) \subset B - C$.

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Line 1. We can check that all sets of such form make an algebra, and that any algebra that contains $\mathcal{C}$ contains all sets of such form.

Line (-3). Choose a countable subfamily $\mathcal{C}$ that generates $\mathcal{A}$, and by the sigma-finiteness, there is such a sequence (B_n). It's easy to check that $\mathcal{C} \cup \{B_1, B_2, ...\}$ generates $\mathcal{A}$.

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"It is not hard to check that $\|\cdot\|$ is a norm on the vector space of all bounded linear operators from V_1 to V_2".

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Line (-1). Suppose in $\mathcal{L}^\infty, g_1 = g_2$ locally a.e., and $f \in \mathcal{L}^1$. We want to show $\int fg_1 = \int fg_2$. Define, for each $n \in \mathbb{N}, E_n = \{x : |f(x)| > 1/n\}$. Since f is integrable, $\mu(E_n) < \infty$. Define $F_N = \cup_{n=1}^N E_n$. Then $F_n \uparrow \cup E_n = \{x : f(x) \neq 0\}$.

Let $S = \{x : g_1(x) \neq g_2(x)\}$ which is locally null. Then $S \cap F_N$ is a null set, for each N, and since S and F_N are measurable, $S \cap F_N$ is a zero-measure set. Therefore $g_1 = g_2$ a.e. on all F_N. So we have $\int_{F_N} (g_1 - g_2)f = 0$. Note $|(g_1 - g_2)f| \leq 2\|g_1\|_\infty |f| \in \mathcal{L}^1$. So the dominated convergence theorem implies

$$\int (g_1 - g_2)f = \int_{\cup E_n} (g_1 - g_2)f = \lim_{N\to\infty} \int_{F_N} (g_1 - g_2)f = 0.$$

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Footnote 12 is not right. If $\mu(X) = 0$ then L^1 contains not only 0.

4 Signed and Complex Measures

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Line (-5). Let $\{A_j\}, \{C_k\}$ be respectively finite partitions of B_1, B_2. Then $\{A_j\} \cup \{C_k\}$ is a finite partition of $B_1 \cup B_2$. So we have, by the definition of $|\mu|$,

$$\sum_j |\mu(A_j)| + \sum_k |\mu(C_k)| \le |\mu|(B_1 \cup B_2).$$

Taking sup twice on the LHS gives the desired result.

Line (-1). How do we get this inequality? Let $\{A_j\}$ be a finite partition of A. Then we have

$$\begin{aligned}
\sum_j |\mu(A_j)| &= \sum_j \sqrt{(\mu_1(A_j) - \mu_2(A_j))^2 + (\mu_3(A_j) - \mu_4(A_j))^2} \\
&\le \sum_j |\mu_1(A_j) - \mu_2(A_j)| + |\mu_3(A_j) - \mu_4(A_j)| \\
&\le \sum_j \mu_1(A_j) + \mu_2(A_j) + \mu_3(A_j) + \mu_4(A_j) \\
&= \mu_1(A) + \mu_2(A) + \mu_3(A) + \mu_4(A).
\end{aligned}$$

Taking sup over all possible finite partitions $\{A_j\}$ gives the inequality (5).

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Line 13. Assuming the conclusion of exercise 6, we only need to check that the total variation gives a norm for $M(X, \mathcal{A}, \mathbb{C})$. To check triangle inequality, we note if $\{A_j\}$ is a finite partition of X, and

$\mu, \nu \in M(X, \mathcal{A}, \mathbb{C})$, then

$$\begin{aligned}\sum_j |\mu(A_j) + \nu(A_j)| &\leq \sum_j |\mu(A_j)| + \sum_j |\nu(A_j)| \\ &\leq \|\mu\| + \|\nu\|.\end{aligned}$$

Taking sup over all possible $\{A_j\}$ for the LHS gives $\|\mu + \nu\| \leq \|\mu\| + \|\nu\|$.

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Line (-3). For $\nu, \mu \in M(X, \mathcal{A}, \mathbb{R})$, and $a, b \in \mathbb{R}_{\geq 0}$, we notice that $a\nu^+ + b\mu^+, a\nu^- + b\mu^-$ is the (unique) Jordan decomposition of $a\nu + b\mu$. For $a < 0, b \geq 0$, for example, the positive and negative parts respectively should be $a\nu^- + b\mu^+, a\nu^+ + b\mu^-$.

We have that $\int \chi_A \, d(a\nu + b\mu) = (a\nu + b\mu)(A) = a \int \chi_A \, d\nu + b \int \chi_A \, d\mu$. Now for $f = \sum_{i=1}^k c_i \chi_{A_i}$, it's straightforward to check

$$\begin{aligned}\int f \, d(a\nu + b\mu) &= \int f \, d(a\nu + b\mu)^+ - \int f \, d(a\nu + b\mu)^- \\ &= \sum c_i \left(\int \chi_{A_i} \, d(a\nu + b\mu)^+ - \int \chi_{A_i} \, d(a\nu + b\mu)^- \right) \\ &= \sum c_i \left(a\nu(A_i) + b\mu(A_i) \right) \\ &= a \int f \, d\nu + b \int f \, d\mu.\end{aligned}$$

Then use Proposition 2.1.8 to deal with an arbitrary f.

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Line 13. I don't know why the author says "uniform limit" but the conclusion still holds. Suppose $|f|$ is bounded by M. Use Proposition 2.1.8 to find sequences of simple functions $g_n^+ \uparrow f^+, g_n^- \uparrow f^-$. Define $g_n = g_n^+ - g_n^-$. We have that $|g_n| \leq g_n^+ + g_n^- \leq f^+ + f^- = |f|$. Applying DCT we have that

$$\left| \int f \, d\mu \right| = \left| \int \lim g_n \, d\mu \right| = \left| \lim \int g_n \, d\mu \right| = \lim \left| \int g_n \, d\mu \right|,$$

but $\left| \int g_n \, d\mu \right| \leq \|g_n\|_\infty \|\mu\| \leq \|f\|_\infty \|\mu\|$. So we have $\left| \int f \, d\mu \right| \leq \|f\|_\infty \|\mu\|$.

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Line 10. Notice the finiteness of ν is needed by Proposition 1.2.5.
Line (-4). After we have $f_1, \ldots$, we define $g_n = f_1 \vee \ldots \vee f_n$. We have $\int f_n \leq \int g_n \leq \sup\{\int f : f \in \mathcal{F}\}$. Therefore, $\int g_n \uparrow \sup\{\int f : f \in \mathcal{F}\}$.

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Line 4. To check countable additivity for ν_0, we take a mutually disjoint sequence $(A_n) \subset \mathcal{A}$, and

$$\begin{aligned}\nu(\cup A_n) - \int_{\cup A_n} g &= \left(\lim_{k\to\infty}\sum_{n=1}^{k}\nu(A_n)\right) - \left(\lim_{k\to\infty}\sum_{n=1}^{k}\int_{A_n} g\right)\\ &= \lim_{k\to\infty}\left(\sum_{n=1}^{k}\nu(A_n) - \int_{A_n} g\right)\\ &= \sum_n \nu(A_n) - \int_{A_n} g.\end{aligned}$$

Line 12. If $\mu(P) = 0$, then $\nu(P) = 0$. Since we have $g \geq 0$ and $\nu_0(A) = \nu(A) - \int_A g\, d\mu$ defines a positive measure, we must have $\int_P g\, d\mu = 0$ and $\nu_0(P) = 0$.

Line (-10). It's easy to check the $\mathcal{A}$-measurable subsets of B_n are exactly those of the form $A \cap B_n, A \in \mathcal{A}$. And denote the measures ν, μ restricted to the space $(B_n, \{A \cap B_n, A \in \mathcal{A}\})$ by respectively ν_n, μ_n. Line (-9) should be more rigorously written as $\nu_n(A) = \int_A g_n\, d\mu_n$.

Define $g = \sum \chi_{B_n} g_n = \lim_{k\to\infty}\sum_{i=1}^{k}\chi_{B_i} g_i$ measurable. For each $A \in \mathcal{A}$, denote by C_n the set $A \cap B_n$. Abuse the notation and define g_n on X by letting it be g_n on B_n and 0 elsewhere. We now have

$$\nu(A) = \sum \nu(C_n) = \sum \nu_n(C_n) = \int_{C_n} g_n\, d\mu_n = \int_{C_n} g_n\, d\mu = \int_{C_n} g\, d\mu = \int_A g\, d\mu.$$

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Example 4.2.3. Suppose such f exists. Take A a countable subset of $[0, 1]$. Then

$$\infty = \nu(A) = \int_A f\, d\mu = 0 \quad \text{(Example 2.3.7. (d))},$$

a contradiction.

Line 15. We only check this for complex measures. The procedure is similar for singed measures. We recall that

$$|\nu|(A) := \sup\left\{\sum |\nu(A_j)|\right\}$$

where the sup is taken over all possible finite partitions of A into measurable sets $\{A_j\}$. We also note that the real and imaginary parts of ν are respectively signed measure $\Re\nu = \nu_1 - \nu_2, \Im\nu = \nu_3 - \nu_4$. And we have that $|\Re\nu| = \nu_1 + \nu_2, |\Im\nu| = \nu_3 + \nu_4$. We claim $|\Re\nu|, |\Im\nu| \leq |\nu|$.

Proof. Recall that for a complex measure ν, its total variation on a measurable set E is

$$|\nu|(E) := \sup\Big\{\sum_{j=1}^{n} |\nu(E_j)| : \{E_j\}_{j=1}^{n} \text{ is a finite measurable partition of } E\Big\}.$$

Similarly, for the signed measure $\Re\nu$,

$$|\Re\nu|(E) := \sup\Big\{\sum_{j=1}^{n} |\Re\nu(E_j)| : \{E_j\}_{j=1}^{n} \text{ is a finite measurable partition of } E\Big\}.$$

Fix a measurable set E and a finite measurable partition $\{E_j\}_{j=1}^{n}$ of E. For each j, we have

$$|\Re\nu(E_j)| \le |\nu(E_j)|$$

because $|\Re z| \le |z|$ for all $z \in \mathbb{C}$. Summing over j gives

$$\sum_{j=1}^{n} |\Re\nu(E_j)| \ \le\ \sum_{j=1}^{n} |\nu(E_j)|.$$

Taking the supremum over all such partitions yields

$$|\Re\nu|(E) \le |\nu|(E).$$

Since this holds for every measurable E, we conclude that

$$|\Re\nu| \le |\nu|.$$

The same argument shows that $|\Im\nu| \le |\nu|$ as well. $\square$

With this, it is easy to see if $|\nu| \ll \mu$ then we have $\nu_i \ll \mu$. Conversely, consider the positive measure

$$\lambda := \nu_1 + \nu_2 + \nu_3 + \nu_4.$$

For every measurable E,

$$\begin{aligned}
|\nu(E)| &= |(\nu_1 - \nu_2)(E) + i(\nu_3 - \nu_4)(E)| \\
&\le |(\nu_1 - \nu_2)(E)| + |(\nu_3 - \nu_4)(E)| \\
&\le (\nu_1 + \nu_2)(E) + (\nu_3 + \nu_4)(E) = \lambda(E).
\end{aligned}$$

By the defining minimality property of total variation—namely, $|\nu|$ is the least positive measure with $|\nu(E)| \le \lambda(E)$ for all E (exercise 4.1.5)—we obtain

$$|\nu| \le \lambda.$$

If each $\nu_k \ll \mu$, then $\lambda \ll \mu$; hence $|\nu| \le \lambda \ll \mu$ implies $|\nu| \ll \mu$.

Let ν be a complex measure and $|\nu|$ its total variation. For a (positive) measure μ, the following are equivalent:

$$|\nu| \ll \mu \quad \iff \quad \big(\forall A \in \mathcal{F}\big)\ \mu(A) = 0 \implies \nu(A) = 0.$$

Proof. ($\Rightarrow$) Suppose $|\nu| \ll \mu$. If $\mu(A) = 0$, then $|\nu|(A) = 0$. By the discussion above,

$$\nu_k(A) = 0,$$

hence $\nu(A) = 0$.

($\Leftarrow$) Suppose instead that $\mu(A) = 0 \implies \nu(A) = 0$ for all measurable A. Let B be any measurable set with $\mu(B) = 0$. For any finite measurable partition $\{B_j\}_{j=1}^n$ of B, we also have $\mu(B_j) = 0$ for each j, hence $\nu(B_j) = 0$ by assumption. Therefore

$$\sum_{j=1}^n |\nu(B_j)| = 0.$$

Taking the supremum over all such partitions gives

$$|\nu|(B) = \sup_{\{B_j\}} \sum_{j=1}^n |\nu(B_j)| = 0.$$

Thus $|\nu| \ll \mu$. □

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Line (-11). We can construct such a sequence (g_n) by $g_n = \chi_{\{f \le -1/n\}} + \chi_{\{f > -1/n\}}$.

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Notice in Theorem 4.3.2, in the case when ν is σ-finite, the constructed ν_a, ν_s are also σ-finite. In fact, if we look at line (-14), page 131, we can take an increasing sequence (D_i) with $\cup D_i = X$, and $\nu(D_i) < \infty$. Then $\nu_a(D_i) = \nu(D_i \cap N^c) < \infty$, and $\nu_s(D_i) = \nu(D_i \cap N) < \infty$.

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At the end of this page, we have seen that the mapping from the collection of right-continuous functions (on $\mathbb{R}$) of finite variation that vanishes at $-\infty$ into the collection of finite signed measures on $\mathbb{R}$ is an onto mapping, just like Proposition 1.3.9.

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Line 6. "It is easy to check that F is bounded". Suppose it is not, then let $x_0 = 0$, and there exist $x_n, |F(x_n)| > |F(x_{n-1})| + 1, \forall n = 1, 2, ...$ Now $V_F(-\infty, \infty) = \infty$.

Consider the strictly increasing finite sequence $(c_1, ..., c_n = a, c_{n+1}, ..., c_m = b) \subset \mathbb{R}$. We have

$$\left(\sum_{i=1}^{n-1} |F(c_{i+1}) - F(c_i)|\right) + \left(\sum_{i=n}^{m-1} |F(c_{i+1}) - F(c_i)|\right) = \sum_{i=1}^{m-1} |F(c_{i+1}) - F(c_i)|.$$

Taking sup over all such sequences for RHS, we see

$$\left(\sum_{i=1}^{n-1} |F(c_{i+1}) - F(c_i)|\right) + \left(\sum_{i=n}^{m-1} |F(c_{i+1}) - F(c_i)|\right) \leq V_F(-\infty, b],$$

and if we take sup twice for the two respective terms inside brackets at LHS, we see $V_F(-\infty, a] + V_F[a, b] \leq V_F(-\infty, b]$. If, instead, we first take sup twice for LHS then for RHS, we see $V_F(-\infty, a] + V_F[a, b] \geq V_F(-\infty, b]$.

Line 14. At $V_F(-\infty, b] - \varepsilon < V_F[a, b] \leq V_F(-\infty, b]$, we let $a \downarrow -\infty$ then notice ε is arbitrary, and conclude $\lim_{a\to-\infty} V_F[a, b] = V_F(-\infty, b]$.

Line 16. Consider $\varepsilon > 0$. There exists a finite increasing sequence $t_1 = a < t_2 < ... < t_n = c$, with

$$V_F[a, c] - \varepsilon < \sum_{i=1}^{n-1} |F(t_{i+1}) - F(t_i)|.$$

Now, find $b \in (a, t_2)$ and we have

$$V_F[a, c] - \varepsilon < \sum_{i=1}^{n-1} |F(t_{i+1}) - F(t_i)| \leq |F(b) - F(t_1)| + |F(t_2) - F(b)| + \sum_{i=2}^{n-1} |F(t_{i+1}) - F(t_i)|,$$

and notice the RHS of the above is $\leq |F(b) - F(t_1)| + V_F[b, c]$, in other words,

$$V_F[a, c] - \varepsilon < \sum_{i=1}^{n-1} |F(t_{i+1}) - F(t_i)| \leq |F(b) - F(t_1)| + V_F[b, c].$$

If F is right-continuous, we let $b \downarrow a$, and because $\lim_{b\downarrow a} |F(b) - F(t_1)| = 0$, we see that

$$V_F[a,c] - \varepsilon \leq \lim_{b\downarrow a} V_F[b,c],$$

and since ε is arbitrary, we have $V_F[a,c] \leq \lim_{b\downarrow a} V_F[b,c]$. We also have that naturally, $V_F[a,c] \geq \lim_{b\downarrow a} V_F[b,c]$, hence the desired equality.

Lemma 4.4.1 (c). For real numbers $a < c$, take $x \in (a,c)$. Then $V_F(c) = V_F(x) + V_F[x,c]$, and taking limit for both sides yields

$$V_F(c) = \lim_{x\downarrow a} V_F(x) + \lim_{x\downarrow a} V_F[x,c] = \lim_{x\downarrow a} V_F(x) + V_F[a,c],$$

and after rearranging we get $V_F(c) - V_F[a,c] = V_F(a) = \lim_{x\downarrow a} V_F(x)$.

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Line 14. By the fact that F_μ (and therefore similarly F_{μ^-}, F_{μ^+}, F_{ν^-}, and F_{ν^+}) are right-continuous functions of finite variation that vanish at $-\infty$, we see that $F_{\mu^-} + F_{\nu^+}$ and $F_{\mu^+} + F_{\nu^-}$ are bounded, increasing (by equation (1)), right-continuous and vanish at $-\infty$. By equation (1), we have

$$F_{\mu^-} + F_{\nu^+} = F_{\mu^- + \nu^+}, F_{\mu^+} + F_{\nu^-} = F_{\mu^+ + \nu^-}.$$

Line (-5). Fix $c > 0$. For (s_i, t_i), we can choose a (finite) partition of $[s_i, t_i]$, say $a_0^i = s_i \leq a_1^i ... \leq a_{k(i)}^i = t_i$, such that $\sum_{j=1}^{k(i)} |F(a_j^i) - F(a_{j-1}^i)| \geq V_F[s_i, t_i] - c/2^i$. The collection $\{(u_j, v_j)\}$ is simply $\cup_i \{(a_0^i, a_1^i), ..., (a_{k(i)-1}^i, a_{k(i)}^i)\}$. And it's easily seen that

$$\varepsilon > \sum_j |F(u_j) - F(v_j)| \geq \sum_i V_F[s_i, t_i] - c.$$

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Line 12. It's easily seen that a scalar multiple of an absolutely continuous function, and the sum of two absolutely continuous functions are both absolutely continuous.

It's also straightforward to check that F_1, F_2 are bounded, right-continuous, and vanish at $-\infty$. It's easily seen that F_1 is nondecreasing. To see F_2 is so as well, take $-\infty < x < y < \infty$, and note (we write F instead of F_μ)

$$F_2(y) - F_2(x) = \frac{V_F(x,y] - (F(y) - F(x))}{2} \geq \frac{|F(x) - F(y)| - (F(y) - F(x))}{2} \geq 0,$$

and so there do exist μ_1, μ_2 corresponding to respectively F_1, F_2 as claimed.

5 Product Measures

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Line 15. $x \mapsto \nu(E_x)$ is measurable due to Proposition 2.1.5.

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Line (-6). I_f is real-valued. Also, $\int\int (f^+)_x \, d\nu \, d\mu, \int\int (f^-)_x \, d\nu \, d\mu \in \mathbb{R}$ and

$$\begin{aligned}
&\int\int (f^+)_x \, d\nu \, d\mu - \int\int (f^-)_x \, d\nu \, d\mu \\
&= \int\int (f^+)_x - (f^-)_x \, d\nu \, d\mu \\
&= \int_{N^c}\int (f^+)_x - (f^-)_x \, d\nu \, d\mu \\
&= \int_{N^c} I_f(x) \, d\mu(x) \\
&= \int I_f(x) \, d\mu(x),
\end{aligned}$$

which means $I_f \in \mathcal{L}^1(X, \mathcal{A}, \mu, \mathbb{R})$.

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Line 1. Let

$$E = \{(x, y) \in X \times \mathbb{R} : 0 \le y < f(x)\}.$$

First write

$$E = (X \times [0, \infty)) \cap \{(x, y) : y < f(x)\}.$$

Clearly $X \times [0, \infty) \in \mathcal{A} \times \mathcal{B}(\mathbb{R})$, so it suffices to show that

$$A := \{(x, y) : y < f(x)\}$$

belongs to $\mathcal{A} \times \mathcal{B}(\mathbb{R})$.

For real numbers y, a we have

$$y < a \quad \Longleftrightarrow \quad \exists\, q \in \mathbb{Q} \text{ such that } y < q < a.$$

Hence

$$A = \bigcup_{q \in \mathbb{Q}} \Big(\{(x, y) : y < q\} \cap \{(x, y) : q < f(x)\} \Big).$$

Now

$$\{(x, y) : y < q\} = X \times (-\infty, q) \in \mathcal{A} \times \mathcal{B}(\mathbb{R}),$$

and since f is $\mathcal{A}$-measurable,

$$\{(x, y) : q < f(x)\} = \{x \in X : f(x) > q\} \times \mathbb{R} \in \mathcal{A} \times \mathcal{B}(\mathbb{R}).$$

Therefore each intersection

$$\{(x, y) : y < q\} \cap \{(x, y) : q < f(x)\}$$

belongs to $\mathcal{A} \times \mathcal{B}(\mathbb{R})$. Since the union is countable (over $q \in \mathbb{Q}$), we conclude that $A \in \mathcal{A} \times \mathcal{B}(\mathbb{R})$.

6 Differentiation

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Claim. Let D be a real-valued function defined on $n \times n$ matrices with columns $A = (A_1, \dots, A_n)$. Suppose that

1. D is multilinear in the columns,
2. D is alternating, i.e., $D(A) = 0$ whenever two columns of A are equal,
3. $D(I) = 1$, where I is the identity matrix.

Then

$$D(A) = \det(A)$$

for every $n \times n$ matrix A. Hence the determinant is the unique function with these properties.

Proof. Let $e_1, \dots, e_n$ be the standard basis of $\mathbb{R}^n$. Write the columns of A as

$$A_j = \sum_{i=1}^{n} a_{ij} e_i, \qquad j = 1, \dots, n.$$

We can expand

$$D(A_1, \dots, A_n) = D\Big(\sum_{i_1=1}^{n} a_{i_1 1} e_{i_1}, \dots, \sum_{i_n=1}^{n} a_{i_n n} e_{i_n}\Big).$$

Applying multilinearity repeatedly gives

$$D(A) = \sum_{i_1,\dots,i_n=1}^{n} a_{i_1 1} \cdots a_{i_n n} D(e_{i_1}, \dots, e_{i_n}).$$

If two indices among $i_1, \dots, i_n$ are equal, then two columns in $(e_{i_1}, \dots, e_{i_n})$ coincide, so by the alternating property

$$D(e_{i_1}, \dots, e_{i_n}) = 0.$$

Hence only those terms remain where $i_1, \ldots, i_n$ are all distinct. Such an n-tuple must be a permutation σ of $\{1, \ldots, n\}$. Therefore

$$D(A) = \sum_{\sigma \in S_n} a_{\sigma(1),1} \cdots a_{\sigma(n),n} D(e_{\sigma(1)}, \ldots, e_{\sigma(n)}).$$

Next we determine $D(e_{\sigma(1)}, \ldots, e_{\sigma(n)})$. Consider two vectors u, v and two adjacent columns. Since two equal columns give zero,

$$0 = D(\ldots, u + v, u + v, \ldots).$$

Using multilinearity we obtain

$$0 = D(\ldots, u, u, \ldots) + D(\ldots, u, v, \ldots) + D(\ldots, v, u, \ldots) + D(\ldots, v, v, \ldots).$$

The first and last terms vanish by the alternating property, so

$$D(\ldots, v, u, \ldots) = -D(\ldots, u, v, \ldots).$$

Thus swapping two columns changes the sign of D.

Any permutation σ can be written as a product of transpositions, so

$$D(e_{\sigma(1)}, \ldots, e_{\sigma(n)}) = \operatorname{sgn}(\sigma)\, D(e_1, \ldots, e_n).$$

Since $(e_1, \ldots, e_n) = I$, we have

$$D(e_1, \ldots, e_n) = D(I) = 1.$$

Therefore

$$D(e_{\sigma(1)}, \ldots, e_{\sigma(n)}) = \operatorname{sgn}(\sigma).$$

Substituting into the expansion for $D(A)$ gives

$$D(A) = \sum_{\sigma \in S_n} \operatorname{sgn}(\sigma) a_{\sigma(1),1} \cdots a_{\sigma(n),n}.$$

But this is exactly the Leibniz formula for the determinant:

$$\det(A) = \sum_{\sigma \in S_n} \operatorname{sgn}(\sigma) \prod_{j=1}^{n} a_{\sigma(j),j}.$$

Hence $D(A) = \det(A)$ for every matrix A. $\square$

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Proposition 6.1.3. "T and T^{-1} are continuous." Recall Baby Rudin Theorem 9.7. In fact, they are uniformly continuous.

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Line 3. Suppose B is closed and of finite measure. By Proposition 1.1.6, B is the limit of a sequence of decreasing open sets (U_n). Also, Proposition 1.4.1 implies there is open $O \supset B$, with $\lambda(O) < \infty$. So up to defining $U'_n = U_n \cap O$, we assume U_n are of finite measure. $\lambda(T_k(U_1)) < \infty$ by (1). We have (notice $T_k(U_n)$ are open sets, $\forall n$)

$$|\det(T_k)|\lambda(B) = |\det(T_k)| \lim_n \lambda(U_n) = \lim_n |\det(T_k)|\lambda(U_n) = \lim_n \lambda(T_k(U_n)) = \lambda(\lim_n T_k(U_n)),$$

which equals (easy to verify)

$$\lambda(T_k(\lim_n U_n)) = \lambda(T_k(B)).$$

Now assume B is closed and $\lambda(B) = \infty$. Define $C_n = B \cap [-n, n]^d$, and apply (1) for C_n.

For a general Borel set B, fix a positive ϵ. By Proposition 1.4.1, there exists compact K and open U with

$$K \subset B \subset U, \lambda(K) \geq \lambda(B) - \epsilon, \lambda(B) \geq \lambda(U) - \epsilon.$$

We have $T_k(K) \subset T_k(B) \subset T_k(U)$,

$$\lambda(T_k(B)) \geq \lambda(T_k(K)) = |\det(T)|\lambda(K) \geq |\det(T)|(\lambda(B) - \epsilon),$$

and

$$\lambda(T_k(B)) \leq \lambda(T_k(U)) = |\det(T)|\lambda(U) \leq |\det(T)|(\lambda(B) + \epsilon).$$

This means $\lambda(T_k(B)) = |\det(T_k)|\lambda(B)$.

Line 10. Denote $\mu = \lambda^{d-1}, \nu = \lambda$. By Theorem 5.1.4,

$$(\mu \times \nu)(B) = \int_{\mathbb{R}^{d-1}} \lambda(B_u)\, \mu d(u),$$

and

$$(\mu \times \nu)(T_k(B)) = \int_{\mathbb{R}^{d-1}} \lambda(T_k(B)_u)\, \mu d(u).$$

Since $\lambda(B_u) = \lambda(T_k(B)_u)$, the above two integrals are equal.

Some of the discussions on the remaining of this page can also be found in Baby Rudin, Theorem 9.12, 9.15.

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Line 10, Lemma 6.1.5, Lemma 6.1.6. See Baby Rudin Theorem 9.17, 9.19, and 9.21.

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Line (-1).

Proof. Since $W \subset U \subset \mathbb{R}^d$ are open, define for each $n \geq 1$,

$$W_n := \left\{ x \in W : \operatorname{dist}\big(x, \mathbb{R}^d \setminus W\big) > \frac{1}{n},\ \operatorname{dist}\big(x, \mathbb{R}^d \setminus U\big) > \frac{1}{n},\ |x| < n \right\}.$$

We claim that $\{W_n\}_{n=1}^{\infty}$ has the desired properties.

First, each W_n is open, because the maps

$$x \mapsto \operatorname{dist}\big(x, \mathbb{R}^d \setminus W\big), \qquad x \mapsto \operatorname{dist}\big(x, \mathbb{R}^d \setminus U\big), \qquad x \mapsto |x|$$

are continuous (recall the fact that the distance function to a closed set is continuous), and W_n is defined by strict inequalities.

Next, the sequence is increasing. Indeed, if $x \in W_n$, then

$$\operatorname{dist}\big(x, \mathbb{R}^d \setminus W\big) > \frac{1}{n} > \frac{1}{n+1},$$

$$\operatorname{dist}\big(x, \mathbb{R}^d \setminus U\big) > \frac{1}{n} > \frac{1}{n+1},$$

and

$$|x| < n < n+1,$$

so $x \in W_{n+1}$. Hence $W_n \subset W_{n+1}$ for all n.

We now show that

$$W = \bigcup_{n=1}^{\infty} W_n.$$

Since each $W_n \subset W$, it is enough to prove the reverse inclusion. Let $x \in W$. Because W is open,

$$\operatorname{dist}\big(x, \mathbb{R}^d \setminus W\big) > 0.$$

Also $x \in W \subset U$, and since U is open,

$$\operatorname{dist}\big(x, \mathbb{R}^d \setminus U\big) > 0.$$

Finally, $|x| < \infty$. Therefore, for all sufficiently large n,

$$\frac{1}{n} < \operatorname{dist}\bigl(x, \mathbb{R}^d \setminus W\bigr), \qquad \frac{1}{n} < \operatorname{dist}\bigl(x, \mathbb{R}^d \setminus U\bigr), \qquad |x| < n.$$

Thus $x \in W_n$ for all sufficiently large n, and so $x \in \bigcup_{n=1}^{\infty} W_n$. Hence

$$W = \bigcup_{n=1}^{\infty} W_n.$$

It remains to show that $\overline{W_n}$ is compact and contained in U. Since $W_n \subset \{x : |x| < n\}$, we have

$$\overline{W_n} \subset \{x : |x| \leq n\},$$

so $\overline{W_n}$ is bounded. It is also closed by definition, hence compact by the Heine–Borel theorem.

Finally, if $x \in \overline{W_n}$, then by continuity of the distance function (there exists $(x_k) \subset W_n, x_k \to x, \lim_k \operatorname{dist}\bigl(x_k, \mathbb{R}^d \setminus U\bigr) = \operatorname{dist}\bigl(x, \mathbb{R}^d \setminus U\bigr)$),

$$\operatorname{dist}\bigl(x, \mathbb{R}^d \setminus U\bigr) \geq \frac{1}{n}.$$

In particular, $x \notin \mathbb{R}^d \setminus U$, so $x \in U$. Therefore

$$\overline{W_n} \subset U.$$

Thus $\{W_n\}_{n=1}^{\infty}$ is an increasing sequence of open sets such that

$$W = \bigcup_{n=1}^{\infty} W_n,$$

and for each n, the closure $\overline{W_n}$ is compact and contained in U. □

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Line 6. Notice that W_b is open and so we can apply (8). This is because $|J_T(x)| = |\det(T'(x))|$, and the mapping $x \mapsto T'(x)$, the determinant, and the absolute value function are all continuous, so each $x \in W_b$ has a neighborhood included in W_b.

Line 9. Since b can be arbitrarily close to a, we have $\lambda(T(B)) \leq a\lambda(W)$. By Proposition 1.4.1, find (U_n) a sequence of open sets containing B such that $\lambda(U_n) \downarrow \lambda(B)$. Up to taking $U'_n = U_n \cap U$ we can assume each $U_n \subset U$. We have $\lambda(T(B)) \leq a\lambda(U_n), \forall n$ so taking limit $n \to \infty$ we have $\lambda(T(B)) \leq a\lambda(B)$.

Line 12. Recall the second line on page 159, and we have

$$(T^{-1})'(y) \circ T'(T^{-1}(y)) = I,$$

and so $|J_{T^{-1}}(y)| \leq 1/a$. amer

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Line (-6). The last inequality also uses Proposition 5.2.1.

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Line (-2). For any $x \in A$, since U is open, there is an open cube $C(x) \subset A$, centered at x, with edges parallel to axes, and with length of edges l. There is a cube from $\mathcal{V}$ containing x that has a length of edges smaller than $l/4$. Such a cube is contained in $C(x)$ and clearly belongs to $\mathcal{V}_0$, so $\mathcal{V}_0$ is a Vitali covering of A.

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Line 5. $(\cup C_k)^c$ si open and contains some $x \notin \cup C_k$ from A. Find an open cube neighborhood of x included in $(\cup C_k)^c$ and there is a cube from $\mathcal{V}_0$ that is inside this neighborhood.

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Line (-6).

Proof. Let $\mathcal{C}$ denote the collection of cubes in $\mathbb{R}^d$ and $\mathcal{U}$ the collection of open cubes whose edges are parallel to the coordinate axes.

Define for $\varepsilon > 0$

$$A(\varepsilon) := \sup\left\{\frac{\mu(C)}{\lambda(C)} : C \in \mathcal{C},\ x \in C,\ e(C) < \varepsilon\right\},$$

and

$$B(\varepsilon) := \sup\left\{\frac{\mu(U)}{\lambda(U)} : U \in \mathcal{U},\ x \in U,\ e(U) < \varepsilon\right\}.$$

Then by definition

$$(\overline{D\mu})(x) = \lim_{\varepsilon \downarrow 0} A(\varepsilon).$$

We prove that $A(\varepsilon)$ and $B(\varepsilon)$ have the same limit as $\varepsilon \downarrow 0$.

Step 1. $A(\varepsilon) \leq B(\varepsilon)$.

Let $C \in \mathcal{C}$ satisfy $x \in C$ and $e(C) < \varepsilon$. There exists a decreasing sequence of open cubes $U_n \in \mathcal{U}$ such that

$$C = \bigcap_{n=1}^{\infty} U_n$$

and $x \in U_n$ for all n. Moreover $e(U_n) \downarrow e(C)$.

Since μ is finite and $U_n \downarrow C$, continuity from above gives

$$\mu(U_n) \downarrow \mu(C).$$

Similarly,

$$\lambda(U_n) \downarrow \lambda(C).$$

Hence

$$\frac{\mu(U_n)}{\lambda(U_n)} \longrightarrow \frac{\mu(C)}{\lambda(C)}.$$

For sufficiently large n we have $e(U_n) < \varepsilon$, so

$$\frac{\mu(U_n)}{\lambda(U_n)} \leq B(\varepsilon).$$

Taking limits yields

$$\frac{\mu(C)}{\lambda(C)} \leq B(\varepsilon).$$

Taking the supremum over all such cubes C gives

$$A(\varepsilon) \leq B(\varepsilon).$$

Step 2. $B(\varepsilon) \leq A(\varepsilon)$.

Let $U \in \mathcal{U}$ satisfy $x \in U$ and $e(U) < \varepsilon$. There exists an increasing sequence of cubes $C_n \in \mathcal{C}$ such that

$$U = \bigcup_{n=1}^{\infty} C_n$$

and $x \in C_n$ for all sufficiently large n. Moreover $e(C_n) \uparrow e(U)$.

By continuity from below,

$$\mu(C_n) \uparrow \mu(U), \qquad \lambda(C_n) \uparrow \lambda(U).$$

Thus

$$\frac{\mu(C_n)}{\lambda(C_n)} \longrightarrow \frac{\mu(U)}{\lambda(U)}.$$

For all n we have $e(C_n) < \varepsilon$, so

$$\frac{\mu(C_n)}{\lambda(C_n)} \leq A(\varepsilon).$$

Passing to the limit yields

$$\frac{\mu(U)}{\lambda(U)} \leq A(\varepsilon).$$

Taking the supremum over all such U gives

$$B(\varepsilon) \leq A(\varepsilon).$$

Step 3. Conclusion.

We have

$$A(\varepsilon) = B(\varepsilon).$$

Letting $\varepsilon \downarrow 0$ implies

$$\lim_{\varepsilon \downarrow 0} A(\varepsilon) = \lim_{\varepsilon \downarrow 0} B(\varepsilon).$$

Therefore

$$(\overline{D\mu})(x) = \limsup_{\varepsilon \downarrow 0} \left\{ \frac{\mu(U)}{\lambda(U)} : U \in \mathcal{U},\ x \in U,\ e(U) < \varepsilon \right\}.$$

$\square$

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Line 1. $x \in$ LHS iff there exists $U \in \mathcal{U}$, with $x \in U, e(U) < \varepsilon, \frac{\mu(U)}{\lambda(U)} > a$, iff x belongs to some U that satisfies these conditions: $U \in \mathcal{U}, e(U) < \varepsilon, \frac{\mu(U)}{\lambda(U)} > a$. Therefore, the union of all such Us are exactly LHS.

Theorem 6.2.3. Now we have a new interpretation (some lecturers may have mentioned this already in elementary probability classes) of the density of a continuous random variable. Suppose random variable X has a density f, i.e.,

$$\Pr(X \in A) = \mathbb{P} \circ X^{-1}(A) = \int_A f(x)\, d\lambda(x), A \in \mathcal{B}(\mathbb{R}),$$

then $f(x) = (D\mu)(x)$ a.e. and I believe it can be shown this actually holds everywhere in the case of a continuous random variable X.

In other words, the density $f(x)$ is the limit of ratio

$$\frac{\mathbb{P} \circ X^{-1}([x - \varepsilon/2, x + \varepsilon/2])}{\varepsilon} = \frac{\Pr(X \in [x - \varepsilon/2, x + \varepsilon/2])}{\varepsilon}$$

as $\varepsilon \to 0$.

Proof of Lemma 6.2.4, line 3. Fix $x \in A$ then there is an open cube centered at x that is included in U. Denote the length of edges of this cube by l. Since $\overline{D\mu}(x) \geq a$, there exists $C \in \mathcal{C}$ with $x \in C, e(C) < l/10$, and $\mu(C)/\lambda(C) > a - \varepsilon$. Because $x \in C, e(C) < l/10$, C must be contained in U. This means such C does exist, i.e., $\mathcal{V}$ is not empty. By the above construction we know that for any $x \in A$, and any $m : m < l/10, m > 0$, there exists some $C \in \mathcal{V}, e(C) < m, x \in C$. Therefore $\mathcal{V}$ is a Vitali covering of A.

Line (-10). $\lambda(A) = \lambda(A \bigcap \cup_n C_n)$. And

$$\lambda(\cup C_n) = \lambda(A \bigcap \cup C_n) + \lambda(\cup C_n - A) \geq \lambda(A \bigcap \cup C_n) = \lambda(A).$$

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Notice that informally speaking, if x is a point of density of E, then as we look at the shrinking cube neighborhoods of x (not necessarily centered at x), the intersection of these neighborhoods with E should "almost be the neighborhoods themselves", in terms of Lebesgue measure.

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Lemma 6.3.1. We can ignore the words inside brackets. Recall $\mu(x_0) = \lambda(x_0) = 0$ so the ratio is equal to

$$\frac{\mu((x, x_0))}{\lambda((x, x_0))}.$$

We then recall the proof of Lemma 6.2.2, which says that the definition of $D\mu$ could have used instead open cubes.

Lemma 6.3.2. Let

$$G(x) := \inf\{F(t) : t > x\},$$

where $F : \mathbb{R} \to \mathbb{R}$ is nondecreasing. We show that G is right continuous.

Fix $x \in \mathbb{R}$, and let $x_n \downarrow x$. Since G is nondecreasing, the sequence

$$G(x_n)$$

is nonincreasing, and moreover $G(x_n) \geq G(x)$ for all n. Hence the limit

$$L := \lim_{n \to \infty} G(x_n)$$

exists and satisfies $L \geq G(x)$.

It remains to show that $L \leq G(x)$. Let $\varepsilon > 0$. By the definition of infimum, there exists $t > x$ such that

$$F(t) < G(x) + \varepsilon.$$

Since $x_n \downarrow x$, we have $x_n < t$ for all sufficiently large n. For such n, the point t belongs to the set $\{s : s > x_n\}$, so

$$G(x_n) \leq F(t) < G(x) + \varepsilon.$$

Passing to the limit as $n \to \infty$, we obtain

$$L \leq G(x) + \varepsilon.$$

Since $\varepsilon > 0$ was arbitrary, it follows that $L \leq G(x)$.

Therefore $L = G(x)$, i.e.

$$\lim_{n\to\infty} G(x_n) = G(x)$$

whenever $x_n \downarrow x$. Hence G is right continuous.

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Line 10. We here show that G and F are continuous at the same points. We first show that for any $x_0 \in \mathbb{R}$

$$\lim_{x\uparrow x_0} G(x) = F(x_0-), \quad \text{where } G(x) = F(x+).$$

First note that for any $x < x_0$, since F is nondecreasing, for all $t \in (x, x_0)$ we have

$$F(t) \leq F(x_0-),$$

hence

$$G(x) = \inf_{t>x} F(t) \leq F(x_0-). \tag{171a}$$

Taking $x \uparrow x_0$, we obtain (recall G is nondecreasing so the limit exists)

$$\lim_{x\uparrow x_0} G(x) \leq F(x_0-).$$

Next, fix $\varepsilon > 0$. By the definition of the left limit $F(x_0-)$, there exists $\delta > 0$ such that whenever $x_0 - \delta < t < x_0$, we have

$$F(t) > F(x_0-) - \varepsilon.$$

Now take x such that $x_0 - \delta < x < x_0$. Then for all $t \in (x, x_0)$,

$$F(t) > F(x_0-) - \varepsilon.$$

Since F is nondecreasing, for all $t > x_0$ we have

$$F(t) \geq F(x_0) \geq F(x_0-) > F(x_0-) - \varepsilon.$$

Therefore, for all $t > x$,

$$F(t) > F(x_0-) - \varepsilon,$$

which implies

$$G(x) = \inf_{t>x} F(t) \geq F(x_0-) - \varepsilon.$$

Taking $x \uparrow x_0$, we obtain

$$\lim_{x \uparrow x_0} G(x) \geq F(x_0-) - \varepsilon.$$

Since $\varepsilon > 0$ is arbitrary,

$$\lim_{x \uparrow x_0} G(x) \geq F(x_0-).$$

Combining the two inequalities, we conclude

$$\lim_{x \uparrow x_0} G(x) = F(x_0-).$$

Now let

$$G(x) = F(x+) := \lim_{t \downarrow x} F(t) = \inf_{t>x} F(t),$$

where $F : \mathbb{R} \to \mathbb{R}$ is nondecreasing. We show that F and G are continuous at the same points.

Fix $x_0 \in \mathbb{R}$. Since F is nondecreasing, the one-sided limits $F(x_0-)$ and $F(x_0+)$ exist. By definition,

$$G(x_0) = F(x_0+).$$

Also, for $x < x_0$,

$$G(x) = F(x+).$$

Also as we have seen, as $x \uparrow x_0$,

$$G(x) \to F(x_0-),$$

so that

$$G(x_0-) = F(x_0-).$$

Now suppose first that F is continuous at x_0. Then

$$F(x_0-) = F(x_0) = F(x_0+).$$

Therefore

$$G(x_0-) = F(x_0-) = F(x_0+) = G(x_0).$$

Since G is right-continuous, it follows that G is continuous at x_0.

Conversely, suppose that G is continuous at x_0. Then

$$F(x_0-) = G(x_0-) = G(x_0),$$

hence

$$F(x_0-) = F(x_0+).$$

But for a nondecreasing function,

$$F(x_0-) \le F(x_0) \le F(x_0+).$$

Therefore

$$F(x_0-) = F(x_0) = F(x_0+),$$

which shows that F is continuous at x_0.

Thus F and G are continuous at the same points.

Line 12. $F(x) \le G(x)$ apparently. Also, by equation (171a) above, we know for $x < x_0$, we have

$$G(x) \le F(x_0-).$$

Therefore, by definition of $G(x_0-)$, we know

$$G(x_0-) \le F(x_0-).$$

As x_0 is arbitrary, we have $G(x-) \le F(x-) \le F(x), \forall x \in \mathbb{R}$

Line 13. Assume that G is differentiable at x_0. We claim that

$$\lim_{x \to x_0} \frac{G(x-) - G(x_0)}{x - x_0} = G'(x_0).$$

Indeed,

$$\frac{G(x-) - G(x_0)}{x - x_0} = \frac{G(x) - G(x_0)}{x - x_0} + \frac{G(x-) - G(x)}{x - x_0}.$$

Since G is differentiable at x_0, we have

$$\frac{G(x) - G(x_0)}{x - x_0} \to G'(x_0).$$

Thus it remains to show that

$$\frac{G(x-) - G(x)}{x - x_0} \to 0.$$

Because G is monotone and differentiable at x_0, its jumps near x_0 are negligible compared with $|x - x_0|$, i.e.

$$G(x) - G(x-) = o(|x - x_0|).$$

Therefore
$$\frac{G(x-)-G(x)}{x-x_0} \to 0.$$
It follows that
$$\lim_{x\to x_0} \frac{G(x-)-G(x_0)}{x-x_0} = \lim_{x\to x_0} \frac{G(x)-G(x_0)}{x-x_0} = G'(x_0).$$
To see $G(x)-G(x-) = o(|x-x_0|)$, let
$$J(x) := G(x) - G(x-) \geq 0.$$
We claim that if G is nondecreasing and differentiable at x_0, then
$$J(x) = o(|x-x_0|),$$
i.e.
$$\frac{G(x)-G(x-)}{|x-x_0|} \to 0 \qquad (x \to x_0).$$
Assume, to the contrary, that this is false. Then there exist $\varepsilon > 0$ and a sequence $x_n \to x_0$, $x_n \neq x_0$, such that
$$J(x_n) \geq \varepsilon |x_n - x_0| \qquad \text{for all } n.$$
Passing to a subsequence, we may assume that $x_n > x_0$ for all n and $x_n \downarrow x_0$.

By the definition of $G(x_n-)$, for each n we can choose $y_n < x_n$ such that
$$x_n - y_n < \frac{1}{n}(x_n - x_0)$$
and
$$G(y_n) \leq G(x_n-) + \frac{1}{n}(x_n - x_0).$$
Then
$$G(x_n) - G(y_n) \geq G(x_n) - G(x_n-) - \frac{1}{n}(x_n - x_0) \geq \left(\varepsilon - \frac{1}{n}\right)(x_n - x_0).$$
Therefore
$$\frac{G(x_n)-G(y_n)}{x_n - y_n} \geq \frac{\left(\varepsilon - \frac{1}{n}\right)(x_n - x_0)}{\frac{1}{n}(x_n - x_0)} = n\left(\varepsilon - \frac{1}{n}\right).$$
Hence
$$\frac{G(x_n)-G(y_n)}{x_n - y_n} \to \infty.$$
On the other hand, since $x_n \to x_0$ and $y_n \to x_0$, and since G is differentiable at x_0, the secant

slopes must converge to $G'(x_0)$:

$$\frac{G(x_n) - G(y_n)}{x_n - y_n} \to G'(x_0),$$

which is impossible. This contradiction proves that

$$\frac{G(x) - G(x-)}{|x - x_0|} \to 0.$$

Thus

$$G(x) - G(x-) = o(|x - x_0|).$$

If, after passing to a subsequence, the case is that $x_n < x_0$ for all n, we may assume $x_n \uparrow x_0$. Suppose

$$G(x_n) - G(x_n-) \geq \varepsilon(x_0 - x_n) \qquad \text{for all } n.$$

By the definition of $G(x_n-)$, for each n we can choose $y_n < x_n$ such that

$$x_n - y_n < \frac{1}{n}(x_0 - x_n)$$

and

$$G(y_n) \leq G(x_n-) + \frac{1}{n}(x_0 - x_n).$$

Then

$$G(x_n) - G(y_n) \geq G(x_n) - G(x_n-) - \frac{1}{n}(x_0 - x_n) \geq \left(\varepsilon - \frac{1}{n}\right)(x_0 - x_n).$$

Therefore

$$\frac{G(x_n) - G(y_n)}{x_n - y_n} \geq \frac{\left(\varepsilon - \frac{1}{n}\right)(x_0 - x_n)}{\frac{1}{n}(x_0 - x_n)} = n\left(\varepsilon - \frac{1}{n}\right) \to \infty.$$

On the other hand, $x_n \to x_0$ and $y_n \to x_0$, so differentiability of G at x_0 implies

$$\frac{G(x_n) - G(y_n)}{x_n - y_n} \to G'(x_0),$$

which is impossible. This contradiction shows that this case cannot occur.

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Line 10. We now complete the proof.

Proof. Now remove the requirement that the functions F_n be right-continuous, while still assuming that they are bounded, non-decreasing, vanishing at $-\infty$ and that F is bounded. Define

$$G_n(x) := F_n(x+), \qquad G(x) := F(x+).$$

By Lemma 6.3.2, each G_n and G is nondecreasing and right-continuous. We claim that

$$G(x) = \sum_{n=1}^{\infty} G_n(x) \qquad (x \in \mathbb{R}).$$

Fix $x \in \mathbb{R}$. For each $t > x$ and each n,

$$G_n(x) = \inf_{s>x} F_n(s) \leq F_n(t).$$

Hence for every $N \geq 1$,

$$\sum_{n=1}^{N} G_n(x) \leq \sum_{n=1}^{N} F_n(t) \leq \sum_{n=1}^{\infty} F_n(t) = F(t),$$

and we notice increasing N to ∞ shows G_n, G are bounded (that each G_n vanishes at $-\infty$ can be proven by contradiction). Since this holds for every $t > x$, we obtain

$$\sum_{n=1}^{N} G_n(x) \leq \inf_{t>x} F(t) = G(x).$$

Letting $N \to \infty$ gives

$$\sum_{n=1}^{\infty} G_n(x) \leq G(x).$$

To prove the reverse inequality, fix $\varepsilon > 0$. Since

$$\sum_{n=1}^{\infty} F_n(x+1) = F(x+1) < \infty,$$

there exists $N \geq 1$ such that

$$\sum_{n>N} F_n(x+1) < \varepsilon.$$

Now for every $m \geq 1$, since $x + 1/m \leq x + 1$ and each F_n is nondecreasing,

$$\sum_{n>N} F_n(x+1/m) \leq \sum_{n>N} F_n(x+1) < \varepsilon.$$

Therefore

$$F(x+1/m) = \sum_{n=1}^{N} F_n(x+1/m) + \sum_{n>N} F_n(x+1/m) \leq \sum_{n=1}^{N} F_n(x+1/m) + \varepsilon.$$

Letting $m \to \infty$, we obtain

$$G(x) \le \sum_{n=1}^{N} G_n(x) + \varepsilon \le \sum_{n=1}^{\infty} G_n(x) + \varepsilon.$$

Since $\varepsilon > 0$ is arbitrary,

$$G(x) \le \sum_{n=1}^{\infty} G_n(x).$$

Thus

$$G(x) = \sum_{n=1}^{\infty} G_n(x) \qquad (x \in \mathbb{R}).$$

The preceding case therefore applies to the functions G_n, G, and yields

$$G'(x) = \sum_{n=1}^{\infty} G_n'(x) \qquad \text{for a.e. } x.$$

Fix n. By Lemma 6.3.2 and the proof of Theorem 6.3.3 (second paragraph), the functions F_n and G_n are continuous at the same points and agree at each point where they are continuous. Again by Theorem 6.3.3 and the proof of Theorem 6.3.3 (end of second paragraph), we know that a.e., G_n is differentiable and $F_n' = G_n'$.

Now we turn to the functions F and G. Note that in proof of Theorem 6.3.3, paragraph 2, we need the vanishing property of F only to have the vanishing property of G in order to apply the result from the first paragraph and conclude "by what we have just proved, differentiable almost everywhere". All remaining discussions in the second paragraph (including our notes before for page 171) do not need the vanishing property. Now that F is bounded and nondecreasing, although it's not necessarily vanishing at $-\infty$, we can still conclude that G is differentiable almost everywhere (by the statement of Theorem 6.3.3), and use the remaining discussion in the 2nd paragraph to conclude that a.e., G is differentiable and $G' = F'$.

Consequently,

$$F'(x) = G'(x) = \sum_{n=1}^{\infty} G_n'(x) = \sum_{n=1}^{\infty} F_n'(x) \qquad \text{for a.e. } x.$$

Finally, suppose that the functions F_n are arbitrary nondecreasing functions such that the series $\sum_n F_n(x)$ converges at each $x \in \mathbb{R}$. It is enough to prove that

$$F'(x) = \sum_{n=1}^{\infty} F_n'(x)$$

holds for almost every x in an arbitrary bounded open interval (a,b). For each n define

$$\widetilde{F}_n(x) := \begin{cases} 0, & x \leq a, \\ F_n(x) - F_n(a), & a < x < b, \\ F_n(b) - F_n(a), & b \leq x. \end{cases}$$

Then each $\widetilde{F}_n$ is bounded and nondecreasing. Also, if we define

$$\widetilde{F}(x) := \sum_{n=1}^{\infty} \widetilde{F}_n(x),$$

then $\widetilde{F}$ is bounded, because

$$\widetilde{F}(x) = \begin{cases} 0, & x \leq a, \\ F(x) - F(a), & a < x < b, \\ F(b) - F(a), & b \leq x. \end{cases}$$

Thus the result already proved applies to the functions $\widetilde{F}_n$ (bounded, nondecreasing and vanishing at $-\infty$), and we obtain

$$\widetilde{F}'(x) = \sum_{n=1}^{\infty} \widetilde{F}_n'(x) \qquad \text{for a.e. } x \in \mathbb{R}.$$

For $x \in (a,b)$ we have

$$\widetilde{F}_n(x) = F_n(x) - F_n(a), \qquad \widetilde{F}(x) = F(x) - F(a),$$

and hence, wherever derivatives exist,

$$\widetilde{F}_n'(x) = F_n'(x), \qquad \widetilde{F}'(x) = F'(x).$$

Therefore

$$F'(x) = \sum_{n=1}^{\infty} F_n'(x) \qquad \text{for a.e. } x \in (a,b).$$

Since (a,b) was arbitrary, the proof is complete. □

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Line 12. It is straightforward to check by definition that this newly defined function (call it G) is absolutely continuous on $\mathbb{R}$, and so by exercise 4.4.5 is of finite variation on $[a,b]$. To see it's of finite variation on $\mathbb{R}$, note that for an increasing sequence $\{t_i\}_{i=0}^{n} \in \mathbb{R}$, with at least two points

$t_j, t_k, j, k \in \{0, \dots, n\}$ in $[a, b]$, suppose all points that fall into $[a, b]$ are $t_{k_0}, ..., t_{k_m}, m \le n$. Suppose the maximum absolute value of G in $[a, b]$ is M (G is continuous). We have

$$\sum_{i=1}^{n} |G(t_i) - G(t_{i-1})| \le 2M + |F(b) - F(a)| + \sum_{a=1}^{m} |F(t_{k_a}) - F(t_{k_{a-1}})|,$$

which in turn is no larger than $2M + |F(b) - F(a)| + V_F[a, b]$. Taking the supremum for LHS, we see

$$V_G(-\infty, \infty) \le 2M + |F(b) - F(a)| + V_F[a, b].$$

Line (-10). If we define $g(t) = F'(t)$ on the subset of $[a, b]$ where F' exists, and 0 elsewhere on $\mathbb{R}$, then we have $G(x) = \int_{-\infty}^{x} g(t)\, dt$, where G is defined as above, and we can apply Proposition 4.4.6 to conclude G is absolutely continuous, and so is F.

In this section, whenever the author says F' is integrable, we should interpret it as saying $F' \in \mathcal{L}^1(\mathbb{R}, \mathcal{M}_{\lambda^*}, \mu, \mathbb{R})$. Notice in Proposition 4.4.6, we are dealing with $g \in \mathcal{L}^1(\mathbb{R}, \mathcal{B}(\mathbb{R}), \mu, \mathbb{R})$. This introduces no problem. We recall how the integral of a measurable function is defined, and the fact that simple functions are measurable both w.r.t. $\mathcal{B}(\mathbb{R})$ and w.r.t. $\mathcal{M}_{\lambda^*}$, and conclude that if a function f is integrable for the space $(\mathbb{R}, \mathcal{B}(\mathbb{R}), \mu, \mathbb{R})$, it is also integrable for the space $(\mathbb{R}, \mathcal{M}_{\lambda^*}, \mu, \mathbb{R})$, and the two integrals are equal over every Borel-measurable set (and of course, over the entire $\mathbb{R}$).

When proving the second direction, we assume $F' \in \mathcal{L}^1(\mathbb{R}, \mathcal{M}_{\lambda^*}, \mu, \mathbb{R})$, and find a function f that is Borel-measurable and satisfies $f = F'$ Borel-λ-a.e. (Proposition 2.2.5). Then f is also Lebesgue-measurable and $f = F'$ Lebesgue-λ-a.e. Therefore on any Borel set, the (Lebesgue space) integral of F' equals the (Lebesgue space) integral of g which equals the (Borel space) integral of g. Therefore, we can rewrite equation (1) as

$$F(x) = F(a) + \int_{a}^{x} g(t)\, dt,$$

where the integral is w.r.t. the Borel measure space. Then we can apply Proposition 4.4.6.

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Line 10. Applying Corollary 6.3.8 to F, G, FG, we know $F', G', (FG)'$ exist and are integrable a.e. By chain rule,

$$(FG)' = F'G + FG' \quad \text{a.e.}$$

Notice Proposition 2.3.9 would only imply $F'G + FG'$ (in fact, a function that is defined on $[a, b]$ everywhere and agrees with $F'G+FG'$ a.e.) is integrable. To show $F'G$ and FG' are both integrable separately, we note

$$\int_{[a,b]} |F'G| \le \int_{[a,b]} N|F'|.$$

Here again we have abused the notation and used F' to mean a function that is defined on $[a, b]$ everywhere and agrees with F' a.e.

Proposition 6.3.10. Notice in the proof we first assume f is Borel measurable. If f is only Lebesgue measurable, then $|f(t) - r|$ is Lebesgue measurable, but not necessarily Borel measurable. Hence the set function

$$\mu_r(A) = \int_{A\cap(a,b)} |f(t) - r|\, dt$$

is defined for Lebesgue measurable sets A, and thus μ_r is a measure on the Lebesgue σ-algebra, but not necessarily a Borel measure.

Problem. The differentiation theorem (Theorem 6.2.3) is stated for finite Borel measures. Therefore, it cannot be directly applied to μ_r, since μ_r need not be a Borel measure when f is only Lebesgue measurable.

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Proof. We first prove that a function $f : \mathbb{R} \to \mathbb{R}$ is continuous if and only if it is both lower semicontinuous and upper semicontinuous.

($\Rightarrow$) Suppose f is continuous. To show that f is lower semicontinuous, fix $x \in \mathbb{R}$ and let $A < f(x)$. Since $f(x) - A > 0$, by continuity of f at x there exists $\delta > 0$ such that

$$|t - x| < \delta \quad \Longrightarrow \quad |f(t) - f(x)| < f(x) - A.$$

Hence for such t,

$$f(t) > f(x) - (f(x) - A) = A.$$

Thus f is lower semicontinuous. Since $-f$ is also continuous, the same argument shows that $-f$ is lower semicontinuous, i.e. f is upper semicontinuous.

($\Leftarrow$) Suppose now that f is both lower semicontinuous and upper semicontinuous. Fix $x \in \mathbb{R}$ and let $\varepsilon > 0$. Since $f(x) - \varepsilon < f(x)$ and $f(x) < f(x) + \varepsilon$, by lower semicontinuity and upper semicontinuity there exist $\delta_1, \delta_2 > 0$ such that

$$|t - x| < \delta_1 \quad \Longrightarrow \quad f(t) > f(x) - \varepsilon,$$

and

$$|t - x| < \delta_2 \quad \Longrightarrow \quad f(t) < f(x) + \varepsilon.$$

Therefore, if $|t - x| < \delta := \min\{\delta_1, \delta_2\}$, then

$$f(x) - \varepsilon < f(t) < f(x) + \varepsilon,$$

so that $|f(t) - f(x)| < \varepsilon$. Hence f is continuous at x. Since x was arbitrary, f is continuous on $\mathbb{R}$.

We now prove claims (a)–(f).

(a) A function $f : \mathbb{R} \to (-\infty, +\infty]$ is lower semicontinuous if and only if for each real number A, the set

$$\{x \in \mathbb{R} : A < f(x)\}$$

is open.

($\Rightarrow$) Assume f is lower semicontinuous, and fix $A \in \mathbb{R}$. Let

$$E_A := \{x \in \mathbb{R} : A < f(x)\}.$$

If $x \in E_A$, then $A < f(x)$. By lower semicontinuity of f at x, there exists $\delta > 0$ such that

$$|t - x| < \delta \quad \Longrightarrow \quad A < f(t).$$

Thus $(x - \delta, x + \delta) \subseteq E_A$, showing that E_A is open.

($\Leftarrow$) Conversely, assume that for each real A, the set $E_A := \{x \in \mathbb{R} : A < f(x)\}$ is open. Fix $x \in \mathbb{R}$ and let $A < f(x)$. Then $x \in E_A$, and since E_A is open, there exists $\delta > 0$ such that

$$(x - \delta, x + \delta) \subseteq E_A.$$

Hence

$$|t - x| < \delta \quad \Longrightarrow \quad A < f(t).$$

Therefore f is lower semicontinuous at x. Since x was arbitrary, f is lower semicontinuous.

(b) A function $f : \mathbb{R} \to [-\infty, +\infty)$ is upper semicontinuous if and only if for each real number A, the set

$$\{x \in \mathbb{R} : f(x) < A\}$$

is open.

This follows immediately from part (a) applied to $-f$. Indeed, f is upper semicontinuous if and only if $-f$ is lower semicontinuous. By part (a), this is equivalent to saying that for each real A, the set

$$\{x \in \mathbb{R} : A < -f(x)\}$$

is open. But

$$\{x \in \mathbb{R} : A < -f(x)\} = \{x \in \mathbb{R} : f(x) < -A\}.$$

Since $-A$ runs through all real numbers as A does, the conclusion follows.

(c) If U is an open subset of $\mathbb{R}$, then the characteristic function χ_U is lower semicontinuous.

We use part (a). Fix $A \in \mathbb{R}$. Since χ_U takes only the values 0 and 1, the set

$$\{x \in \mathbb{R} : A < \chi_U(x)\}$$

has one of the following forms:

$$\{x : A < \chi_U(x)\} = \begin{cases} \varnothing, & A \geq 1, \\ U, & 0 \leq A < 1, \\ \mathbb{R}, & A < 0. \end{cases}$$

Each of these sets is open, because U is open. Hence by part (a), χ_U is lower semicontinuous.

(d) If C is a closed subset of $\mathbb{R}$, then the characteristic function χ_C is upper semicontinuous.

Again we use part (b). Fix $A \in \mathbb{R}$. Since χ_C takes only the values 0 and 1, the set

$$\{x \in \mathbb{R} : \chi_C(x) < A\}$$

has one of the following forms:

$$\{x : \chi_C(x) < A\} = \begin{cases} \varnothing, & A \leq 0, \\ \mathbb{R} \setminus C, & 0 < A \leq 1, \\ \mathbb{R}, & A > 1. \end{cases}$$

Because C is closed, $\mathbb{R} \setminus C$ is open. Thus each of these sets is open, and part (b) implies that χ_C is upper semicontinuous.

(e) If f and g are lower semicontinuous, then $f + g$ is lower semicontinuous.

Fix $x \in \mathbb{R}$ and let $A < (f + g)(x) = f(x) + g(x)$. Set

$$\eta := f(x) + g(x) - A > 0.$$

Then

$$A = \big(f(x) - \eta/2\big) + \big(g(x) - \eta/2\big).$$

Since f is lower semicontinuous at x and $f(x) - \eta/2 < f(x)$, there exists $\delta_1 > 0$ such that

$$|t - x| < \delta_1 \quad \Longrightarrow \quad f(t) > f(x) - \eta/2.$$

Likewise, since g is lower semicontinuous at x and $g(x) - \eta/2 < g(x)$, there exists $\delta_2 > 0$ such that

$$|t - x| < \delta_2 \quad \Longrightarrow \quad g(t) > g(x) - \eta/2.$$

Hence, if $|t-x| < \delta := \min\{\delta_1, \delta_2\}$, then

$$(f+g)(t) = f(t) + g(t) > f(x) - \eta/2 + g(x) - \eta/2 = f(x) + g(x) - \eta = A.$$

Therefore $f+g$ is lower semicontinuous at x. Since x was arbitrary, $f+g$ is lower semicontinuous on $\mathbb{R}$.

(f) If $\{f_n\}$ is an increasing sequence of lower semicontinuous functions, then $\lim_n f_n$ is lower semicontinuous.

Let

$$f(x) := \lim_{n\to\infty} f_n(x) = \sup_{n\in\mathbb{N}} f_n(x),$$

where the limit exists in $(-\infty, +\infty]$ because $\{f_n(x)\}$ is increasing for each x.

To show that f is lower semicontinuous, we use part (a). Fix a real number A. Then

$$\{x \in \mathbb{R} : A < f(x)\} = \left\{x \in \mathbb{R} : A < \sup_n f_n(x)\right\}.$$

For a fixed x, the inequality $A < \sup_n f_n(x)$ holds if and only if there exists some n such that $A < f_n(x)$. Hence

$$\{x \in \mathbb{R} : A < f(x)\} = \bigcup_{n=1}^{\infty} \{x \in \mathbb{R} : A < f_n(x)\}.$$

Since each f_n is lower semicontinuous, part (a) shows that each set

$$\{x \in \mathbb{R} : A < f_n(x)\}$$

is open. Therefore their union is open. By part (a) again, f is lower semicontinuous. $\square$

Line (-11), (-9). Recall Proposition 2.3.1 basically says $f \le g \implies \int f \le \int g$, so to be nitpicking, it should be that $\sum a_k \lambda(U_k) \le \sum a_k \lambda(A_k) + \epsilon/2$ and $\int_a^b f^\infty(t)\,dt \le \int \sum a_k \chi_{A_k}\,dt$.

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Line 4. Suppose H is nondecreasing, but not strictly increasing. Then we can find $x < y, x, y \in [a, b]$, such that $f(x) = f(y)$. But by assumption, there exists $x_1 \in [x, y]$ and $\delta_1 > 0$ such that $H > H(x_1)$ on $(x_1, x_1 + \delta_1)$. So now we have

$$H(x_1 + \varepsilon) > H(x_1) \ge H(x) = H(y),$$

for some small ε, but since the nondecreasing property implies $H(x_1 + \varepsilon) \le H(y)$, we have a contradiction.

Line 19. Notice by Proposition 4.4.6, G is (absolutely) continuous, and so we can, as said in line (-6), apply Lemma 6.3.13 to $G - F$.

Line 21. Notice by definition of lower semicontinuity, for each positive ε, we can find a neighborhood of x such that for all y in this neighborhood with $y > x$, we have $g(y) > g(x) - \varepsilon$. For such y,

$$\frac{G(y) - G(x)}{y - x} = \frac{\int_x^y g(t)\, dt}{y - x} > g(x) - \varepsilon.$$

Therefore the $\liminf$ of LHS is $\geq g(x) - \varepsilon$. Also recall ε is arbitrary.

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Line 19. See section 4.2, exercise 3.

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Line (-4). See Corollary 1.6.3. Notice $\lambda(T(\cdot))$ defines a measure on the Borel measurable space.

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Line 1. $P_Y(A) = P \circ Y^{-1}(A) = P \circ X^{-1}\big(T^{-1}(A)\big) = P_X\big(T^{-1}(A)\big)$.

For line 2, notice

$$\chi_{T^{-1}(A)} \circ T^{-1}(t) = 1 \text{ iff } T^{-1}(t) \in T^{-1}(A) \text{ iff } t \in A.$$

Also see Theorem 6.1.7. Here the mapping T^{-1} is taken as the T in 6.1.7.

Line (-3).

Theorem. Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space and let $(X_i)_{i\in I}$ be a collection of random variables. Then

$$\{X_i : i \in I\} \text{ are independent} \quad \Longleftrightarrow \quad \{\sigma(X_i) : i \in I\} \text{ are independent.}$$

Proof. Recall that $(X_i)_{i\in I}$ are independent if for every finite collection of distinct indices $i_1, \ldots, i_n \in I$ and every sets $B_1, \ldots, B_n \in \mathcal{B}$,

$$\mathbb{P}\big(X_{i_1} \in B_1, \ldots, X_{i_n} \in B_n\big) = \prod_{k=1}^{n} \mathbb{P}(X_{i_k} \in B_k).$$

The σ-algebras $(\sigma(X_i))_{i\in I}$ are independent if for every such finite collection and every $A_k \in \sigma(X_{i_k})$,

$$\mathbb{P}\Big(\bigcap_{k=1}^{n} A_k\Big) = \prod_{k=1}^{n} \mathbb{P}(A_k).$$

(⇐) Suppose $\{\sigma(X_i)\}_{i\in I}$ are independent. For any $B_k \in \mathcal{B}$,

$$\{X_{i_k} \in B_k\} \in \sigma(X_{i_k}),$$

hence

$$\mathbb{P}\big(X_{i_1} \in B_1, \dots, X_{i_n} \in B_n\big) = \mathbb{P}\Big(\bigcap_{k=1}^{n} \{X_{i_k} \in B_k\}\Big) = \prod_{k=1}^{n} \mathbb{P}(X_{i_k} \in B_k).$$

Thus (X_i) are independent.

(⇒) Suppose $(X_i)_{i\in I}$ are independent. Fix distinct $i_1, \dots, i_n \in I$. For each k, define

$$\mathcal{C}_k := \{X_{i_k}^{-1}(B) : B \in \mathcal{B}\}.$$

Then each $\mathcal{C}_k$ is a π-system and it can be checked that

$$\sigma(\mathcal{C}_k) = \sigma(X_{i_k})$$

(because if a sigma algebra contains $\mathcal{C}_k$, it makes X_{i_k} measurable, so it contains $\sigma(X_{i_k})$, and therefore $\sigma(X_{i_k}) = \sigma(\mathcal{C}_k)$). By independence of the random variables,

$$\mathbb{P}\Big(\bigcap_{k=1}^{n} C_k\Big) = \prod_{k=1}^{n} \mathbb{P}(C_k), \qquad C_k \in \mathcal{C}_k.$$

We extend this to $\sigma(X_{i_k})$ using the π-λ theorem.

Step 1. Fix $A_2 \in \mathcal{C}_2, \dots, A_n \in \mathcal{C}_n$, and define

$$\mathcal{L}_1 := \left\{ A \in \sigma(X_{i_1}) : \mathbb{P}\Big(A \cap \bigcap_{k=2}^{n} A_k\Big) = \mathbb{P}(A) \prod_{k=2}^{n} \mathbb{P}(A_k) \right\}.$$

One can check that $\mathcal{L}_1$ is a λ-system (or d-system).

Since $\mathcal{C}_1 \subset \mathcal{L}_1$ and $\mathcal{C}_1$ is a π-system generating $\sigma(X_{i_1})$, the π-λ theorem gives

$$\mathcal{L}_1 \supset \sigma(X_{i_1}).$$

Thus the desired factorization holds for all $A_1 \in \sigma(X_{i_1})$ and $A_2, \dots, A_n \in \mathcal{C}_2, \dots, \mathcal{C}_n$.

Step 2. Fix $A_1 \in \sigma(X_{i_1})$ and $A_3 \in \mathcal{C}_3, \ldots, A_n \in \mathcal{C}_n$, and define

$$\mathcal{L}_2 := \left\{ A \in \sigma(X_{i_2}) : \mathbb{P}\Big(A_1 \cap A \cap \bigcap_{k=3}^{n} A_k\Big) = \mathbb{P}(A_1)\mathbb{P}(A) \prod_{k=3}^{n} \mathbb{P}(A_k) \right\}.$$

Again $\mathcal{L}_2$ is a λ-system containing $\mathcal{C}_2$, hence

$$\mathcal{L}_2 = \sigma(X_{i_2}).$$

Step 3. Continuing inductively, we obtain

$$\mathbb{P}\Big(\bigcap_{k=1}^{n} A_k\Big) = \prod_{k=1}^{n} \mathbb{P}(A_k) \quad \text{for all } A_k \in \sigma(X_{i_k}).$$

Since the indices were arbitrary, the family $(\sigma(X_i))_{i \in I}$ is independent. □

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Line 9. Take for example $n = 2$. We first use Theorem 5.2.1 to justify that $X_1 X_2$ is integrable ($|X_1 X_2|$ is the f in 5.2.1 and 5.2.1 (b) tells us $\int |X_1 X_2| < \infty$), and then apply 5.2.2

Line 16. Note that it is obvious (but may be tedious to prove following the framework in this book) that if X_i, X_j are independent, then $X_i - a$ and $X_j - b$ are independent for any real a, b. Also note that by exercise 3.3.9, if X_1 has a finite second moment, then X_1 is integrable.

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Line 13. For the first equation, note that

$$\nu_1(A - y) = \int_{A-y} f(x)\, d\lambda(x) = \int \chi_A(x + y) f(x)\, d\lambda(x),$$

while the second equation follows from Theorem 6.1.7 (5), with $T(x) = x - y$, that is,

$$\int \chi_A(x + y) f(x)\, d\lambda(x) = \int \chi_A(x) f(x - y)\, d\lambda(x).$$

Note that $f(x-y)$ is a measurable function on $\mathbb{R}^2$ and Proposition 5.2.1 (a) implies $\int f(x-y)\, \nu_2(dy)$ is a measurable function of x.

Line (-1). Why the variables Y_n are measurable, independent, and Bernoulli.

Let $X \sim$ Uniform$[0, 1]$, and define $Y_n(\omega)$ as the n-th digit in the binary expansion of $X(\omega)$ (choosing the expansion that ends in infinitely many 0's when ambiguity arises).

For each n, the variable Y_n can be written as

$$Y_n(\omega) = \mathbf{1}_{A_n}(X(\omega)),$$

where $A_n \subset [0,1]$ is a finite union of intervals of the form

$$\left[\frac{2i-1}{2^n}, \frac{2i}{2^n}\right), \quad i = 1, \dots, 2^{n-1}.$$

Hence A_n is a Borel set. Since X is measurable and indicator functions of Borel sets are measurable, it follows that each Y_n is measurable.

For each n, exactly half of the dyadic intervals correspond to $Y_n = 1$ and half correspond to $Y_n = 0$. Therefore,

$$\mathbb{P}(Y_n = 1) = \frac{1}{2}, \quad \mathbb{P}(Y_n = 0) = \frac{1}{2}.$$

Thus, $Y_n \sim \text{Bernoulli}(1/2)$.

Consider any finite collection $Y_1, \dots, Y_k$. Fix $(y_1, \dots, y_k) \in \{0,1\}^k$. Then the event

$$\{Y_1 = y_1, \dots, Y_k = y_k\}$$

corresponds to the set of ω such that $X(\omega)$ lies in a single dyadic interval of length 2^{-k}:

$$\left[\frac{m}{2^k}, \frac{m+1}{2^k}\right)$$

for some integer m.

Since X is uniformly distributed on $[0,1]$, we have

$$\mathbb{P}(Y_1 = y_1, \dots, Y_k = y_k) = 2^{-k}.$$

On the other hand,

$$\mathbb{P}(Y_1 = y_1) \cdots \mathbb{P}(Y_k = y_k) = \left(\frac{1}{2}\right)^k = 2^{-k}.$$

Thus,

$$\mathbb{P}(Y_1 = y_1, \dots, Y_k = y_k) = \prod_{i=1}^{k} \mathbb{P}(Y_i = y_i),$$

which shows that $\{Y_n\}$ are independent.

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Line 24. I don't understand why 1.5.6 helps. But I think with Corollary 1.6.3 we can also reach the conclusion. Notice first that all half open half closed cubes of that form within $[0,1]$ are a π

system. Denote this system by $\mathcal{A}$.

By taking the intersection with $[0,1/2)\cup[1/2,1)$ for all sets involved in the statement of Lemma 1.4.2 (recall by definition, an open subset of $[0,1)$ under the subspace topology is precisely the intersection of an open subset of $\mathbb{R}$ with the set $[0,1)$), we see that each open subset of $[0,1)$ is the union of a countably disjoint cubes from $\mathcal{A}$. Now we use Corollary 1.6.3. Suppose a sigma algebra $\sigma \supset \mathcal{A}$, then it contains all countable unions of sets in $\mathcal{A}$ so contains all open sets in $[0,1)$, and therefore it contains $\mathcal{B}\big([0,1)\big)$. This means $\sigma(\mathcal{A}) = \mathcal{B}\big([0,1)\big)$ and Lemma 1.6.3 implies $\mathbb{P}_X = \lambda$ on the Borel space of $[0,1)$. So X is uniformly distributed on $[0,1)$ and if we apply the argument on page 315, line (-3) again, we can also view X as uniformly distributed on $[0,1]$.

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Corollary 10.1.16. Let $X_1,\dots,X_k$ be independent random variables and let f be a Borel measurable function. For any Borel sets $A_1,\dots,A_k$, we have

$$\{f(X_i)\in A_i\} = \{X_i \in f^{-1}(A_i)\}.$$

Since $f^{-1}(A_i)$ are Borel sets, the independence of $X_1,\dots,X_k$ implies

$$\mathbb{P}(X_1\in f^{-1}(A_1),\dots,X_k\in f^{-1}(A_k)) = \prod_{i=1}^{k}\mathbb{P}(X_i\in f^{-1}(A_i)).$$

Therefore,

$$\mathbb{P}(f(X_1)\in A_1,\dots,f(X_k)\in A_k) = \prod_{i=1}^{k}\mathbb{P}(f(X_i)\in A_i),$$

which shows that $f(X_1),\dots,f(X_k)$ are independent.

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Example 10.2.4. Let $\{X_n\}_{n\ge 1}$ be a sequence of independent random variables, and define

$$S_n = \sum_{k=1}^{n} X_k.$$

Consider the event

$$A := \left\{\omega : \lim_{n\to\infty} S_n(\omega) \text{ exists in } \mathbb{R}\right\}.$$

We claim that A belongs to the tail σ-algebra

$$\mathcal{T} := \bigcap_{k=1}^{\infty}\sigma(X_k, X_{k+1},\dots).$$

Fix $k \geq 1$. Define

$$T_n^{(k)} := \sum_{j=k}^{n} X_j = S_n - S_{k-1}.$$

Then for all $n \geq k$,

$$S_n = S_{k-1} + T_n^{(k)}.$$

Observe that for each ω,

$$\lim_{n\to\infty} S_n(\omega) \text{ exists} \qquad \Longleftrightarrow \qquad \lim_{n\to\infty} T_n^{(k)}(\omega) \text{ exists.}$$

Indeed, adding the constant $S_{k-1}(\omega)$ does not affect convergence.

Therefore,

$$A = \left\{\omega : \lim_{n\to\infty} T_n^{(k)}(\omega) \text{ exists}\right\}.$$

But for each $n \geq k$, $T_n^{(k)}$ is measurable with respect to $\sigma(X_k, X_{k+1}, \ldots, X_n)$, hence the event

$$\left\{\lim_{n\to\infty} T_n^{(k)} \text{ exists}\right\}$$

belongs to $\sigma(X_k, X_{k+1}, \ldots)$.

Thus $A \in \sigma(X_k, X_{k+1}, \ldots)$ for every $k \geq 1$, and therefore

$$A \in \bigcap_{k=1}^{\infty} \sigma(X_k, X_{k+1}, \ldots) = \mathcal{T}.$$

Hence A is a tail event.

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Line (-9). Let

$$S_j := \sum_{i=j}^{\infty} \frac{1}{i^2}.$$

Since the function $x \mapsto x^{-2}$ is decreasing on $(0, \infty)$, we have

$$\sum_{i=j+1}^{\infty} \frac{1}{i^2} \leq \int_j^{\infty} \frac{dx}{x^2} = \frac{1}{j}.$$

Therefore,

$$S_j = \frac{1}{j^2} + \sum_{i=j+1}^{\infty} \frac{1}{i^2} \leq \frac{1}{j^2} + \frac{1}{j}.$$

Since $j \geq 1$, we have $\frac{1}{j^2} \leq \frac{1}{j}$, and hence $S_j \leq \frac{2}{j}$. Thus, one can take $C = 2$.

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Line 14. Note that the function $d(x, F)$ is 1-Lipschitz in x. To see this, for any $x, y \in \mathbb{R}^d$ and any $z \in F$, by the triangle inequality,

$$|x - z| \leq |x - y| + |y - z|.$$

Taking the infimum over $z \in F$, we obtain

$$d(x, F) = \inf_{z \in F} |x - z| \leq |x - y| + \inf_{z \in F} |y - z| = |x - y| + d(y, F).$$

Hence,

$$d(x, F) - d(y, F) \leq |x - y|.$$

By symmetry (interchanging x and y), we also have

$$d(y, F) - d(x, F) \leq |x - y|.$$

Therefore,

$$|d(x, F) - d(y, F)| \leq |x - y|.$$

This shows that $d(\cdot, F)$ is 1-Lipschitz. From here it is easy to see f_k is k-Lipschitz, and hence uniformly continuous.

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Line (-5). Here it means C_i is a subinterval of U_i.

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Lemma. Let $z, z_1, z_2, \ldots \in \mathbb{C}$ and suppose that $z_n \to z$. Then

$$\lim_{n \to \infty} \left(1 + \frac{z_n}{n}\right)^n = e^z.$$

Proof. By the binomial theorem,

$$\left(1 + \frac{z_n}{n}\right)^n = \sum_{k=0}^{n} \binom{n}{k} \frac{z_n^k}{n^k}.$$

Define

$$a_{n,k} = \binom{n}{k} \frac{z_n^k}{n^k}, \qquad k \geq 0,$$

with the convention that $a_{n,k} = 0$ whenever $k > n$. Then

$$\left(1 + \frac{z_n}{n}\right)^n = \sum_{k=0}^{\infty} a_{n,k}.$$

Fix $k \geq 0$. Since

$$\binom{n}{k} = \frac{n(n-1)\cdots(n-k+1)}{k!},$$

we may write

$$a_{n,k} = \frac{1}{k!}\left(1 - \frac{1}{n}\right)\left(1 - \frac{2}{n}\right)\cdots\left(1 - \frac{k-1}{n}\right) z_n^k.$$

Because $z_n \to z$, we have $z_n^k \to z^k$, and each factor

$$1 - \frac{j}{n} \to 1 \qquad (j = 1, \dots, k-1).$$

Therefore,

$$a_{n,k} \longrightarrow \frac{z^k}{k!} \qquad \text{as } n \to \infty.$$

Since $z_n \to z$, the sequence $\{z_n\}$ is bounded. Choose $M > 0$ such that

$$|z_n| \leq M, \qquad |z| \leq M$$

for all n. Then

$$|a_{n,k}| = \left|\binom{n}{k}\frac{z_n^k}{n^k}\right| \leq \frac{n^k}{k!}\frac{M^k}{n^k} = \frac{M^k}{k!}.$$

The dominating sequence

$$b_k = \frac{M^k}{k!}$$

is summable, since

$$\sum_{k=0}^{\infty} \frac{M^k}{k!} = e^M < \infty.$$

Hence the dominated convergence theorem on $(\mathbb{N}, \#)$, where # denotes counting measure, implies that

$$\lim_{n\to\infty} \sum_{k=0}^{\infty} a_{n,k} = \sum_{k=0}^{\infty} \lim_{n\to\infty} a_{n,k}.$$

Consequently,

$$\lim_{n\to\infty}\left(1 + \frac{z_n}{n}\right)^n = \sum_{k=0}^{\infty} \frac{z^k}{k!} = e^z.$$

This completes the proof. $\square$

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Line 18. It says $\phi_\mu^{(0)}$ has the required form. It actually denotes $1 \equiv x^0$ for consistency of notation, without worrying the definition of 0^0.

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Line 11. We begin with

$$\phi_P'(t) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} ix\, e^{ixt} e^{-x^2/2}\, dx.$$

Observe that

$$\frac{d}{dx}\Big(-e^{-x^2/2}\Big) = xe^{-x^2/2},$$

so that

$$ix\, e^{ixt} e^{-x^2/2} = ie^{ixt}\, d\Big(-e^{-x^2/2}\Big)\,.$$

Hence

$$\phi_P'(t) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} ie^{ixt}\, d\Big(-e^{-x^2/2}\Big)\,.$$

Applying integration by parts with

$$u(x) = ie^{ixt}, \qquad dv = d\Big(-e^{-x^2/2}\Big)\,,$$

we have

$$v(x) = -e^{-x^2/2}, \qquad du = i(it)e^{ixt}\, dx = -t\, e^{ixt}\, dx.$$

Therefore,

$$\int_{-\infty}^{\infty} ie^{ixt}\, d\Big(-e^{-x^2/2}\Big) = \Big[-ie^{ixt}e^{-x^2/2}\Big]_{-\infty}^{\infty} - \int_{-\infty}^{\infty} (-e^{-x^2/2})(-te^{ixt})\, dx.$$

Since

$$|e^{ixt}| = 1,$$

we have

$$\Big|-ie^{ixt}e^{-x^2/2}\Big| = e^{-x^2/2} \to 0 \qquad (|x| \to \infty),$$

and thus the boundary term vanishes. Consequently,

$$\int_{-\infty}^{\infty} ie^{ixt}\, d\Big(-e^{-x^2/2}\Big) = -t \int_{-\infty}^{\infty} e^{ixt}e^{-x^2/2}\, dx.$$

Substituting this into the expression for $\phi_P'(t)$ yields

$$\phi_P'(t) = -\frac{t}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{ixt}e^{-x^2/2}\, dx.$$

Recognizing the integral as $\phi_P(t)$, we obtain

$$\boxed{\phi_P'(t) = -t\,\phi_P(t).}$$

Thus ϕ_P satisfies the differential equation

$$\phi_P'(t) + t\phi_P(t) = 0.$$

Multiplying by the integrating factor $e^{t^2/2}$ gives

$$\frac{d}{dt}\Big(e^{t^2/2}\phi_P(t)\Big) = e^{t^2/2}\big(\phi_P'(t) + t\phi_P(t)\big) = 0.$$

Hence

$$e^{t^2/2}\phi_P(t) = C$$

for some constant C. Since every characteristic function satisfies

$$\phi_P(0) = 1,$$

we find

$$C = e^0\phi_P(0) = 1.$$

Therefore

$$\boxed{\phi_P(t) = e^{-t^2/2}.}$$

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Line (-3). To justify the claim that

$$\gamma_\sigma * \mu \Rightarrow \mu \quad \text{as } \sigma \to 0,$$

let X be a random variable with distribution μ, and let

$$Z_\sigma \sim N(0, \sigma^2)$$

be independent of X. Then $\gamma_\sigma * \mu$ is the distribution of

$$X + Z_\sigma.$$

We first show that $X + Z_\sigma \to X$ in probability. For any $\varepsilon > 0$,

$$\mathbb{P}(|X + Z_\sigma - X| > \varepsilon) = \mathbb{P}(|Z_\sigma| > \varepsilon)\,.$$

Since $Z_\sigma = \sigma Z$ with $Z \sim N(0,1)$,

$$\mathbb{P}(|Z_\sigma| > \varepsilon) = \mathbb{P}\Big(|Z| > \frac{\varepsilon}{\sigma}\Big)\,.$$

As $\sigma \to 0$, we have $\varepsilon/\sigma \to \infty$, and hence

$$\mathbb{P}\Big(|Z| > \frac{\varepsilon}{\sigma}\Big) \to 0.$$

Therefore,

$$X + Z_\sigma \xrightarrow{P} X.$$

By Exercise 7, convergence in probability implies convergence in distribution of the corresponding laws. Hence,

$$\gamma_\sigma * \mu \Rightarrow \mu.$$

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Line (-3). To justify the existence of t_0, note that G is assumed to be continuous at x. Hence

$$\lim_{y \uparrow x} G(y) = G(x).$$

Therefore, for any $\varepsilon > 0$, we may choose $y < x$ sufficiently close to x so that

$$G(y) > G(x) - \varepsilon.$$

Since D is dense in $\mathbb{R}$, there exists $t_0 \in D$ such that

$$y < t_0 < x.$$

By the definition of G,

$$G(y) = \inf\{G_0(t) : t \in D,\ t > y\}.$$

Since $t_0 > y$, the value $G_0(t_0)$ is one of the elements over which the infimum is taken. Consequently,

$$G(y) \le G_0(t_0).$$

Combining this with $G(y) > G(x) - \varepsilon$, we obtain

$$G(x) - \varepsilon < G(y) \leq G_0(t_0).$$

Thus there exists $t_0 \in D$ with $t_0 < x$ such that

$$G(x) - \varepsilon < G_0(t_0),$$

as required.

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Line (-12). It is easy to check all sets that are a finite disjoint union of sets of the form $\{Y = y_j\}$ is indeed a sigma algebra $\mathcal{S}$. It's also straightforward to check that if Y is measurable w.r.t. a sigma algebra $\mathcal{A}$, then $\mathcal{A} \supset \mathcal{S}$. So, $\sigma(Y) = \mathcal{S}$.

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Line 12. We notice $\int |X| = \int_{A_+} |X| + \int_{A_-} |X| \geq \int_{A_+} X + \int_{A_-} (-X)$.

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Proposition 10.4.4 Monotone and Dominated Convergence Theorems for Conditional Expectations. Let $(\Omega, \mathcal{A}, P)$ be a probability space, let $\mathcal{B}$ be a sub-σ-algebra of $\mathcal{A}$, and let $X_1, X_2, \ldots$ be random variables with finite expected values such that $\lim_n X_n$ exists almost surely. If

1. $\{X_n\}$ is an increasing sequence such that $\lim_n \mathbb{E}(X_n)$ is finite, or
2. there exists a random variable Y with finite expected value such that each X_n satisfies $|X_n| \leq Y$ almost surely,

then $\lim_n X_n$ has a finite expected value and

$$\mathbb{E}\Big(\lim_n X_n \,\Big|\, \mathcal{B}\Big) = \lim_n \mathbb{E}(X_n \mid \mathcal{B}) \quad \text{almost surely.}$$

We need X_i to be of finite expected values because the conditional expectation is only defined for integrable random variables. We need that $\lim_n X_n$ exists almost surely, because we need to write $\mathbb{E}(\lim_n X_n \,|\, \mathcal{B})$. But is it legit to define the expectation for a random variable Z that's only defined almost surely? Recall we never defined the integral for a random variable that's only defined almost everywhere. I guess if Z is only defined a.s., then $\mathbb{E}[Z], \mathbb{E}[Z|\mathcal{B}]$ are calculated for the extended version of Z (define arbitrary values on the null set where Z isn't defined)?

Notice by 10.4.3. (b), $\mathbb{E}[X_k|\mathcal{B}] \leq \mathbb{E}[X_{k+1}|\mathcal{B}]$ a.s. for all $k \in \mathbb{N}$. Therefore the set $\{\omega : [X_1|\mathcal{B}] \leq \mathbb{E}[X_2|\mathcal{B}] \leq ...\}$ has probability 1.

When we assume (a) holds and that all $X_k \geq 0$, we have that by corollary 2.3.13, $\mathbb{E}[X_k|\mathcal{B}] \geq 0$ a.s. So up to modifying $\mathbb{E}[X_k|\mathcal{B}]$ on the null sets where they are < 0, to make them ≥ 0 everywhere, we are able to apply Theorem 2.4.1.

For the case where (a) holds and X_k are not necessarily nonnegative, we have that X_1 is finite a.s. due to integrability, so $X_n - X_1$ is well defined for all n, a.s. Notice when we reach

$$\mathbb{E}\Big(\lim_n (X_n - X_1) \,\Big|\, \mathcal{B}\Big) = \lim_n \mathbb{E}(X_n - X_1 \mid \mathcal{B}) \quad \text{almost surely,}$$

we add $\mathbb{E}[X_1 \mid \mathcal{B}]$, which is finite almost surely due to its integrability, to both sides. With some tedious arguments that deal with "almost surely", we can reach the desired conclusion. For example, following the second to fourth lines in the original proof, we know that $\mathbb{E}(X_n - X_1 \mid \mathcal{B}) \uparrow \lim \mathbb{E}(X_n - X_1 \mid \mathcal{B})$ a.s. But we also have that $\mathbb{E}(X_n - X_1 \mid \mathcal{B}) = \mathbb{E}(X_n \mid \mathcal{B}) - \mathbb{E}(X_1 \mid \mathcal{B})$ a.s. due to 10.4.3. And because of the second to fourth lines in the original proof, we know that $\lim \mathbb{E}(X_n \mid \mathcal{B})$ exists a.s. We also know $\lim \mathbb{E}(X_1 \mid \mathcal{B}) = \mathbb{E}(X_1 \mid \mathcal{B}) \in \mathbb{R}$ a.s. Therefore by the additivity of sequential limits,

$$\lim\{\mathbb{E}(X_n \mid \mathcal{B}) - \mathbb{E}(X_1 \mid \mathcal{B})\} = \lim\{\mathbb{E}(X_n \mid \mathcal{B})\} - \mathbb{E}(X_1 \mid \mathcal{B}) \quad \text{a.s.}$$

Recall the fact (we will prove this later) that if $f_n = g_n$ a.s., $\forall n$ and if $\lim f_n, \lim g_n$ both are well defined a.s., then $\lim f_n = \lim g_n$ a.s.

Now take $f_n = \mathbb{E}(X_n - X_1 \mid \mathcal{B}), g_n = \mathbb{E}(X_n \mid \mathcal{B}) - \mathbb{E}(X_1 \mid \mathcal{B})$. So we have

$$\lim \mathbb{E}(X_n - X_1 \mid \mathcal{B}) = \lim\{\mathbb{E}(X_n \mid \mathcal{B})\} - \mathbb{E}(X_1 \mid \mathcal{B}) \quad \text{a.s.}$$

and if we add $\mathbb{E}(X_1 \mid \mathcal{B})$ to $\lim \mathbb{E}(X_n - X_1 \mid \mathcal{B})$, it becomes $\lim\{\mathbb{E}(X_n \mid \mathcal{B})\}$ a.s.

Claim: Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space and let $\{f_n\}_{n\geq 1}$, $\{g_n\}_{n\geq 1}$ be measurable functions such that $f_n = g_n$ a.s. for every n. If $\lim_{n\to\infty} f_n$ and $\lim_{n\to\infty} g_n$ both are well defined (either in $\mathbb{R}$ or $+/-\infty$) a.s., then

$$\lim_{n\to\infty} f_n = \lim_{n\to\infty} g_n \quad \textbf{a.s.}$$

Proof. For each n, let $N_n = \{\omega \in \Omega : f_n(\omega) \neq g_n(\omega)\}$. Then $\mathbb{P}(N_n) = 0$ by hypothesis. Set

$$A := \Omega \setminus \bigcup_{n=1}^{\infty} N_n,$$

so $\mathbb{P}(A) = 1$ and $f_n(\omega) = g_n(\omega)$ for all n whenever $\omega \in A$.

Let

$$E_f := \{\omega : (f_n(\omega)) \text{ converges}\}, \qquad E_g := \{\omega : (g_n(\omega)) \text{ converges}\}.$$

By assumption, $\mathbb{P}(E_f) = \mathbb{P}(E_g) = 1$.

Fix $\omega \in A \cap E_f \cap E_g$ (which has probability 1). Then the sequences $(f_n(\omega))$ and $(g_n(\omega))$ are termwise equal and both converge (to either a real number or infinity), so their limits agree:

$$\lim_{n\to\infty} f_n(\omega) = \lim_{n\to\infty} g_n(\omega).$$

Hence

$$\{\lim f_n \neq \lim g_n\} \subseteq A^c \cup E_f^c \cup E_g^c,$$

whose probability is 0. Thus $\lim f_n = \lim g_n$ a.s. $\square$

Suppose (b) holds. We are assuming $|X_n| \leq Y$ a.s. for each n. Say this holds for each n on S_n, with $\Pr(S_n) = 1$. So $\cap_n S_n$ is of probability 1. Say $\lim_n X_n$ exists on the set Q of probability 1, so $\cap_n S_n \cap Q$ is of probability 1, on which $|\lim_n X_n| \leq Y$. All the remaining discussions in this paragraph, if treated rigorously, should be seen as omitting the tedious "a.s." argument.

"If we apply the first half of the proposition to the sequence $\{Y_n\}$": do we have that $\lim \mathbb{E}[Y_n]$ is finite? Yes. Recall $|X_n| \leq Y$ a.s., so we have that $|Y_n| \leq Y, \forall n$ a.s., and therefore, Y_n is integrable for each n, a.s. Up to redefining all Y_n so that at points ω where $Y_n(\omega) = \infty$ or $-\infty$, we let $Y_n(\omega) = 0$, we can apply Proposition 2.3.6 and conclude $\{\mathbb{E}[Y_n]\}$ is increasing. This increasing sequence is bounded by $\mathbb{E}[Y] \in \mathbb{R}$ so its limit is a real number.

For the well-definedness of sequence $\{Y - Z_n\}$, we notice because $|X_n| \leq Y, \forall n$ a.s., and $|Y| \in \mathbb{R}$ a.s. due to integrability, we have a.s. $Z_n \in \mathbb{R}$. So the sequence $\{Y - Z_n\}$ is well defined and $\lim(Y - Z_n)$ exists a.s.

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Line 4. For example, if we checked that for each $n \in \mathbb{N}_0$, and for each $B \in \mathcal{F}_n$, we have $\int_B X_n \leq \int_B X_{n+1}$, then by the definition of conditional expectation, we know

$$\int_B X_n \leq \int_B X_{n+1} = \int_B \mathbb{E}[X_{n+1} \mid \mathcal{F}_n],$$

and by Theorem 2.3.13, we know $X_n \leq \mathbb{E}[X_{n+1} \mid \mathcal{F}_n]$ a.s.

Exercises

1.1.1

Intuition says the answer may be the sigma algebra $\mathcal{A}$ given by (f) in examples 1.1.1. Now let's try to prove it. Suppose $\mathcal{B}$ is a sigma algebra on $\mathbb{R}$ that contains all one-point sets. Then by closure under the formation of countable unions, $\mathcal{B}$ contains all countable subsets of $\mathbb{R}$. By closure of complementation, $\mathcal{B}$ contains all subsets of $\mathbb{R}$ whose complement is countable. Therefore, $\mathcal{B} \supset \mathcal{A}$.

1.1.2

Consider $(-\infty, b]$, where b is some rational number. By Proposition 1.1.4, $(-\infty, b] \in \mathcal{B}(\mathbb{R})$. Now suppose $\mathcal{S}$ is a sigma-algebra that contains all such $(-\infty, b]$. For any real number c, there is a sequence of rational numbers $(b_n) \downarrow c$, meaning $(-\infty, c] = \cap_n(-\infty, b_n]$. Therefore, $(-\infty, c] \in \mathcal{S}$. By Proposition 1.1.4, $\mathcal{S} \supset \mathcal{B}(\mathbb{R})$.

1.1.4

By Proposition 1.1.7, we only need to show that $\cup_n^\infty A_n \in \mathcal{A}$ if (A_n) is an increasing sequence of sets. Let (A_n) be such a sequence. Define $B_m := A_m - A_{m-1} \in \mathcal{A}, m \geq 2$ with $B_1 := A_1$, then $\cup_{n=1}^\infty A_n = \cup_{m=1}^\infty B_m$ (if not feeling comfortable, you can show this by arguing that LHS contains RHS, and RHS contains LHS). But each two different B_i and B_j are disjoint, so $\cup_{n=1}^\infty A_n = \cup_{m=1}^\infty B_m \in \mathcal{A}$.

1.1.5

Consider $X = \{1, 2, 3\}$. And let the two sigma algebras be $\{\{3\}, \{1, 2\}, \{1, 2, 3\}, \emptyset\}$ and $\{\{2\}, \{1, 3\}, \{1, 2, 3\}, \emptyset\}$. The union of these two is not a sigma algebra.

1.1.7

As it says that $\mathcal{S}$ is a "collection of subsets", we assume $\mathcal{S} \neq \emptyset$, i.e., there exists $Y \subset X, Y \in \mathcal{S}$. Let $\mathcal{A}$ be defined as in the hint. We first show it is a sigma algebra. Since $\{Y\}$ is a countable subfamily of $\mathcal{S}$, $\mathcal{A} \supset \sigma(\{Y\})$ which includes X and therefore $X \in \mathcal{A}$. If $A \in \mathcal{A}$, then by the definition of $\mathcal{A}$, there is some countable subfamily $\mathcal{C}$ of $\mathcal{S}$ such that $A \in \sigma(\mathcal{C})$, so $A^c \in \sigma(\mathcal{C}) \subset \mathcal{A}$. Now assume $A_1, A_2, ... \in \mathcal{A}$, i.e., $A_1 \in \sigma(\mathcal{C}_1), A_2 \in \sigma(\mathcal{C}_2), ...$ where $\mathcal{C}_i$ are countable subfamilies of $\mathcal{S}$, and $\mathcal{C}_i$ might be identical for different values of i. Now we define $\mathcal{C}_\infty := \cup_i \mathcal{C}_i$ which is also a countable subfamily. We notice for all $i \in \mathbb{N}$, $A_i \in \sigma(\mathcal{C}_i) \subset \sigma(\mathcal{C}_\infty)$, and so $\cup_i A_i \in \sigma(\mathcal{C}_\infty) \subset \mathcal{A}$. The above shows $\mathcal{A}$ is a sigma algebra.

Since $\mathcal{A} =$ the union of elements in $\{\sigma(\mathcal{C}) : \mathcal{C}$ a countable subfamily of $\mathcal{S}\}$, and $\sigma(\mathcal{C}) \subset \sigma(\mathcal{S})$ for all such $\mathcal{C}$, we know $\mathcal{A} \subset \sigma(\mathcal{S})$. For any element $Y \subset \mathcal{S}$, $\sigma(\{Y\}) \subset \mathcal{A}$. Therefore, the union of all such $\sigma(\{Y\})$ is a subset of $\mathcal{A}$. But the union of all such $\sigma(\{Y\})$ includes the union of all such $\{Y\}$ which is $\mathcal{S}$ itself. This means $\mathcal{S} \subset \mathcal{A}$. Now we have $\mathcal{S} \subset \mathcal{A} \subset \sigma(\mathcal{S}) \Rightarrow \mathcal{A} = \sigma(\mathcal{S})$. Therefore, for each A contained in $\sigma(\mathcal{S}) = \mathcal{A}$, A is contained in $\sigma(\mathcal{C}_0)$ for some $\mathcal{C}_0$ which is a countable subfamily of $\mathcal{S}$.

1.2.2

Let $(q_1, q_2, ...)$ be an enumeration of rational numbers, and let $\mathbb{I}$ denote the set of irrational numbers. Define $A_n := \{q_n\} \cup \mathbb{I}$. Then considering the sequence (A_n), we see μ is sigma-finite.

1.2.3
Take $A_n := \{n, n+1, n+2, ...\}$.

1.2.5
Assume $\delta_x = \delta_y$, i.e., for any $A \in \mathcal{A}, \delta_x(A) = \delta_y(A)$. Suppose it is not that x, y belong to exactly the same sets in $\mathcal{A}$. The case that neither of x, y belongs to any set of $\mathcal{A}$ counts as "belonging to exactly the same sets" so this case is not possible. Case 1: $x \in B \in \mathcal{A}$, and y belongs to no set in $\mathcal{A}$ (or, with roles of x and y switching). Then we have $\delta_x(B) \neq \delta_y(B)$, a contradiction. Case 2: both of x, y belong to some sets in $\mathcal{A}$, but the sets that either belongs to are not identical. Without loss of generality, suppose there is a $B \in \mathcal{A}$ s.t. $x \in B, y \notin B$. Then this again implies $\delta_x(B) \neq \delta_y(B)$.

Conversely, if x, y belong to exactly the same sets in $\mathcal{A}$, it is trivial to see that $\delta_x = \delta_y$.

1.2.6
(b). Once we have proven (a), consider the measures $\nu_k := \sum_{n=1}^{k} \mu_n$ and apply (a), then (b) follows.
(a). That $\mu(\emptyset) = 0$ is easy to see. For each $A \in \mathcal{A}, \mu(A)$ is nonnegative and well-defined as $\mu_n(A)$ is increasing in n. Now we check sigma additivity. Consider a disjoint sequence of sets $(A_k) \subset \mathcal{A}$. $\mu(\cup A_k) := \lim_{n\to\infty} \mu_n(\cup A_k) = \lim_{n\to\infty} \sum_{k=1}^{\infty} \mu_n(A_k)$. This actually equals (which we will prove shortly) $\sum_{k=1}^{\infty} \lim_{n\to\infty} \mu_n(A_k) = \sum_{k}^{\infty} \mu(A_k)$, which means countable additivity holds. Now we prove the claim used before.
Claim. Let $(a_{m,n})_{m,n\geq 0}$ be an array of nonnegative numbers such that, for all $n \geq 0, (a_{m,n})_{m\geq 0}$ is a nondecreasing sequence. Then, we have that $\lim_{m\to\infty} \sum_{n=0}^{\infty} a_{m,n} = \sum_{n=0}^{\infty} \lim_{m\to\infty} a_{m,n}$ in $[0, \infty]$.
Proof. We recall the fact that $\sum_i \sum_j a_{i,j} = \sum_j \sum_i a_{i,j}$ for a nonnegative doubly-indexed sequence $a_{i,j}$. Consider the array $A_{m,n} \geq 0$ given by $A_{0,n} = a_{0,n}$ and $A_{m,n} = a_{m,n} - a_{m-1,n}, m \geq 1$. Now, $\sum_{n=0}^{\infty} a_{m,n} = \sum_{n=0}^{\infty} \sum_{k=0}^{m} A_{k,n} = \sum_{k=0}^{m} \sum_{n=0}^{\infty} A_{k,n} \Rightarrow \lim_{m\to\infty} \sum_{n=0}^{\infty} a_{m,n} = \sum_{k=0}^{\infty} \sum_{n=0}^{\infty} A_{k,n}$. But $\sum_{k=0}^{\infty} \sum_{n=0}^{\infty} A_{k,n} = \sum_{n=0}^{\infty} \sum_{k=0}^{\infty} A_{k,n} = \sum_{n=0}^{\infty} \lim_{m\to\infty} \sum_{k=0}^{m} A_{k,n} = \sum_{n=0}^{\infty} \lim_{m\to\infty} a_{m,n}$.
Now back to the problem. If $\mu_n(A_k)$ is finite $\forall n, k$, then we have $\lim_{n\to\infty} \sum_{k=1}^{\infty} \mu_n(A_k)$ equals $\sum_{k=1}^{\infty} \lim_{n\to\infty} \mu_n(A_k)$ just as in the claim above. If some $\mu_n(A_k) = \infty$, we cannot resort to the claim as the difference of two infinities is not defined. But it is obvious that we still have the same $\lim_{n\to\infty} \sum_{k=1}^{\infty} \mu_n(A_k) = \sum_{k=1}^{\infty} \lim_{n\to\infty} \mu_n(A_k)$ as both sides are infinity.

1.2.9
In fact, the limsup of such a sequence (A_n) is defined as $\limsup_n A_n := \cap_{n\geq 1} \cup_{k\geq n} A_k$. Notice that $\limsup_n A_n$ contains exactly points that appear in infinitely many A_k's. To see this, suppose x belongs to infinitely many A_k's, then x belongs to $\cup_{k\geq n} A_k$ for every n, and therefore belongs to $\cap_{n\geq 1} \cup_{k\geq n} A_k$. Conversely, suppose x belongs to only finitely many A_k's, then there exists $N \in \mathbb{N}$ such that for all $n > N, x \notin A_n$. So for all $n > N, x \notin \cup_{k\geq n} A_k \Rightarrow x \notin \cap_{n\geq 1} \cup_{k\geq n} A_k$.

Fix $\epsilon > 0$, there exists $N \in \mathbb{N}, \sum_{n\geq N} \mu(A_n) < \epsilon$. Following the hint, we see that $\mu(\limsup A_n) \leq$

$\mu(\cup_{n\geq N} A_n) \leq \sum_{n\geq N} \mu(A_n) < \epsilon$. As ϵ is arbitrary, $\mu(\limsup A_n) = 0$.

This theorem is the Borel–Cantelli lemma. Here I give an intuitive explanation. Recall $\limsup A_n = \{x \in X : x \in A_n \text{ for infinitely many } n\}$. Suppose $\mu(\limsup A_n) > 0$. For some $x \in \limsup A_n$, the measure mass that x occupies is counted (countably) infinitely many times during the calculation $\sum_k \mu(A_k)$. That is because, for those A_k that contains x, their measure $\geq \mu(\{x\})$. If we allow some non-rigor and let i denote "countably infinitely many", then we know $\sum_k \mu(A_k) \geq i\mu(\{x\})$. For another $y \in \limsup A_n$, its measure is also counted (countably) infinitely many times during the calculation $\sum_k \mu(A_k)$. Therefore, considering x, y together, we have $\sum_k \mu(A_k) \geq i\mu(\{x, y\})$. We make this argument for all elements in $\limsup A_n$ (not very rigorous, because if there are uncountably many such elements, how would we "argue" for each of them? In what order?). Then we would have $\sum_k \mu(A_k) \geq i\mu(\limsup A_n)$. Since i denotes countable infinity and $\mu(\limsup A_n) > 0$, the RHS is ∞, meaning the LHS must be so as well.

1.3.3

If $\lambda^*(A) = \infty$, take $B = \mathbb{R}$. Now suppose $\lambda^*(A) < \infty$. For $n \in \mathbb{N}$, find a sequence of open intervals $\{(a_k^n, b_k^n)\}_{k=1}^\infty$ that covers A with $\sum_k b_k^n - a_k^n < \mu^*(A) + 1/n$. Define $B^n = \cup_k (a_k^n, b_k^n)$. Note that B^n is Borel, contains A, and $\mu(B^n) \leq \sum_k b_k^n - a_k^n < \mu^*(A) + 1/n$. Now $B := \cap_n B^n$ is also Borel, contains A, and satisfies for any $n \in \mathbb{N}$, $\mu(B) \leq \mu(B^n) < \mu^*(A) + 1/n$. Therefore, $\mu(B) \leq \mu^*(A)$ and so $\mu(B) = \mu^*(A)$

1.4.1

(a). We show $A :=$ the x-axis (and analogously y-axis) has zero measure. Consider any positive number c and the sequence of open rectangles $\{R_n = (-2^n, 2^n) \times (-c2^{-2n}, c2^{-2n})\}, n = 1, 2, ...$ Apparently $\cup_n R_n \supset A \Rightarrow \lambda^*(A) \leq \lambda^*(\cup_n R_n) \leq \sum_n \lambda^*(R_n) = \sum_n \mathrm{vol}(R_n) = 4c$. c can be arbitrarily small, meaning $\lambda^*(A) = 0$. By proposition 1.3.5, $\lambda(A) = 0$. If we consider the rotation and translation of the x-axis and use exercise 1.4.3, we can see that any straight line of $\mathbb{R}^2$ has Lebesgue measure zero.

1.4.3

(a). Let T denote the rotation in $\mathbb{R}^2$, which is obviously a bijection of $\mathbb{R}^2$. Notice T^{-1} is also a rotation. We follow the hint. $\mathcal{A} := \{A \subset \mathbb{R}^2 : T(A) \in \mathcal{B}(\mathbb{R}^2)\}$ is easily seen to be a sigma algebra: $\mathbb{R}^2 \in \mathcal{A}$; if $A \in \mathcal{A}$, then $A^c \in \mathcal{A}$ because $T(A^c) = T(A)^c$ as T is a bijection; if $(A_n)_n \subset \mathcal{A}$, then $\cup_n A_n \in \mathcal{A}$ because $T(\cup_n A_n) = \cup_n T(A_n)$ (easy to check inclusion of both directions).

Also, $\mathcal{A}$ includes all open sets: Let A be open. Since all open balls are a basis for the general topology on the Euclidean space, A is a union of some open balls. Now, $T(A)$ is the union of the rotation about origin of these open balls, which are still open balls. So, $T(A)$ is open and so $A \in \mathcal{A}$. Therefore, $\mathcal{A} \supset \mathcal{B}(\mathbb{R}^2)$. Similarly we can show $\mathcal{C} := \{A \subset \mathbb{R}^2 : T^{-1}(A) \in \mathcal{B}(\mathbb{R}^2)\} \supset \mathcal{B}(\mathbb{R}^2)$.

Now we show the converse inclusion. Let $A \in \mathcal{A} : T(A) \in \mathcal{B}(\mathbb{R}^2)$. Therefore, $T(A) \in \mathcal{C}$, which means $A = T^{-1} \circ T(A) \in \mathcal{B}(\mathbb{R}^2)$. This means $\mathcal{A} \subset \mathcal{B}(\mathbb{R}^2)$ and so the two sets are equal.
(b). Define μ by $\mu(A) = \lambda(T(A))$ on $\mathcal{B}(\mathbb{R}^2)$. It's easy to check μ is a nonzero measure. Then μ satisfies all conditions in proposition 1.4.5. For translation invariance, $\mu(A + x) = \lambda(T(A + x))$. But $T(A + x)$ is a translation of $T(A)$ so $\lambda(T(A + x)) = \lambda(T(A)) = \mu(A)$, by proposition 1.4.4. Therefore, μ is translation invariant. Now by proposition 1.4.5, we have $\mu(A) = c\lambda(A), \forall A \in \mathcal{B}(\mathbb{R}^2)$.

Now let A be the unit disc. A is closed so is measurable for the Borel sigma algebra. Also, A has positive Lebesgue measure as it includes a square, whose volume is positive. So $c = \lambda(A)/\mu(A) = \lambda(A)/\lambda(T(A)) = \lambda(A)/\lambda(A) = 1$.

1.4.4

Look up "Smith–Volterra–Cantor set" or "fat Cantor set". The following is from the Wikipedia.

In general, one can remove r_n from each remaining subinterval at the n-th step of the algorithm, and end up with a Cantor-like set. The resulting set will have positive measure if and only if the sum of the sequence is less than the measure of the initial interval. For instance, suppose the middle intervals of length a^n are removed from $[0, 1]$ for the n-th iteration, for some $0 \leq a \leq 1/3$. Then, the resulting set has Lebesgue measure

$$1 - \sum_{n=0}^{\infty} 2^n a^{n+1} = \frac{1 - 3a}{1 - 2a}.$$

which goes from 0 to 1 as a goes from $1/3$ to 0.

1.5.1

By Proposition 1.5.1, we see $(\mathcal{A}_\mu)_{\overline{\mu}} \supset \mathcal{A}_\mu$. To show $(\mathcal{A}_\mu)_{\overline{\mu}} \subset \mathcal{A}_\mu$, take $A \in (\mathcal{A}_\mu)_{\overline{\mu}}$, and notice by definition there exist $E, F \in \mathcal{A}_\mu$ with $E \subset A \subset F, \overline{\mu}(F - E) = 0$. By the definition again, there exist $E_-, E_+, F_-, F_+ \in \mathcal{A}$ with $E_- \subset E \subset E_+, F_- \subset F \subset F_+, \mu(E_+ - E_-) = \mu(F_+ - F_-) = 0$. Since by Proposition 1.5.1, the restriction of $\overline{\mu}$ on $\mathcal{A}$ is μ, the last relation can also be written as $\overline{\mu}(E_+ - E_-) = \overline{\mu}(F_+ - F_-) = 0$. So, $\overline{\mu}(E - E_-) = \overline{\mu}(F_+ - F) = 0$

Recall the sets E_-, E, F, F_+ all belong to $\mathcal{A}_\mu$. We have

$$\begin{aligned}
\mu(F_+ - E_-) &= \overline{\mu}(F_+ - E_-) \\
&= \overline{\mu}\big((F_+ - F) \cup (F - E) \cup (E - E_-)\big) \\
&\leq \overline{\mu}(F_+ - F) + \overline{\mu}(F - E) + \overline{\mu}(E - E_-) = 0,
\end{aligned}$$

which means we have found E_-, F_+ that belong to $\mathcal{A}$ and "squeeze" A with $\mu(F_+ - E_-) = 0$. Therefore, $A \in \mathcal{A}_\mu$.

1.5.3

(a). Consider $X = \{0,1,2\}$ and $\mathcal{A} = \{\emptyset, \{0,1,2\}, \{0\}, \{1,2\}\}$. Let μ be the counting measure, and let $\nu = 0$. Then $\mathcal{A}_\nu$ is the power set of X, which is different than $\mathcal{A}_\mu$.

(b). This is wrong. If $\mathcal{A}$ is already the power set of X, then no matter what μ and ν are, $\mathcal{A}_\mu = \mathcal{A}_\nu$.

1.5.4

Let X denote a (Lebesgue) non-measurable set. Consider $X \times \{0\}$. Notice this set is contained in $\mathbb{R} \times \{0\}$ which has Lebesgue measure zero (see exercise 1.4.1). Since Lebesgue measure is complete on the sigma algebra of all Lebesgue measurable sets, $X \times \{0\}$ is measurable and has measure zero.

1.5.5

By the definition of inner measure, there exists a sequence of sets $(B_n) \subset \mathcal{A}$ such that $B_n \subset A, \forall n$ and $\mu(B_n) > \mu_*(A) - 1/n$. Take $A_0 = \cup_n B_n$. The monotonicity of measure implies that $\mu(A_0) \geq \mu_*(A)$. Also since $A_0 \subset A$, the definition of inner measure implies $\mu(A_0) \leq \mu_*(A)$. Therefore, $\mu(A_0) = \mu_*(A)$. By a symmetric argument using the definition of outer measure, there is an $A_1 \supset A, \mu(A_1) = \mu^*(A)$.

1.5.8

Take $B = [0,1] \cap A$ where A is the set constructed in Proposition 1.4.11. By the remark after Proposition 1.5.4, we know $\lambda_*(B) \leq \lambda_*(A) = 0 \Rightarrow \lambda_*(B) = 0$. Now because $B \subset [0,1]$, by the monotonicity of outer measure we know $\lambda^*(B) \leq 1$. Suppose $\lambda^*(B) < 1$. Then by the definition of outer measure, there is a cover of B by countably many bounded open intervals, $\{(a_i, b_i)\}$, such that $\sum_{i=1}^{\infty} b_i - a_i < 1$. This means, even if those open intervals do not overlap with each other, they still cannot cover the whole interval $(0,1)$ which has volume 1, and when this is the case, the set $[0,1] - \cup_i(a_i, b_i)$ contains some small open interval I (some drawing might help understand). When some of the intervals of this open cover do overlap, the part of $[0,1]$ that can be covered by $\cup_i(a_i, b_i)$ is even less, and again $[0,1] - \cup_i(a_i, b_i)$ contains some small open interval I. Now in 1.4.11, as $A := E + G_0$ and G_0 is dense in $\mathbb{R}$, we know B is dense in $[0,1]$. Therefore, these is some element of B in I, which isn't covered by $\cup_i(a_i, b_i)$, a contradiction. So $\lambda^*(B) = 1$.

1.5.11

(a). Trivial.

(b). By definition of outer measure, since $A_1 \cap C_1 \supset A_1 \cap C$, we have $\mu^*(A_1 \cap C) \leq \mu(A_1 \cap C_1)$. We now show the inequality is actually an equality. Suppose it is instead a strict inequality, i.e., LHS $<$ RHS, then there exists $B \supset A_1 \cap C, \mu(B) < \mu(A_1 \cap C_1)$. Note $C \subset B \cup (C - A_1) \subset B \cup (C_1 - A_1)$. But $\mu(B \cup (C_1 - A_1)) \leq \mu(B) + \mu(C_1 - A_1) < \mu(A_1 \cap C_1) + \mu(C_1 - A_1) = \mu(C_1)$, which contradicts that $\mu(C_1) = \mu^*(C)$. Therefore, $\mu^*(A_1 \cap C) = \mu(A_1 \cap C_1)$. The same argument for $A_2 \cap C_1$ shows that $\mu^*(A_2 \cap C) = \mu(A_2 \cap C_1)$.

(c). Say $B = A \cap C$ for some $A \in \mathcal{A}$. We want to show, for some C_1 as in part (b), $\mu(A \cap C_1) = \mu^*(A \cap C)$. But this is done in the beginning of the proof for part (b).

(d). The only nontrivial property to check is countable additivity. For a sequence $(B_n) \subset \mathcal{A}_C$ of disjoint sets, suppose $B_n = A_n \cap C$ for some $A_n \in \mathcal{A}, \forall n$. Then for some C_1 as in (b), (c), $\mu_C(\cup_n B_n) = \mu_C((\cup_n A_n) \cap C) := \mu((\cup_n A_n) \cap C_1) = \sum_n \mu(A_n \cap C_1) =: \sum_n \mu_C(A_n \cap C)$, but RHS is exactly $\sum_n \mu_C(B_n)$.

1.5.12

(a). Define $\mathcal{B} := \{(A_1 \cap C) \cup (A_2 \cap C^c) : A_1, A_2 \in \mathcal{A}\}$. We first check $\mathcal{B}$ is a sigma-algebra. Take $X = A_1 = A_2$, we see $X \in \mathcal{B}$. It is also easy to check $\mathcal{B}$ is closed under countable union. For $S = (A_1 \cap C) \cup (A_2 \cap C^c)$, notice

$$\begin{aligned} S^c &= (A_1^c \cup C^c) \cap (A_2^c \cup C) \\ &= (A_1^c \cap C) \cup (A_2^c \cap C^c) \cup (A_1^c \cap A_2^c) \\ &= (A_1^c \cap C) \cup (A_2^c \cap C^c) \cup (A_1^c \cap A_2^c \cap C) \cup (A_1^c \cap A_2^c \cap C^c) \\ &= (A_1^c \cap C) \cup (A_2^c \cap C^c) \end{aligned}$$

(b). Notice we have justified the existence of such C_0 and C_1 in exercise 1.5.5. We show only that μ_0 is a measure, as μ_1 is similar. Nonnegativity and that $\mu_0(\emptyset) = 0$ are obvious. For a sequence of disjoint sets $(A^n) \subset \sigma(\mathcal{A} \cup \{C\})$, we recall the fact shown in (a). Then we have that

$$\begin{aligned} \mu_0(\cup_n A^n) &= \mu\big((\cup_n A_1^n) \cap C_0\big) + \mu_{C^c}\big((\cup_n A_2^n) \cap C^c\big) \\ &= \big(\sum_n \mu(A_1^n \cap C_0)\big) + \big(\sum_n \mu_{C^c}(A_2^n \cap C^c)\big) \\ &= \sum_n \mu(A_1^n \cap C_0) + \mu_{C^c}(A_2^n \cap C^c) \\ &= \sum_n \mu(A^n \cap C_0) + \mu_{C^c}(A^n \cap C^c) = \sum_n \mu_0(A^n). \end{aligned}$$

This means countable additivity holds.

Now we want to show μ_0 agrees with μ on $\mathcal{A}$. For $A \in \mathcal{A}$,

$$\mu_0(A) = \mu(A \cap C_0) + \mu^*(A \cap C^c)$$

by exercise 1.5.11 (c). To show this equals $\mu(A)$, we can show $\mu(A \cap C_0^c) = \mu^*(A \cap C^c)$. We recall the definition of outer measure, which is the approximation "from above" by measure. Therefore, since $C_0^c \supset C^c$, we have $\mu(A \cap C_0^c) \geq \mu^*(A \cap C^c)$. Now we need to show $\mu(A \cap C_0^c) \leq \mu^*(A \cap C^c)$. How to show this hard direction?

The rest is easy to prove.

1.6.6

(a). Consider a collection $\{\mathcal{C}_i\}_{i\in I}$ where I is an index set, and $\mathcal{C}_i$ are monotone classes that include $\mathcal{A}$. Notice first that the power set of X is a monotone class of X that includes $\mathcal{A}$, therefore, the collection $\{\mathcal{C}_i\}_{i\in I}$ is not empty. It is then easy to argue that $\cap_{i\in I}\mathcal{C}_i$ is what we look for, i.e., $\cap_{i\in I}\mathcal{C}_i = m(\mathcal{A})$.
(b). See online resources.

2.1.3

Define functions $g_n(x) = \frac{f(x+1/n)-f(x)}{1/n}, n \in \mathbb{N}$. Each g_n is measurable as the difference of two continuous functions is continuous therefore measurable. $f' = \lim_n g_n$ is measurable by Proposition 2.1.5.

2.1.4

The set considered is $\{x \in X : \liminf f_n(x) = \limsup f_n(x)\} \cap \{x \in X : \limsup f_n(x) \in \mathbb{R}\}$. The former set appearing in this intersection is measurable because of Propositions 2.1.3, 2.1.5. The latter set equals $\cup_{m\in\mathbb{N}}\{x \in X : \limsup f_n(x) \in (-m, m)\} \in \mathcal{A}$.

2.1.7

(a). We use the notation from example 2.1.10. This is obvious by recalling the definition of f on the set $[0,1] - K$. Take the interval $I = [2/9, 3/9]$ for example. $\mu(I) = \mu((-\infty, 3/9]) - \mu((-\infty, 2/9))$, which by continuity is $F(3/9) - \lim_{y\uparrow 2/9} F(y) = \lim_{y\downarrow 3/9} F(y) - \lim_{y\uparrow 2/9} F(y) = 1/2 - 1/4 = 1/4$.
(b). Recall $K_n \supset K_{n+1}$ and that $\mu(K_n) = 2^n \times 1/2^n = 1$. $\mu(K) = \mu(\cap_n K_n) = \mu(\lim_n K_n) = 1$ by Proposition 1.2.5., the upper continuity of measure.
(c). Fix x. $\mu(\{x\}) = \mu((-\infty, x]) - \mu((-\infty, x)) = F(x) - \lim_{y\uparrow x} F(y) = 0$ because Cantor function is continuous.

2.1.9

Suppose f has the form $f_1\chi_C + f_2\chi_{C^c}$. Notice since $\sigma(\mathcal{A}\cup\{C\}) \supset \mathcal{A}$, both f_1 and f_2 are $\sigma(\mathcal{A}\cup\{C\})$-measurable. As $C \in \sigma(\mathcal{A}\cup\{C\}), \chi_C$ is $\sigma(\mathcal{A}\cup\{C\})$-measurable. Therefore, $f = f_1\chi_C + f_2\chi_{C^c}$ is so.

What about the hard direction?

2.2.1

Let f, g be constant and agree on $\mathbb{Q}$ but constant and disagree on $\mathbb{I}$. Note that singletons are closed sets so are all Borel, and so $\mathbb{Q}$ is Borel. By example 2.1.2. (d), f, g are Borel-measurable.

2.2.3

Suppose $f(x) < g(x)$. Then by continuity of $f - g$, there is a neighborhood (a, b) of x where $f - g < 0$. The interval (a, b) has positive measure, a contradiction.

2.2.5

There exist a negligible set E and a zero-measure $N \supset E$ such that $f_n \to f$ on N^c. For all n, define g_n as f_n on N^c and f on N. $g_n = f_n\chi_{N^c} + f\chi_N$ so are measurable.

2.3.1

Consider $\max(f, g) - g$, which equals $\max(f - g, 0) = (f - g)^+ \in \mathcal{L}^1$. Since $g \in \mathcal{L}^1$ we have that $\max(f, g) = g + (f - g)^+ \in \mathcal{L}^1$. Notice $\min(f, g) = -\max(-f, -g)$.

2.3.2

Take $f = g = \frac{1}{\sqrt{x}}\chi_{(0,1]}(x)$. Then $fg = \frac{1}{x}\chi_{(0,1]}(x)$. f is Riemann integrable on $(0, 1]$ with Riemann integral equal to 2, so its Lebesgue integral is also 2 (see section 2.5). To see fg is not Lebesgue integrable, we define $f_1 = 1\chi_{(0,1]}$, $f_2 = 2\chi_{(0,1/2]} + 1\chi_{(1/2,1]}$, $f_3 = 4\chi_{(0,1/4]} + 2\chi_{(1/4,1/2]} + 1\chi_{(1/2,1]}$, $f_4 = 8\chi_{(0,1/8]} + 4\chi_{(1/8,1/4]} + 2\chi_{(1/4,1/2]} + 1\chi_{(1/2,1]}$... We see that $fg \geq f_n, n = 1, 2, ...$, therefore by Proposition 2.3.4, $\int fg \geq \int f_n = \frac{1+n}{2} \Rightarrow \int fg = \infty$.

2.3.3

We follow the hint. Let A be the Lebesgue-nonmeasurable subset of $(0, 1)$, and let B be $(0, 1) - A$. To see that f is not Lebesgue integrable, note that $f^+ = \chi_A$ which is not event measurable (hence not integrable) by Examples 2.1.2 (d).

2.4.2

We notice from the relation $f_1 \leq f_2 \leq ...$ and $f = \lim f_n$ both holding a.e., we have that a.e., $0 \leq f_1^+ \leq f_2^+ \leq ... \leq f^+$, $f_1^- \geq f_2^- \geq ... \geq f^- \geq 0$, and $f^+ = \lim_n f_n^+, f^- = \lim_n f_n^-$. We have that

$$\int f^+ = \lim_n \int f_n^+,$$

by Theorem 2.4.1. We also have

$$\infty > \int f^- = \lim_n \int f_n^-$$

by Theorem 2.4.5, where f_1^- serves as the function g in 2.4.5. Also, by Theorem 2.4.5, each f_n^- is integrable, hence a finite integral. Now, we have that

$$\int f = \int f^+ - \int f^- = \left(\lim_n \int f_n^+\right) - \left(\lim_n \int f_n^-\right) = \lim_n \int f_n^+ - \int f_n^- := \lim_n \int f_n.$$

2.4.6

(a). Notice $f(x) = \sum_{n=1}^{\infty} \chi_{\{y:f(y)\geq n\}}(x)$. Define $f_m(x) = \sum_{n=1}^{m} \chi_{\{y:f(y)\geq n\}}(x)$, then the integral of each f_m is $\sum_{n=1}^{m} \mu(\{x : f(x) \geq n\})$. By Theorem 2.4.1 The Monotone Convergence Theorem, we have that $\int f d\mu = \sum_{n=1}^{\infty} \mu(\{x : f(x) \geq n\})$.

(b). Suppose f is integrable. By Corollary 2.3.14, $f < \infty$ a.e. We define f' as a function that agrees with f except that $f'(x)$ takes 0 if $f(x) = \infty$. It's easy to check that f' is measurable. By Proposition 2.3.9, $\int f' = \int f < \infty$. We define $g(x)$ as the floor function $\lfloor f'(x) \rfloor$. Notice that $g = \sum_{n=1}^{\infty} n\chi_{f'(x)\in[n,n+1)}$ and is easily seen to be measurable. We have that $0 \leq g \leq f$. By Proposition 2.3.4, we know $\int g$ is finite, and since g is integer-valued, by part (a) we know

$$\begin{aligned}\infty > \int g d\mu = \sum_{n=1}^{\infty} \mu(\{x : g(x) \geq n\}) &\geq \sum_{n=1}^{\infty} \mu(\{x : f(x) - 1 \geq n\}) \\ &= \sum_{n=2}^{\infty} \mu(\{x : f(x) \geq n\}) \Rightarrow \\ \sum_{n=1}^{\infty} \mu(\{x : f(x) \geq n\}) < \infty, &\end{aligned}$$

where the last inequality is because μ is a finite measure so $\mu(\{x : f(x) \geq 1\}) < \infty$.

Now we turn to the other direction. Suppose the series $\sum_{n=1}^{\infty} \mu(\{x : f(x) \geq n\})$ is convergent. We define $h = \lceil f \rceil$ the ceiling function composed with f, and notice that $f \leq h \leq f + 1$. Then since μ is finite, we know $\int h = \sum_{n=1}^{\infty} \mu(\{x : h(x) \geq n\}) \leq \sum_{n=1}^{\infty} \mu(\{x : f(x) + 1 \geq n\}) = \sum_{n=0}^{\infty} \mu(\{x : f(x) \geq n\})$, which is convergent. Therefore, $\int f \leq \int h < \infty$.

2.4.7

(a) That this mapping is nonnegative and assigns zero to Ø is easy to check. The countable additivity of this mapping can be checked applying the Monotone Convergence Theorem.

(b), (c). First suppose $f = \chi_B$ for some $B \in \mathcal{B}$. Then the mapping in (b) is simply $K(x, B)$ which is $\mathcal{A}$-measurable so (b) holds. It's also trivial to check that (c) holds. Now suppose f is a nonnegative simple function that is finite, i.e., $f = \sum_{n=1}^{m} \alpha_n \chi_{B_n}, \alpha_n > 0, \alpha_n \in \mathbb{R}, B_n \in \mathcal{B}$. By Proposition 2.3.4, Proposition 2.1.6, (b) holds. By Proposition 2.3.4, we can also check that (c) holds. Finally suppose $f \in [0, \infty]$. By Proposition 2.1.8, find a sequence (f_n) of nonnegative simple functions that are finite. By Proposition 2.4.1, Proposition 2.1.5, we know that (b) holds. By Proposition 2.4.1, it can also be seen that (c) holds.

2.4.8

(a), (b). First suppose $f = \chi_B$ for some $B \in \mathcal{B}$. By (ii), the measurability holds. By Proposition

2.3.4, $\int \chi_B K(x,dy) \leq \int \chi_Y K(x,dy) = K(x,Y) \geq 0$. Also by the assumption that $\sup_x\{K(x,Y)\} < \infty$, we know $\int \chi_B K(x,dy)$ is uniformly bounded for all x so (a) holds. It's easy to check (b) holds. Then suppose f is a real valued simple function, i.e., $f = \sum_{n=1}^m \alpha_n \chi_{B_n}, \alpha_n \neq 0, \alpha_n \in \mathbb{R}, B_n \in \mathcal{B}$. By Proposition 2.3.6, we know $\int f(y)K(x,dy) = \sum_{n=1}^m \alpha_n \int \chi_{B_n} K(x,dy)$, and by Proposition 2.1.7, the mapping from x into $\int f(y)K(x,dy)$ is measurable. To see boundedness, note that

$$\sum_{n=1}^m \alpha_n \int \chi_{B_n} K(x,dy) \leq \sum_{n=1}^m \alpha_n \int \chi_Y K(x,dy) = \sum_{n=1}^m \alpha_n K(x,Y),$$

which is bounded by the assumption $\sup_x\{K(x,Y)\} < \infty$. This means (a) holds. Now we check for (b). Recall that we have shown for any $B \in \mathcal{B}$,

$$\int \chi_B K(x,dy) = K(x,B)$$

is uniformly bounded for all x. Therefore,

$$\int \chi_B \nu(dy) = \nu(B) = \int K(x,B)\mu(dx) < \infty,$$

because of the uniform boundedness and that μ is finite. In other words, characteristic functions are integrable under the measure ν. Therefore, by Proposition 2.3.6,

$$\int f(y)\nu(dy) = \sum_{n=1}^m \alpha_n \int \chi_{B_n} \nu(dy) = \sum_{n=1}^m \alpha_n \int K(x,B_n)\mu(dx),$$

while recall $\int f(y)K(x,dy) = \sum_{n=1}^m \alpha_n \int \chi_{B_n} K(x,dy)$ so

$$\int \Big(\int f(y)K(x,dy) \Big) \mu(dx) = \int \sum_{n=1}^m \alpha_n \int \chi_{B_n} K(x,dy)\mu(dx) = \int \sum_{n=1}^m \alpha_n K(x,B_n)\mu(dx).$$

Now we only need to justify the exchange of summation and integration. But recall we already showed that $K(x,B_n) = \int \chi_B K(x,dy) \leq \int \chi_Y K(x,dy) = K(x,Y) \leq \sup_x\{K(x,Y)\} < \infty$, and so $K(x,B_n)$ is actually uniformly bounded for all x, B_n, and therefore integrable under the finite μ. Now the exchange of integration and summation is justified by Proposition 2.3.6.

Now suppose f is a measurable function bounded by $M > 0$. Notice since μ is finite, the constant function M is integrable. By the remark after Proposition 2.1.8, there is a sequence of real valued simple functions (f_n) whose pointwise limit is f. Without loss of generality, M can be chosen to be large so that $M \geq |f_n|, \forall n$. By Theorem 2.4.5, Proposition 2.1.5, (a) holds. We skip the details in checking (b), but it's worth noticing that by Theorem 2.4.5 (DCT), we have

$\int f_n K(x,dy) \to \int f K(x,dy)$, and by DCT again,

$$\int \lim\Big(\int f_n K(x,dy)\Big)\mu(dx) = \lim \int \Big(\int f_n K(x,dy)\Big)\mu(dx).$$

2.4.9

Take any sequence $(t_n) \in [0,\infty), (t_n) \uparrow \infty$. By DCT, it can be seen $\int f = \lim_n f_{t_n}$. Therefore, $\int f = \lim_{t\to\infty} f_t$.

2.4.10

Let $g(x,t) = e^{tx} f(x)$. We claim that $\partial_t g(x,t) = x\, e^{tx} f(x) \in L^1$. By the hint, for all $u \in \mathbb{R}$,

$$|e^u - 1| \le |u|\, e^{|u|},$$

which is trivial to check by writing down the Taylor expansion of e^u. Fix $\delta > 0$ small such that $t \pm 2\delta \in I$ and apply this with $u = \delta|x| > 0$ to get

$$\delta|x| \le e^{\delta|x|} - 1 \le \delta|x|\, e^{\delta|x|} \quad \Longrightarrow \quad |x| \le \frac{e^{\delta|x|}-1}{\delta} \le \frac{1}{\delta}\, e^{\delta|x|}.$$

Then, for any $t \in \mathbb{R}$,

$$\begin{aligned} |x|\, e^{tx}\, |f(x)| &\le \frac{1}{\delta}\, e^{\delta|x|}\, e^{tx}\, |f(x)| \\ &= \frac{1}{\delta}\Big(e^{(t+\delta)x}\, \mathbf{1}_{\{x\ge 0\}} + e^{(t-\delta)x}\, \mathbf{1}_{\{x<0\}}\Big)\, |f(x)|. \end{aligned}$$

By assumption $e^{(t\pm\delta)x} f(x) \in L^1$, hence the right-hand side is integrable. Therefore $x\, e^{tx} f(x) \in L^1$.

Differentiating under the integral Consider the difference quotient

$$\Phi_h(x) := e^{tx}\, \frac{e^{hx}-1}{h}\, f(x) \quad \Longrightarrow \quad \Phi_h(x) \xrightarrow[h\to 0]{} x\, e^{tx} f(x) \quad \text{pointwise.}$$

For $0 < |h| \le \delta$, the hint gives

$$\Big|\frac{e^{hx}-1}{h}\Big| \le |x|\, e^{|h||x|} \le \frac{1}{\delta}\, e^{\delta|x|}\, e^{|h||x|} \le \frac{1}{\delta}\, e^{2\delta|x|},$$

hence

$$|\Phi_h(x)| \le \frac{1}{\delta}\, e^{tx}\, e^{2\delta|x|}\, |f(x)| = \frac{1}{\delta}\Big(e^{(t+2\delta)x}\, \mathbf{1}_{\{x\ge 0\}} + e^{(t-2\delta)x}\, \mathbf{1}_{\{x<0\}}\Big)\, |f(x)|.$$

Again the right-hand side is integrable by the assumption with $s = t \pm 2\delta$. Thus by the dominated convergence theorem,

$$\frac{d}{dt}\int_{\mathbb{R}} e^{tx} f(x)\,d\lambda = \int_{\mathbb{R}} x\,e^{tx} f(x)\,d\lambda.$$

2.4.11

Since it's written as $|f_n - f|$, we have to assume that f_n, f never both take the value ∞ at the same x. By Corollary 2.3.14, and some simple arguments, it's easily seen that $f_n \to f, f_n < \infty, \forall n, f < \infty$ hold a.e. Assume they don't all hold on possible a null set E^c, so they all hold on E. Notice

$$|f - f_n|^+ = \max\{0, f - f_n\} \le f$$

holds on E, and notice $|f - f_n|^+ \to 0$ on E. By DCT,

$$\lim_n \int |f - f_n|^+ = \int \lim_n |f - f_n|^+ = \int 0 = 0.$$

It's given that $\int f = \lim_n \int f_n$, and since $f_n, f \in \mathcal{L}^1$, we can subtract $\int f$ from both sides and get

$$0 = \lim_n \int f_n - \int f = \lim_n \int f_n - f$$

so $0 = \lim_n \int f - f_n = \lim_n \int |f - f_n|^+ - |f - f_n|^-$. Now recall $\lim_n \int |f - f_n|^+ = 0$, and we know $\lim_n |f - f_n| = 0$.

2.5.3

We follow the hint and consider such B. In Proposition 2.1.11 we saw B is not a Borel set and so χ_B is not Borel measurable (Examples 2.1.2 (d)). Recall we denote the Cantor set by K, and χ_B is constant, i.e., zero, on the open set $\mathbb{R} - K$. In other words, for any $x \in \mathbb{R} - K$, find an open neighborhood of x that is contained in $\mathbb{R} - K$, then χ_B is constant in this neighborhood, so is continuous at x. Therefore, if we consider the interval $[0, 1]$, then χ_B is continuous almost everywhere there (except for possibly K, which has measure zero). By Theorem 2.5.4, it is Riemann integrable on $[0, 1]$.

2.5.5

Consider $f_n := f\chi_{[a_n, b_n]}$ where $(a_n) \searrow -\infty, (b_n) \nearrow \infty$. Then $f_n \to f, |f_n| \le |f|$. By DCT, we have that $\int f d\lambda = \lim_n \int f_n d\lambda$.

2.5.8

$\frac{n}{n+j} = \frac{1}{1+j/n}$. The mean of these numbers as j ranges from 1 to n equals the Riemann sum of this integral, with subinterval lengths of $[a_{i-1}, a_i]$ being uniformly $1/n$ and tags x_i being i/n. By

Proposition 2.5.7, we have the desired result.

2.6.1

By Example 2.6.5, we know the mapping $x \mapsto (f(x), g(x))$ is measurable. Consider the map from $\mathbb{R}^2$ into $\mathbb{R}$ defined by $h((x,y)) = x + y$, which is easily seen to be continuous, and hence the measurability by Example 2.1.2 (a). The composition map $x \mapsto h(f(x), g(x)) = f(x) + g(x)$ is therefore measurable. Similar arguments hold for fg.

2.6.2

Identify the mapping $x \mapsto f(x) = f_1(x) + if_2(x)$ with $x \mapsto (f_1(x), f_2(x))$, which is measurable by Example 2.6.5. Consider the map from $\mathbb{R}^2$ into $\mathbb{R}$ defined by $h((x,y)) = \sqrt{x^2 + y^2}$, which is continuous so is measurable. The composition map $x \mapsto |f(x)| = h(f(x)) = \sqrt{f_1^2(x) + f_2^2(x)}$ is therefore measurable.

2.6.3

Suppose $f(x) = f_1(x) + if_2(x), g(x) = g_1(x) + ig_2(x), f_1, f_2, g_1, g_2 \in \mathbb{R}$. Then $(f/g)(x) = \frac{f_1 + if_2}{g_1 + ig_2}(x) = \frac{(f_1+if_2)(g_1-ig_2)}{g_1^2+g_2^2}(x) = (\frac{f_1g_1+f_2g_2}{g_1^2+g_2^2}(x), \frac{f_2g_1-f_1g_2}{g_1^2+g_2^2}(x))$, which is measurable by Example 2.6.5, Propositions 2.1.6, 2.1.7.

2.6.4 (a)

We first check that $\mathcal{B}(\overline{\mathbb{R}})$ includes all sets in this form. In the following, $t \in \mathbb{R}$. $[-\infty, t] = (-\infty, t] \cup \{-\infty\}$, and so by the definition of $\mathcal{B}(\overline{\mathbb{R}})$ on page 74, is included in $\mathcal{B}(\overline{\mathbb{R}})$. Consider any sigma algebra $\mathcal{S}$ on $\overline{\mathbb{R}}$ that includes all sets of the form $[-\infty, t]$, we will check that $\mathcal{S} \supset \mathcal{B}(\overline{\mathbb{R}})$. Consider the sequence $[-\infty, -n], n = 1, 2, ...$ and their intersection, we can see that $\{-\infty\} \in \mathcal{S}$. Consider the union of such sequence, we see $[-\infty, \infty)$ and therefore $\{\infty\}$, is in $\mathcal{S}$. Then, $(-\infty, t] = [-\infty, t] \cap \{-\infty\}^c \in \mathcal{S}$. And, since $\mathcal{B}(\mathbb{R})$ is generated by sets of the form $(-\infty, t]$, we know $\mathcal{S} \supset \mathcal{B}(\mathbb{R})$. Now we have that $\mathcal{S} \supset \mathcal{B}(\mathbb{R})$ and $\mathcal{S} \supset \{-\infty, \infty\}$, so by definition on page 74 again, we know $\mathcal{S} \supset \mathcal{B}(\overline{\mathbb{R}})$.

2.6.5

Since f is a function and it doesn't make sense for a domain to be empty, we assume X is nonempty. Therefore, Y is also nonempty.

First, notice some identities that are easy to verify by logic (i.e., showing LHS includes RHS and RHS includes LHS): $f^{-1}(Y) = X$, $f^{-1}(\cap_{i\in I} B_i) = \cap_{i\in I} f^{-1}(B_i)$, $f^{-1}(\cup_{i\in I} B_i) = \cup_{i\in I} f^{-1}(B_i)$, and $f^{-1}(B^c) = \left(f^{-1}(B)\right)^c$, where I is an arbitrary index set and here $B_i, B \subset Y$.

Once we've established the above identities, (a) and (b) are trivial.

(c). See more discussions here. We here follow the top rated answer. We denote $\{f^{-1}(C) : C \in \mathcal{C}\}$ by $f^{-1}(\mathcal{C})$ and $\{f^{-1}(B) : B \in \sigma(\mathcal{C})\}$ by $f^{-1}(\sigma(\mathcal{C}))$. That the RHS $f^{-1}(\sigma(\mathcal{C}))$ includes $f^{-1}(\mathcal{C})$ is obvious. Therefore, $f^{-1}(\sigma(\mathcal{C})) \supset \sigma\big(f^{-1}(\mathcal{C})\big)$. Now we turn to the other direction of inclusion.

Let $\mathcal{D} := \{A \subset Y : f^{-1}(A) \in \sigma\big(f^{-1}(\mathcal{C})\big)\}$. By this definition, if a subset A of Y is in $\mathcal{D}$, then $f^{-1}(A) \in \sigma\big(f^{-1}(\mathcal{C})\big)$. Notice $\mathcal{D} \supset \mathcal{C}$. Recalling the identities we showed before, it is easy to check that $\mathcal{D}$ is a sigma-algebra. Therefore, $\mathcal{D} \supset \sigma(\mathcal{C})$. Taking inverse image, we see $f^{-1}(\mathcal{D}) \supset f^{-1}\big(\sigma(\mathcal{C})\big)$. But $f^{-1}(\mathcal{D}) = \{f^{-1}(A) : A \in \mathcal{D}\} \subset \sigma\big(f^{-1}(\mathcal{C})\big)$. This means $f^{-1}\big(\sigma(\mathcal{C})\big) \subset \sigma\big(f^{-1}(\mathcal{C})\big)$.

2.6.6

(a). Recall in chapter 1 we saw such F is bounded, nondecreasing and right-continuous, and that $\lim_{x\to-\infty} F(x) = 0$. That g is nondecreasing is easy to see. For any $x \in (0, \lim_{x\to\infty} F(x))$, considering the limit of F as t goes to infinity, we see there is $t \in \mathbb{R} : F(t) \geq x$. So the set $\{t \in \mathbb{R} : F(t) \geq x\}$ is not empty. We also recall the limit of F as t goes to minus infinity is 0, so there is a threshold $t_0 \in \mathbb{R}$ such that for all $t < t_0$, we have $F(t) < x$. Therefore, the set $\{t \in \mathbb{R} : F(t) \geq x\}$ is not empty and bounded below, so it has a finite infimum, which means $g(x)$ is finite.

Now we show g is Borel-Borel measurable. We claim that for x, the infimum of the set $\{t \in \mathbb{R} : F(t) \geq x\}$, denoted l_x, satisfies $F(l_x) \geq x$, i.e., the infimum itself belongs to this set. To see this, take a sequence (t_n) from this set that decreases to l_x, and recall $F(t_n) := \mu((-\infty, t_n]) \geq x$. Therefore, by the upper continuity of measure (Proposition 1.2.5), we have $F(l_x) \geq x$. So $g(x) = l_x$.

Now, for any real number $t, x \in g^{-1}((t,\infty))$ iff $l_x = g(x) > t$ iff $F(t) < x$. This means $g^{-1}((t,\infty)) = (F(t),\infty)$ which is a Borel set. Therefore, g is (Borel-Borel) measurable.

(b). Consider a set $B = (-\infty, b], b \in \mathbb{R}$. Now $g^{-1}(B) = g^{-1}\big((b,\infty)^c\big) = \big(g^{-1}((b,\infty))\big)^c = (-\infty, F(b)]$. Now consider $A = (-\infty, a], a \in \mathbb{R}, a < b$. Then $B - A = (a, b]$, and $g^{-1}(B - A) = (F(a), F(b)]$ (see the discussion above about operations inside inverse mapping in solving exercise 2.6.5). Therefore, $\lambda g^{-1}(B-A) = F(b) - F(a)$. Also, it is easy to see that $\mu(B-A) = F(b) = F(a)$. By Corollary 1.6.3 and Proposition 1.1.4, we have the desired result.

3.1.2

Suppose $f_n \to f$ in measure. Then for all positive ϵ, there exists N such that

$$n > N \implies \mu(\{x : |f_n(x) - f(x)| > \epsilon\}) < 1,$$

which must be the case that $|f_n(x) - f(x)| \leq \epsilon, \forall x \in \mathbb{N}$, hence uniform convergence. The converse is similarly easy to show.

3.1.3

There are 0-measure sets S_1, S_2 such that outside S_1, $f_n \to f$, and outside S_2, g is continuous. Take $x \in (S_1 \cup S_2)^c$, then $f_n(x) \to f(x)$, and since g is continuous at $f(x)$, we have $g(f_n(x)) \to g(f(x))$.

3.1.4

Take $\epsilon > 0$ and define $A_{n,\epsilon} = \{x \in X : |f_n(x) - f(x)| \geq \epsilon\}$. By Markov's inequality, Proposition 2.3.10, $\mu(A_{n,\epsilon}) \leq \frac{1}{\epsilon}\int |f_x - f|$. Therefore, $\sum_n \mu(A_{n,\epsilon}) \leq \frac{1}{\epsilon}\sum_n \int |f_x - f| < \infty$. By Borel-Cantelli Lemma, i.e., exercise 1.2.9, we have $\mu(\limsup_n A_{n,\epsilon}) = 0$. This means for almost all $x \in X$, $|f_n(x) - f(x)| \geq \epsilon$ holds for at most finitely many n. Define

$$E_k := \{x : \exists N(x,k) \text{ s.t. } |f_n(x) - f(x)| \leq \frac{1}{k}, \forall n \geq N(x,k)\}.$$

We know E_k^c is a null set for all k, and so is $\cup E_k^c$. Therefore, almost all x belongs to $\cap_k E_k$, i.e., $|f_n(x) - f(x)| \to 0$ a.e.

3.2.2

(a). Notice for $f = 1, \|f\| = 0$. (b). If $\|f\| = 0$, then we must have $f(0) = 0$ and $\int_0^1 |f'(x)|\,dx = 0$. If $f(y) \neq 0$ for some $y \in (0,1]$, then by Baby Rudin theorem 6.13 (that if f is integrable on $[a,b]$ then so is $|f|$) and the fundamental theorem of calculus (Baby Rudin 6.21), we have $\int_0^y |f'(x)|\,dx \geq |\int_0^y f'(x)\,dx| = |f(y)|$, a contradiction.

3.2.3

a

3.2.4

a

3.2.7

(a). The hint is enough to explain. (b). Notice

$$\|ax\| = \sqrt{(ax,ax)} = \sqrt{a(x,ax)} = \sqrt{a(ax,x)} = \sqrt{a^2(x,x)} = a\|x\|.$$

For triangle inequality, we show $\|x+y\|^2 \leq (\|x\| + \|y\|)^2$ by expanding it.

3.2.8, 3.2.9

Straightforward.

3.2.10

(a). The hint helps showing the closure under addition. Other axioms of a vector space are trivial to check.

(b). The only nontrivial part is to check that the range of the function is $\mathbb{R}$. To show that $(\{x_n\},\{y_n\})$ is convergent (i.e., $\in \mathbb{R}$) for any $\{x_n\},\{y_n\} \in l^2$, we note that $(\{x_n\},\{y_n\})$ is even absolutely convergent. In fact, $|x_n y_n| \leq \max\{x_n^2, y_n^2\} \leq x_n^2 + y_n^2$. (c). We know every absolutely convergent series in $\mathbb{R}$ is convergent.

3.2.11

Following the hint, since C is nonempty, we have that $0 \leq d < \infty$. Pick a sequence $(z_n) \subset C$ with $\|z_n\| \downarrow d$. For any $\epsilon > 0$, there exists $N \in \mathbb{N}$ such that $n > \mathbb{N} \implies \|z_n\|^2 < d^2 + \epsilon/4$. By exercise 8 (a), we have that for $n, m > N$,

$$\|z_n - z_m\|^2 = 2\|z_n\|^2 + 2\|z_m\|^2 - \|z_n + z_m\|^2 < 4d^2 + \epsilon - 4d^2 = \epsilon.$$

Since the space H is complete and C is closed, there is $z := \lim z_n \in C$. Note that the norm is a continuous function on H: if $x_n \to x$, i.e., $d(x_n, x) := \|x_n - x\| \to 0$, then by triangle inequality, $\|x_n - x\| \geq |\|x_n\| - \|x\||$, so we have $\|x_n\| \to \|x\|$.

Now if $\|z\| > d$, there is a contradiction since $\|z_n\| \to \|z\|$. The uniqueness follows from the hint.

3.2.12

(a) follows from the hint. (b). Notice $y + tz \in H_0$, therefore the unique minimizer of f is $t = 0$.

3.2.13

(a). Since it's a Banach space (complete), we know $\sum v_k \to v$ for some $v \in V$. Denote by s_n the partial sums $\sum_{k=1}^{n} v_k$. For $\epsilon > 0$, there is integer N_1 such that $n > N_1 \implies \|v - s_n\| < \epsilon/2$. There is integer $N_2 > N_1$ such that $n > N_2 \implies \|v_n\| + \|v_{n+1}\| + ... < \epsilon/2$. Now take $F_\epsilon = \{1, 2, ..., N_2\}$.
(b). It's easy to check unconditional convergence implies convergence (Take $N := \max F_\epsilon$ and consider F as $\{1, ..., N+1\}, \{1, ..., N+2\}...$). We first assume (v_k) are all nonnegative. Then it's easy to see if $\sum v_k$ converges unconditionally to v, $\sum |v_k| = \sum v_k = v$. Similarly, if (v_k) are all negative, then unconditional convergence of $\sum v_k$ to v implies $\sum |v_k| = -\sum v_k = -v$. Now for a general sequence (v_k), if the sequence has only finitely many nonnegative (or negative) terms, it's easy to check unconditional convergence implies absolute convergence.

Now assume (v_k) has infinitely many nonnegative terms and infinitely many negative terms and $\sum v_k$ converges unconditionally to v. Suppose $\sum v_k$ does not converge absolutely, then without loss of generality, assume the sum of all nonnegative terms of (v_k), denoted by $\sum_i v_{k_i}, k_1 < k_2 < ...$, is ∞. Also, denote the negative terms of (v_k) by $v_{h_i}, h_1 < h_2 < ...$

According to the definition of unconditional convergence, there exists integer N such that all finite sets $F \supset \{1, ..., N\}$ are such that $|\sum_{k \in F} v_k - v| < 1 \implies |\sum_{k \in F} v_k| < |v| + 1$. Let J be the smallest integer such that $k_J > N$. Define $F_n = \{1, 2, ..., N, k_J, k_{J+1}, ..., k_{J+n}\}, \forall n \geq 1$, and we have that $|\sum_{k \in F_n} v_k| < |v| + 1$, which is a contradiction.
(c). Consider c^0. Take $x_n = (x_n^1, x_n^2, ...) = (0, 0, ..., 0, 1/n, 0, 0, ...) \in c^0$ as the sequence whose n-th term is $1/n$ and whose other terms are zero. Take the norm as the sup norm as in exercise 4. Then we can check $\sum_n x_n$ converges to the sequence $(1, 1/2, 1/3, ...)$ unconditionally under the sup norm.

However, the series does not converge absolutely.

3.3.9

(a). We note that by Hölder's inequality, and the fact that $|f|^{p_1} \in L^{p_2/p_1}$,

$$\int |f|^{p_1} \times 1 \leq \left(\int (|f|^{p_1})^{p_2/p_1} \right)^{p_1/p_2} \left(\int 1^{p_2/(p_2-p_1)} \right)^{(p_2-p_1)/p_2} = \left(\int |f|^{p_2} \right)^{p_1/p_2}.$$

(b). By assumption, $\lim_{n\to\infty} \int |f_n - f|^{p_2} = 0$, meaning $\lim_{n\to\infty} \left(\int |f_n - f|^{p_2} \right)^{1/p_2} = 0$. And by the conclusion of part (a), the result follows.

5.1.3

Wlog suppose $d \geq 2, a_d \neq 0$. Denote the hyperplan by H. Then by Theorem 5.1.4,

$$\begin{aligned}
\lambda^d(H) &= \int_{\mathbb{R}^{d-1}} \lambda(H_{(x_1,\ldots,x_{d-a})})\lambda^{d-1}(d(x_1, \ldots, x_{d-1})) \\
&= \int_{\mathbb{R}^{d-1}} \lambda(\frac{b - a_1x_1 - \ldots - a_{d-1}x_{d-1}}{a_d})\lambda^{d-1}(d(x_1, \ldots, x_{d-1})) \\
&= \int_{\mathbb{R}^{d-1}} 0\, \lambda^{d-1}(d(x_1, \ldots, x_{d-1})) \\
&= 0.
\end{aligned}$$

5.2.4 (by Chat GPT)

Let $(X, \mathcal{A})$ and $(Y, \mathcal{B})$ be measurable spaces. Assume μ_1, μ_2 are finite measures on $(X, \mathcal{A})$ and ν_1, ν_2 are finite measures on $(Y, \mathcal{B})$ with

$$\mu_2 \ll \mu_1, \qquad \nu_2 \ll \nu_1.$$

By the Radon–Nikodym theorem there exist nonnegative measurable functions

$$f = \frac{d\mu_2}{d\mu_1} \in L^1(X, \mathcal{A}, \mu_1, \mathbb{R}), \qquad g = \frac{d\nu_2}{d\nu_1} \in L^1(Y, \mathcal{B}, \nu_1, \mathbb{R})$$

such that

$$\mu_2(A) = \int_A f\, d\mu_1, \qquad \nu_2(B) = \int_B g\, d\nu_1.$$

Step 1: Absolute continuity of product measures.

For $A \in \mathcal{A}$ and $B \in \mathcal{B}$,

$$\begin{aligned}
(\mu_2 \times \nu_2)(A \times B) &= \mu_2(A)\, \nu_2(B) \\
&= \left(\int_A f \, d\mu_1\right) \left(\int_B g \, d\nu_1\right) \\
&= \int_{A \times B} f(x) g(y) \, d(\mu_1 \times \nu_1)(x, y),
\end{aligned}$$

where we used Tonelli's theorem in the last step. To see $fg(x, y) := f(x)g(y)$ is $\mathcal{A} \times \mathcal{B}$-measurable, let $\pi_X : X \times Y \to X$ and $\pi_Y : X \times Y \to Y$ be the coordinate projections. Then π_X is $(\mathcal{A} \times \mathcal{B}, \mathcal{A})$-measurable and π_Y is $(\mathcal{A} \times \mathcal{B}, \mathcal{B})$-measurable. Since $f : (X, \mathcal{A}) \to (\mathbb{R}, \mathcal{B}(\mathbb{R}))$ and $g : (Y, \mathcal{B}) \to (\mathbb{R}, \mathcal{B}(\mathbb{R}))$ are measurable, the functions

$$F(x, y) := f(\pi_X(x, y)) = f(x), \qquad G(x, y) := g(\pi_Y(x, y)) = g(y)$$

are $(\mathcal{A} \times \mathcal{B}, \mathcal{B}(\mathbb{R}))$-measurable. Consider the multiplication map $m : \mathbb{R}^2 \to \mathbb{R}$ given by $m(u, v) = uv$. This map is continuous, hence $(\mathcal{B}(\mathbb{R}^2), \mathcal{B}(\mathbb{R}))$-measurable. Moreover, the map $H : X \times Y \to \mathbb{R}^2$ defined by $H(x, y) = (F(x, y), G(x, y))$ is measurable. Therefore the composition

$$(x, y) \longmapsto m(H(x, y)) = m(F(x, y), G(x, y)) = f(x)g(y)$$

is $(\mathcal{A} \times \mathcal{B}, \mathcal{B}(\mathbb{R}))$-measurable.

Now define a measure ρ on $\mathcal{A} \times \mathcal{B}$ by

$$\rho(E) = \int_E f(x)g(y) \, d(\mu_1 \times \nu_1)(x, y).$$

Then ρ agrees with $\mu_2 \times \nu_2$ on measurable rectangles. Since rectangles form a π-system generating $\mathcal{A} \times \mathcal{B}$, Corollary 1.6.4 implies

$$\mu_2 \times \nu_2 = \rho.$$

Hence

$$\mu_2 \times \nu_2 \ll \mu_1 \times \nu_1.$$

Step 2: Radon–Nikodym derivative.

From the definition of ρ we conclude that for every $E \in \mathcal{A} \times \mathcal{B}$,

$$(\mu_2 \times \nu_2)(E) = \int_E f(x)g(y) \, d(\mu_1 \times \nu_1)(x, y).$$

Therefore, by uniqueness of the Radon–Nikodym derivative,

$$\boxed{\frac{d(\mu_2 \times \nu_2)}{d(\mu_1 \times \nu_1)}(x, y) = \frac{d\mu_2}{d\mu_1}(x)\,\frac{d\nu_2}{d\nu_1}(y)} \qquad (\mu_1 \times \nu_1)\text{-a.e.}$$

Related derivatives.
Similarly,

$$\frac{d(\mu_2 \times \nu_1)}{d(\mu_1 \times \nu_1)}(x, y) = \frac{d\mu_2}{d\mu_1}(x), \qquad \frac{d(\mu_1 \times \nu_2)}{d(\mu_1 \times \nu_1)}(x, y) = \frac{d\nu_2}{d\nu_1}(y).$$